

Fourier Analysis: A Comprehensive Introduction

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Abstract

This comprehensive lecture series covers the fundamental theory and applications of Fourier analysis. Beginning with Fourier series for periodic functions and progressing through Fourier transforms, sampling theorems, the discrete Fourier transform (DFT), and the fast Fourier transform (FFT), this material provides a rigorous yet accessible treatment of transform methods essential for signal processing, differential equations, and engineering applications. The notes include both theoretical foundations and practical computational methods, with detailed proofs and worked examples throughout. This lecture note builds on *An Introduction to Fourier Analysis* by Ralph N. Morrison.

Notation Conventions

Throughout these notes, $i = \sqrt{-1}$ and differentials are written with the simple italic d , as in dt and $d\omega$. The continuous-time Fourier transform of $f(t)$ is denoted by $F(\omega)$. The complex Fourier-series coefficient of a periodic signal is denoted by $F(n)$ when only one periodic signal is under discussion. In the DFT/FFT chapters, subscript notation F_n is reserved for discrete DFT bins. When pulse spectra and periodic line spectra appear side by side, labels such as $\text{PUL}(\omega)$, $\text{PER}(n)$, or $F_p(n)$ are used only to keep the two continuous spectra visually distinct.

The notes use the unnormalized engineering sinc function

$$\text{sinc}(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

This convention is different from the normalized signal-processing convention $\sin(\pi x)/(\pi x)$; whenever sinc appears below, it means the unnormalized function defined here.

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2 Fourier Series for Periodic Functions

2.1 Orthogonality of Vectors and Functions

Two vectors are said to be orthogonal if their inner product is equal to zero. Let $\mathbf{u}$ and $\mathbf{v}$ be two real n -vectors. Their inner product is

$$(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n u_i v_i.$$

The vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if

$$(\mathbf{u}, \mathbf{v}) = 0.$$

Let

$$\mathcal{S} = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots\}$$

be a set of vectors. If

$$(\mathbf{u}_n, \mathbf{u}_m) = 0 \quad \text{whenever } n \neq m,$$

then the elements of $\mathcal{S}$ are mutually orthogonal and form an orthogonal basis for the space spanned by $\mathcal{S}$. Any vector in that space can be expressed as a linear combination of those basis vectors.

The concept of orthogonality and basis extends naturally to sets of functions. Consider the set

$$\mathcal{S} = \{f_1(t), f_2(t), f_3(t), \dots\}.$$

We say that the members of $\mathcal{S}$ form an orthogonal set over the interval $a < t < b$ if

$$\int_a^b f_n(t) f_m^*(t) dt = 0 \quad \text{whenever } n \neq m.$$

The integral above is the inner product of the pair of functions.

In what follows we will make repeated use of orthogonality. In particular, we shall show that the set

$$\mathcal{S} = \{\dots, e^{-i2\omega_0 t}, e^{-i\omega_0 t}, 1, e^{i\omega_0 t}, e^{i2\omega_0 t}, \dots\}$$

forms an orthogonal set over one period and hence forms a basis for functions lying in the space it spans. This leads to representing periodic current or voltage waveforms as linear combinations of these functions.

We begin with periodic functions; later we shall extend the techniques to nonperiodic waveforms, such as one-time pulses.

2.2 The Complex Exponentials

We recall several elementary facts from complex algebra that will be used frequently. Let

$$z = x + iy$$

be any point in the complex plane, where $i = \sqrt{-1}$. Then

$$z^* = x - iy,$$

$$\text{mod}(z) = |z| = \sqrt{x^2 + y^2} = \sqrt{zz^*},$$

$$\begin{aligned}\arg(z) &= \operatorname{atan2}(y, x), \\ z + z^* &= 2\Re(z) = 2x, \\ z - z^* &= 2i\Im(z) = 2iy, \\ e^{i\theta} &= \cos \theta + i \sin \theta, \\ \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2}, \\ \sin \theta &= \frac{e^{i\theta} - e^{-i\theta}}{2i}.\end{aligned}$$

2.3 Orthogonality of the Complex Exponentials

We consider the complex exponentials

$$e^{in\omega_0 t}, \quad n \in \mathbb{Z},$$

where $\omega_0 = 2\pi/T_0$ is a fundamental angular frequency. Taken together, they form the set

$$\{\dots, e^{-i2\omega_0 t}, e^{-i\omega_0 t}, 1, e^{i\omega_0 t}, e^{i2\omega_0 t}, \dots\}.$$

Theorem 1 (Orthogonality of the Complex Exponentials). *The complex exponentials $e^{in\omega_0 t}$ satisfy*

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{in\omega_0 t} e^{-im\omega_0 t} dt = \delta_{mn} = \begin{cases} 0, & n \neq m, \\ 1, & n = m, \end{cases}$$

where $T_0 = 2\pi/\omega_0$.

Proof. We have

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{in\omega_0 t} e^{-im\omega_0 t} dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{i(n-m)\omega_0 t} dt.$$

If $n \neq m$, let $p = n - m \in \mathbb{Z} \setminus \{0\}$. Then

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{ip\omega_0 t} dt = \frac{1}{T_0} \left[\frac{e^{ip\omega_0 t}}{ip\omega_0} \right]_{-T_0/2}^{T_0/2} = \frac{1}{T_0} \frac{e^{ip\omega_0 T_0/2} - e^{-ip\omega_0 T_0/2}}{ip\omega_0}.$$

Since $\omega_0 T_0 = 2\pi$,

$$e^{ip\omega_0 T_0/2} = e^{ip\pi}, \quad e^{-ip\omega_0 T_0/2} = e^{-ip\pi}.$$

Both exponentials are equal to the same number, namely $(-1)^p$. Their difference in the numerator is therefore zero, and the integral is zero.

If $n = m$, the integrand is 1, and we obtain

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} 1 dt = 1.$$

□

The proof above uses the symmetric interval $[-T_0/2, T_0/2]$. The same result holds for any interval of length T_0 .

2.4 Complex Fourier Series for Periodic Functions

Let $f(t)$ be a periodic function with period T_0 and assume that it can be expressed as an infinite sum of complex exponentials:

$$f(t) = \sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}.$$

The coefficients $F(n)$ are the complex Fourier coefficients and are to be determined. This assumption is reasonable for periodic functions: each exponential term has period T_0 or a divisor of T_0 , and thus the sum is periodic with period T_0 .

To determine $F(n)$, multiply both sides by $e^{-im\omega_0 t}$ and integrate over a full period:

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-im\omega_0 t} dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \left[\sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t} \right] e^{-im\omega_0 t} dt.$$

Interchanging summation and integration,

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-im\omega_0 t} dt = \sum_{n=-\infty}^{\infty} F(n) \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{i(n-m)\omega_0 t} dt.$$

By orthogonality of the complex exponentials, all terms with $n \neq m$ vanish, and the term with $n = m$ equals 1. Hence

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-im\omega_0 t} dt = F(m).$$

Relabeling m as n , we obtain the analysis formula

$$F(n) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-in\omega_0 t} dt.$$

Theorem 2 (Complex Fourier coefficients and reconstruction). *Let $f(t)$ be T_0 -periodic and integrable over one period. Then its complex Fourier coefficients are*

$$F(n) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-in\omega_0 t} dt.$$

The associated formal series is

$$\sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t}.$$

If $f \in L^2$ over one period, this series converges to f in the mean-square sense; under Dirichlet conditions, it converges pointwise as stated in the convergence subsection below.

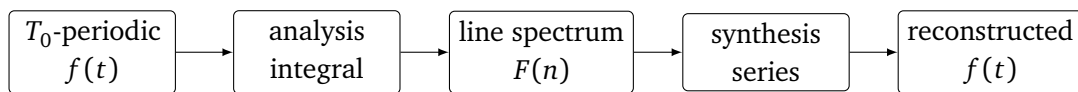


Figure 1: Fourier-series analysis converts one period of a waveform into coefficients $F(n)$; synthesis rebuilds the waveform from those coefficients.

The coefficients $F(n)$ are called the complex Fourier coefficients. The term $F(0)$ is called the average, or dc, value of the waveform (also the “zeroth harmonic”). Pairs of coefficients correspond to harmonics. Define

$$h(1) = F(1)e^{i\omega_0 t} + F(-1)e^{-i\omega_0 t},$$

which is the fundamental (first harmonic); the second harmonic is

$$h(2) = F(2)e^{i2\omega_0 t} + F(-2)e^{-i2\omega_0 t},$$

and, in general, the n th harmonic is

$$h(n) = F(n)e^{in\omega_0 t} + F(-n)e^{-in\omega_0 t}.$$

Definition 1. *The notation*

$$f(t) \longleftrightarrow F(n)$$

means that $f(t)$ has complex Fourier coefficients $F(n)$ given by the analysis formula above, and $F(n)$ synthesizes $f(t)$ via the corresponding Fourier series.

Example 1 (Complex Fourier Series of a Piecewise Constant Waveform). *Consider the periodic waveform $f(t)$, shown over one period in Figure 2, with period $T_0 = 4$, defined analytically over one period by*

$$f(t) = \begin{cases} 0, & -2 < t < -1, \\ 1, & -1 < t < 1, \\ 0, & 1 < t < 2, \end{cases} \quad \text{and} \quad f(t + T_0) = f(t).$$

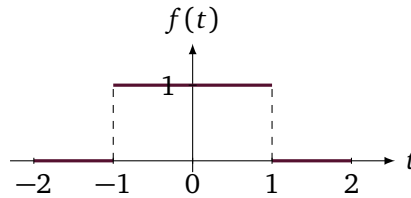


Figure 2: One period of the periodic rectangular waveform used in the example.

For this waveform, the fundamental frequency is

$$\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{2}.$$

The Fourier coefficients are

$$F(n) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-in\omega_0 t} dt = \frac{1}{4} \int_{-1}^1 e^{-in(\pi/2)t} dt.$$

Carrying out the integral,

$$\begin{aligned} F(n) &= \frac{1}{4} \left[\frac{e^{-in(\pi/2)t}}{-in(\pi/2)} \right]_{t=-1}^{t=1} = \frac{1}{4} \cdot \frac{2i}{n\pi} (e^{-in\pi/2} - e^{in\pi/2}) \\ &= \frac{i}{2\pi n} (e^{-in\pi/2} - e^{in\pi/2}). \end{aligned}$$

Since

$$e^{-in\pi/2} - e^{in\pi/2} = -2i \sin\left(\frac{n\pi}{2}\right),$$

we obtain

$$F(n) = \frac{1}{\pi n} \sin\left(\frac{n\pi}{2}\right), \quad n \neq 0.$$

For $n = 0$, we have

$$F(0) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) dt = \frac{1}{4} \int_{-1}^1 1 dt = \frac{1}{2}.$$

Thus,

$$F(n) = \begin{cases} \frac{1}{\pi n} \sin\left(\frac{n\pi}{2}\right), & n \neq 0, \\ \frac{1}{2}, & n = 0. \end{cases}$$

Using the sinc convention fixed in Notation Conventions, this can be written compactly as

$$F(n) = \frac{1}{2} \operatorname{sinc}\left(\frac{n\pi}{2}\right).$$

The corresponding series is

$$f(t) = \sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t} = \frac{1}{2} \sum_{n=-\infty}^{\infty} \operatorname{sinc}\left(\frac{n\pi}{2}\right) e^{in(\pi/2)t}.$$

This is the complex Fourier series representation of the periodic pulse waveform.

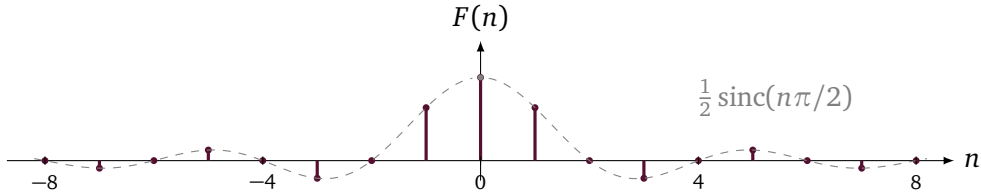


Figure 3: The Fourier coefficients of the rectangular waveform are samples of a sinc envelope.

2.5 Real and Imaginary Parts of the Fourier Coefficients

In general, the complex coefficients $F(n)$ can be written in Cartesian form

$$F(n) = A(n) + iB(n),$$

or in polar form

$$F(n) = |F(n)| e^{i\Theta(n)}.$$

The magnitude and phase are related to $A(n)$ and $B(n)$ by

$$|F(n)| = \sqrt{A(n)^2 + B(n)^2},$$

$$\Theta(n) = \arg(F(n)) = \operatorname{atan2}(B(n), A(n)),$$

where the phase is defined modulo 2π and is undefined when $F(n) = 0$.

The inverse relations are

$$A(n) = |F(n)| \cos[\Theta(n)],$$

$$B(n) = |F(n)| \sin[\Theta(n)].$$

When $f(t)$ is real, the coefficients satisfy additional symmetry properties.

Theorem 3. If $f(t)$ is real-valued and $F(n)$ are its complex Fourier coefficients, then

$$F(n)^* = F(-n) \quad \text{for all } n. \quad (2.1)$$

Proof. If $f(t)$ is real,

$$F(n)^* = \left[\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-in\omega_0 t} dt \right]^* = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{in\omega_0 t} dt.$$

Using $e^{in\omega_0 t} = e^{-i(-n)\omega_0 t}$,

$$F(n)^* = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-i(-n)\omega_0 t} dt = F(-n).$$

□

From $F(n)^* = F(-n)$ and $F(n) = A(n) + iB(n)$ we obtain, for real-valued $f(t)$:

$$|F(-n)| = |F(n)|,$$

$$A(-n) = A(n), \quad B(-n) = -B(n),$$

and, after choosing a consistent phase branch away from zeros,

$$\Theta(-n) \equiv -\Theta(n) \pmod{2\pi}.$$

Thus $A(n)$ and $|F(n)|$ are even functions of n , whereas $B(n)$ is odd. The phase is odd modulo 2π , with the usual caveat that phase is undefined where $F(n) = 0$.

2.6 Even and Odd Functions

Definition 2 (Even function). A function $g(n)$ is even if

$$g(-n) = g(n) \quad \text{for all } n.$$

In the continuous case, $g(t)$ is even if $g(-t) = g(t)$. Even functions have mirror symmetry about the vertical axis. Examples (as functions of t) include

$$g(t) = 3, \quad t^2, \quad \cos t, \quad t^4 \cos t, \quad e^t + e^{-t}.$$

An important property is

$$\int_{-a}^a g(t) dt = 2 \int_0^a g(t) dt \quad \text{if } g(t) \text{ is even.}$$

Definition 3 (Odd function). A function $g(n)$ is odd if

$$g(-n) = -g(n) \quad \text{for all } n.$$

Similarly, $g(t)$ is odd if $g(-t) = -g(t)$. Odd functions are antisymmetric with respect to the origin and satisfy $g(0) = 0$. Examples (as functions of t) include

$$g(t) = t, \quad \sin t, \quad t^3 \sin t, \quad e^t - e^{-t}.$$

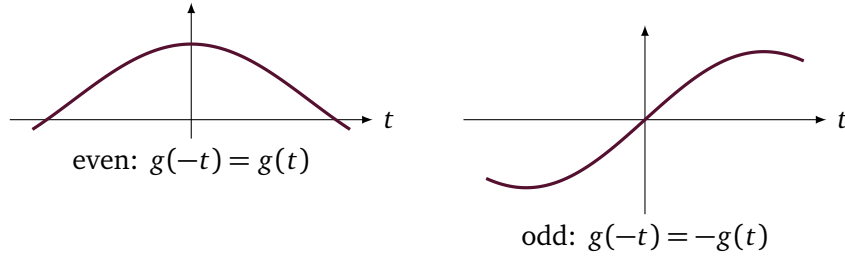


Figure 4: Even functions mirror across the vertical axis; odd functions rotate by 180° about the origin.

For odd functions,

$$\int_{-a}^a g(t) dt = 0 \quad \text{if } g(t) \text{ is odd.}$$

Products of even and odd functions satisfy:

$$\text{even} \times \text{even} = \text{even}, \quad \text{odd} \times \text{odd} = \text{even}, \quad \text{even} \times \text{odd} = \text{odd}.$$

Any function $f(t)$ can be decomposed into even and odd parts:

$$f_e(t) = \frac{1}{2}[f(t) + f(-t)],$$

$$f_o(t) = \frac{1}{2}[f(t) - f(-t)],$$

so that $f(t) = f_e(t) + f_o(t)$, with f_e even and f_o odd.

Theorem 4 (Even/odd decomposition of Fourier series coefficients). *Let $f(t)$ be a real-valued, T -periodic function with $f(t) \longleftrightarrow F(n)$, where $F(n) = A(n) + iB(n)$. Then*

$$f_e(t) \longleftrightarrow A(n), \quad f_o(t) \longleftrightarrow iB(n).$$

Proof. Let $F_e(n)$ and $F_o(n)$ denote the complex Fourier coefficients of $f_e(t)$ and $f_o(t)$, respectively:

$$F_e(n) = \frac{1}{T} \int_{t_0}^{t_0+T} f_e(t) e^{-in\omega_0 t} dt = \frac{1}{2T} \int_{t_0}^{t_0+T} [f(t) + f(-t)] e^{-in\omega_0 t} dt,$$

$$F_o(n) = \frac{1}{T} \int_{t_0}^{t_0+T} f_o(t) e^{-in\omega_0 t} dt = \frac{1}{2T} \int_{t_0}^{t_0+T} [f(t) - f(-t)] e^{-in\omega_0 t} dt.$$

We compute the integral involving $f(-t)$ by the change of variable $\tau = -t$. Since f is T -periodic, the choice of integration interval of length T is immaterial, and we may write

$$\begin{aligned} \int_{t_0}^{t_0+T} f(-t) e^{-in\omega_0 t} dt &= \int_{t_0}^{t_0+T} f(-t) e^{-in\omega_0 t} dt \quad (\tau = -t, d\tau = -dt) \\ &= \int_{-t_0-T}^{-t_0} f(\tau) e^{in\omega_0 \tau} d\tau \\ &= \int_{t_0}^{t_0+T} f(\tau) e^{in\omega_0 \tau} d\tau \\ &= \int_{t_0}^{t_0+T} f(t) e^{in\omega_0 t} dt. \end{aligned}$$

By direct substitution:

$$F_e(n) = \frac{1}{2T} \left[\int_{t_0}^{t_0+T} f(t) e^{-in\omega_0 t} dt + \int_{t_0}^{t_0+T} f(t) e^{in\omega_0 t} dt \right] = \frac{F(n) + F(-n)}{2}.$$

Similarly,

$$F_o(n) = \frac{1}{2T} \left[\int_{t_0}^{t_0+T} f(t) e^{-in\omega_0 t} dt - \int_{t_0}^{t_0+T} f(t) e^{in\omega_0 t} dt \right] = \frac{F(n) - F(-n)}{2}.$$

Because $f(t)$ is real-valued, its Fourier coefficients satisfy the conjugate symmetry (2.1), hence

$$\begin{aligned} F_e(n) &= \frac{F(n) + F^*(n)}{2} = \frac{[A(n) + iB(n)] + [A(n) - iB(n)]}{2} = A(n), \\ F_o(n) &= \frac{F(n) - F^*(n)}{2} = \frac{[A(n) + iB(n)] - [A(n) - iB(n)]}{2} = iB(n). \end{aligned}$$

Therefore $f_e(t) \longleftrightarrow A(n)$ and $f_o(t) \longleftrightarrow iB(n)$, as claimed. $\square$

Corollary 1. For a real periodic function $f(t)$ with coefficients $F(n)$:

- $F(n)$ is real and even if and only if $f(t)$ is even.
- $F(n)$ is purely imaginary and odd if and only if $f(t)$ is odd.

2.7 Power and Parseval's Theorem for Periodic Waveforms

Suppose $f(t)$ is a periodic waveform applied as a voltage across a resistor of resistance R ohms. The instantaneous power is

$$P(t) = \frac{[f(t)]^2}{R}.$$

More generally, with complex-valued waveforms it is convenient to write

$$P(t) = \frac{|f(t)|^2}{R}.$$

For $R = 1 \Omega$, we simply write

$$P(t) = |f(t)|^2.$$

The total energy in one period is then

$$E = \int_{-T_0/2}^{T_0/2} |f(t)|^2 dt,$$

and the average power over one period is

$$P_{\text{ave}} = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |f(t)|^2 dt.$$

Definition 4 (Power signal). If the average power P_{ave} defined above is finite, then $f(t)$ is called a power signal.

There is an important relation between P_{ave} and the Fourier coefficients $F(n)$.

Theorem 5 (Parseval's theorem for periodic waveforms). *Let $f(t)$ be a periodic power signal with Fourier coefficients $F(n)$. Then*

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |f(t)|^2 dt = \sum_{n=-\infty}^{\infty} |F(n)|^2.$$

Proof. For notational convenience, allow $f(t)$ to be complex. Then $|f(t)|^2 = f(t)f^*(t)$. Using the synthesis expression

$$f(t) = \sum_{n=-\infty}^{\infty} F(n)e^{in\omega_0 t}, \quad f^*(t) = \sum_{m=-\infty}^{\infty} F^*(m)e^{-im\omega_0 t},$$

we have

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |f(t)|^2 dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \left[\sum_n F(n)e^{in\omega_0 t} \right] \left[\sum_m F^*(m)e^{-im\omega_0 t} \right] dt.$$

Interchanging summation and integration,

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |f(t)|^2 dt = \sum_n \sum_m F(n)F^*(m) \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{i(n-m)\omega_0 t} dt.$$

By orthogonality, the integral is zero if $n \neq m$ and equal to 1 if $n = m$. Hence only the terms with $n = m$ survive:

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |f(t)|^2 dt = \sum_{n=-\infty}^{\infty} |F(n)|^2.$$

□

Definition 5 (Power spectrum). *For a periodic function $f(t)$ with complex Fourier coefficients $F(n)$, the power spectrum is*

$$P(n) = |F(n)|^2, \quad n \in \mathbb{Z}.$$

Each term $|F(n)|^2$ represents the contribution of the n th coefficient to the total average power.

2.8 Convergence of Fourier Series

Given an analytical definition of $f(t)$, we can compute its Fourier coefficients $F(n)$ via the analysis integral, and then form the series

$$\sum_{n=-\infty}^{\infty} F(n)e^{in\omega_0 t}.$$

A natural question is: under what conditions does this series converge to the original $f(t)$, and how fast does it converge?

A complete characterization of necessary and sufficient conditions is an advanced topic, but the standard sufficient conditions below cover the waveforms normally used in circuit and signal models: piecewise-smooth periodic signals with finitely many jumps per period.

Square-integrability criterion

Assume that $f(t)$ is periodic with period T_0 and

$$\int_{-T_0/2}^{T_0/2} |f(t)|^2 dt < \infty.$$

Then:

1. The integrals defining $F(n)$ exist and are finite.
2. If we define

$$g(t) = \sum_{n=-\infty}^{\infty} F(n)e^{in\omega_0 t},$$

then the difference $e(t) = g(t) - f(t)$ may be nonzero on a set of measure zero, but

$$\int_{-T_0/2}^{T_0/2} |e(t)|^2 dt = 0.$$

Thus, in the mean-square sense, $g(t)$ and $f(t)$ are identical over one period. In engineering language, this says that any mismatch is invisible to average-power measurements.

Dirichlet conditions and convergence at discontinuities

A classical sufficient condition is given by the Dirichlet conditions.

Definition 6 (Dirichlet conditions). *A periodic function $f(t)$ is said to satisfy the Dirichlet conditions if, over any one period:*

1. $f(t)$ is bounded.
2. $f(t)$ has at most a finite number of jump discontinuities, each with finite left and right limits.
3. $f(t)$ has at most a finite number of local maxima and minima, so it is piecewise monotone.

Most idealized engineering waveforms used in practice satisfy these conditions. Under the Dirichlet conditions, one has the following standard result.

Theorem 6 (Convergence under Dirichlet conditions). *If $f(t)$ satisfies the Dirichlet conditions, then:*

1. At any point of continuity of $f(t)$, the Fourier series converges to $f(t)$.
2. At a point t_0 where $f(t)$ has a jump discontinuity with left and right limits $f(t_0^-)$ and $f(t_0^+)$, the Fourier series converges to the half-sum

$$\frac{f(t_0^-) + f(t_0^+)}{2}.$$

The value $\frac{f(t_0^-) + f(t_0^+)}{2}$ at a discontinuity is often called the *half-value*. If a plotted or sampled waveform assigns some other value exactly at the jump, the Fourier series still converges to the half-value, because the one-point assignment does not change the coefficients.

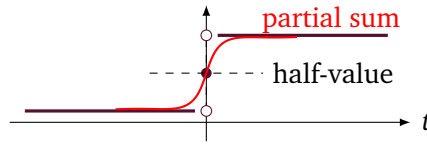


Figure 5: At a jump discontinuity, the Fourier series converges to the half-value; the local overshoot is the Gibbs phenomenon.

Rate of convergence

The decay rate of the complex Fourier coefficients controls the speed of convergence of the series. The following estimates use the same coefficient $F(n)$ as the rest of these notes. The mechanism is integration by parts. For

$$F(n) = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) e^{-in\omega_0 t} dt,$$

each integration by parts introduces a factor $1/(in\omega_0)$. If a boundary term appears because f or one of its derivatives jumps, that boundary term sets the leading decay rate. If no boundary term appears, one can integrate by parts again and gain another factor $1/n$.

- If $f(t)$ has discontinuities, then

$$|F(n)| \lesssim \frac{K}{|n|},$$

for large $|n|$, where K is a positive constant.

- If $f(t)$ is continuous but its first derivative has discontinuities, then

$$|F(n)| \lesssim \frac{K}{|n|^2}.$$

- More generally, if $f(t)$ and its first m derivatives are continuous, but the $(m+1)$ -st derivative has discontinuities, then

$$|F(n)| \lesssim \frac{K}{|n|^{m+2}}$$

for large $|n|$.

Hence, smoother functions have more rapidly decaying coefficients and require fewer terms for accurate approximation.

Note also that if Fourier coefficients decay like K/n , then the power spectrum terms $|F(n)|^2$ decay like K^2/n^2 , so the first few harmonics typically contain most of the power.

Notes on the Gibbs Phenomenon

Gibbs phenomenon. Near jump discontinuities, partial sums of the Fourier series exhibit overshoots and oscillations. As the number of terms increases, the oscillations become more localized, but the maximum overshoot approaches a fixed fraction (approximately 9%) of the jump height. This behavior is known as the Gibbs phenomenon. The oscillations do not vanish with more terms; instead, they become narrower and more concentrated around the discontinuity, while the series converges in the sense described by the Dirichlet and square-integrability criteria.

3 The Fourier Integral

3.1 Introduction

Up to this point the waveforms under consideration have all been periodic. For each of them there exists a Fourier series representation. In many applications, however, the waveforms of interest are not periodic. More commonly they are single pulses that occur once and do not repeat. Fourier analysis also applies to such single pulses, which motivates the study of the Fourier integral.

Periodic waveforms can also be regarded as single, infinitely long pulses that occur once in time. The Fourier integral developed for one-time pulses then applies to them as well. In this sense the Fourier integral provides a unifying framework that covers both classes of waveform: pulses and periodic signals.

3.2 The Fourier Integral

We wish to apply Fourier analysis to a single pulse $f(t)$. From the periodic case we know how to proceed, which suggests the following construction:

Embed the single pulse in one period of a periodic waveform and then let the period T_0 tend to infinity. The resulting limit should yield a pair of analysis and synthesis equations for $f(t)$.

Consider a single pulse $f(t)$ and the same pulse embedded in a periodic waveform $f_T(t)$ that repeats every T_0 seconds. Analytically,

$$f_T(t) = f(t), \quad -\frac{T_0}{2} < t < \frac{T_0}{2},$$

$$f_T(t + T_0) = f_T(t).$$

Thus $f(t)$ specifies one period of $f_T(t)$. For finite T_0 , we can form the complex Fourier series of $f_T(t)$. By subsequently letting $T_0 \rightarrow \infty$, only the single pulse remains, and the Fourier series representation is expected to converge to a Fourier-type representation for $f(t)$.

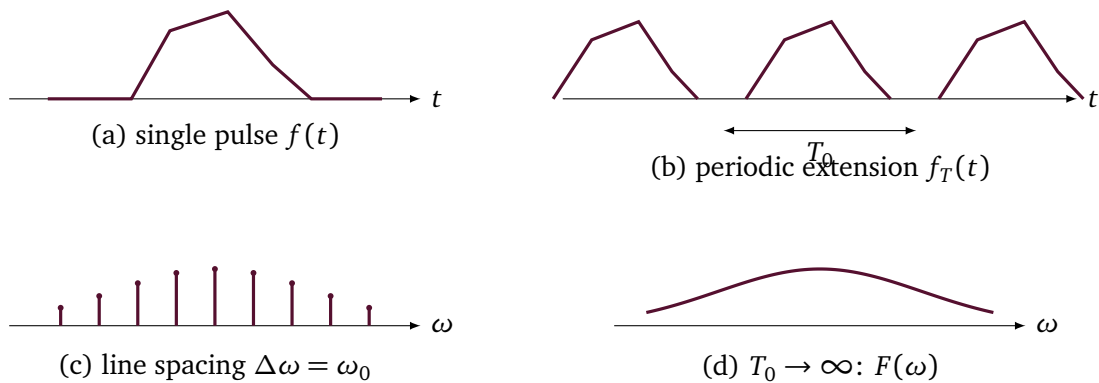


Figure 6: A nonperiodic pulse can be embedded in a periodic waveform. As T_0 grows, the Fourier-series line spacing $\omega_0 = 2\pi/T_0$ shrinks and the line spectrum approaches a continuous Fourier spectrum.

The complex Fourier series coefficient of order n for $f_T(t)$ is

$$F(n) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f_T(t) e^{-in\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{T_0}.$$

Since n and ω_0 appear only through the product $n\omega_0$, it is natural to write $F(n\omega_0)$ in place of $F(n)$:

$$F(n\omega_0) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f_T(t) e^{-in\omega_0 t} dt.$$

On the interval $[-T_0/2, T_0/2]$, $f_T(t)$ coincides with $f(t)$, so

$$F(n\omega_0) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-in\omega_0 t} dt.$$

We intend to let $T_0 \rightarrow \infty$ while keeping $f(t)$ fixed. The right-hand side is the average value of $f(t)e^{-in\omega_0 t}$ over $[-T_0/2, T_0/2]$. As $T_0 \rightarrow \infty$, this average tends to zero, which would trivialize the limit. To avoid this, we rewrite

$$T_0 F(n\omega_0) = \int_{-T_0/2}^{T_0/2} f(t) e^{-in\omega_0 t} dt.$$

For each finite T_0 , this defines a line spectrum $\{T_0 F(n\omega_0)\}$ on the ω -axis at $\omega = n\omega_0$.

Introduce

$$\Delta\omega = \omega_0, \quad \omega_n = n\Delta\omega, \quad F(\omega_n) = T_0 F(n\omega_0).$$

Then

$$F(\omega_n) = \int_{-T_0/2}^{T_0/2} f(t) e^{-i\omega_n t} dt.$$

As $T_0 \rightarrow \infty$,

$$\Delta\omega \rightarrow 0, \quad \omega_n \rightarrow \omega, \quad F(\omega_n) \rightarrow F(\omega),$$

and it is natural to write

$$F(\omega) = \lim_{T_0 \rightarrow \infty} \int_{-T_0/2}^{T_0/2} f(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

This is the *analysis* equation for a single pulse:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

For the synthesis equation, start from the Fourier series representation of the periodic function $f_T(t)$:

$$f_T(t) = \sum_{n=-\infty}^{\infty} F(n\omega_0) e^{in\omega_0 t}, \quad -\frac{T_0}{2} < t < \frac{T_0}{2}.$$

On this interval, the left-hand side equals $f(t)$, so we write

$$f(t) = \sum_{n=-\infty}^{\infty} F(n\omega_0) e^{in\omega_0 t}, \quad -\frac{T_0}{2} < t < \frac{T_0}{2}.$$

Using $F(n\omega_0) = F(\omega_n)/T_0$,

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T_0} F(\omega_n) e^{i\omega_n t}.$$

Since

$$\frac{1}{T_0} = \frac{\omega_0}{2\pi} = \frac{\Delta\omega}{2\pi},$$

we get

$$f(t) = \sum_{n=-\infty}^{\infty} F(\omega_n) e^{i\omega_n t} \frac{\Delta\omega}{2\pi}.$$

Letting $T_0 \rightarrow \infty$ implies $\Delta\omega \rightarrow 0$ and $\omega_n \rightarrow \omega$. Under suitable conditions, this Riemann sum converges to

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega.$$

We summarize the result as follows:

Theorem 7 (Fourier Transform for a Single Pulse). *Suppose $f(t)$ is such that the integrals below exist. The function $f(t)$ and its Fourier transform $F(\omega)$ are related by the transform pair*

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad (3.1)$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega. \quad (3.2)$$

The function $F(\omega)$ in (3.1) is called the Fourier transform or Fourier spectral density of $f(t)$. The integral representation (3.2) is called the inverse Fourier transform or the Fourier integral representation of $f(t)$.

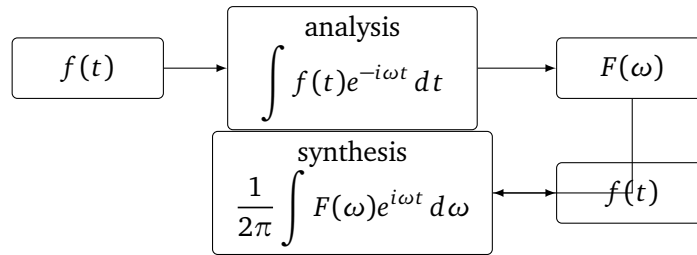


Figure 7: The Fourier integral is an analysis-synthesis pair: the analysis integral finds the continuous frequency description, and the synthesis integral rebuilds the time-domain signal.

The limits of integration in (3.1) extend over all time, so the pulses that can be transformed may have arbitrary duration, finite or infinite. For example, the following pulses are admissible:

$$f(t) = e^{-t^2}, \quad t \in \mathbb{R},$$

$$g(t) = \begin{cases} 1, & 0 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

As will be seen later, periodic waveforms of infinite duration can also be treated within the same framework.

Definition 7 (Fourier Transform Notation). *The shorthand*

$$f(t) \longleftrightarrow F(\omega)$$

will be used to indicate that $F(\omega)$ is the Fourier transform of $f(t)$ as in (3.1), and that $f(t)$ is the inverse transform of $F(\omega)$ as in (3.2).

3.3 Interpretation of the Fourier Integral

For a periodic function with fundamental frequency ω_0 , the Fourier series representation

$$f(t) = \sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t}$$

expresses $f(t)$ as a linear combination of complex exponentials at discrete frequencies $n\omega_0$. The coefficients $F(n)$ are determined by the corresponding analysis equation.

The Fourier integral representation (3.2),

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega,$$

is the counterpart of the Fourier series for non-periodic (or more general) signals. It expresses $f(t)$ as a superposition of complex exponentials at all real frequencies ω , weighted by the coefficient function $F(\omega)$. Conceptually, the Fourier integral is the limit of the Fourier series as the fundamental period T_0 tends to infinity and the line spectrum becomes a continuous spectrum.

Remark 1. The Fourier integral can be viewed as a continuous-frequency linear decomposition of $f(t)$ into the basis $\{e^{i\omega t}\}_{\omega \in \mathbb{R}}$. The transform $F(\omega)$ specifies the contribution of each frequency to the overall waveform.

Example 2 (Fourier Transform of the Rectangular Pulse). The following pulse will be used frequently.

Definition 8 (Rectangular Pulse). The rectangular pulse of width τ and unit height is denoted by $\text{Rect}(t/\tau)$ and defined by

$$\text{Rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1, & |t| < \frac{\tau}{2}, \\ \frac{1}{2}, & |t| = \frac{\tau}{2}, \\ 0, & |t| > \frac{\tau}{2}. \end{cases}$$

The half-height endpoint value matches the inverse Fourier integral convention at jump discontinuities.

Let $f(t) = \text{Rect}(t/\tau)$. Using the analysis equation,

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = \int_{-\tau/2}^{\tau/2} e^{-i\omega t} dt \\ &= \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{t=-\tau/2}^{t=\tau/2} = \frac{e^{-i\omega\tau/2} - e^{i\omega\tau/2}}{-i\omega} = \frac{2 \sin(\omega\tau/2)}{\omega}. \end{aligned}$$

Using the sinc convention fixed in Notation Conventions, we may write

$$F(\omega) = \tau \text{sinc}\left(\frac{\omega\tau}{2}\right).$$

Thus

$$\text{Rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right).$$

The inverse transform gives the Fourier integral representation:

$$\text{Rect}\left(\frac{t}{\tau}\right) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tau \text{sinc}\left(\frac{\omega\tau}{2}\right) e^{i\omega t} d\omega.$$

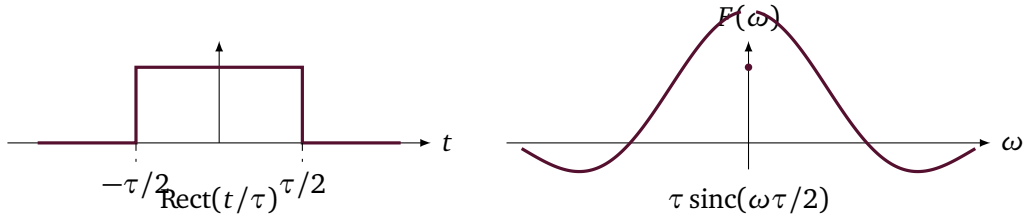


Figure 8: The rectangular pulse and its sinc-shaped Fourier transform. The removable value at $\omega = 0$ is $F(0) = \tau$.

3.4 Basic Properties for Real Signals

In general, the Fourier transform $F(\omega)$ is complex-valued. It can be written in Cartesian form

$$F(\omega) = A(\omega) + iB(\omega),$$

or in polar form

$$F(\omega) = |F(\omega)|e^{i\Theta(\omega)},$$

where

$$|F(\omega)| = \sqrt{A^2(\omega) + B^2(\omega)}, \quad \Theta(\omega) = \text{atan2}(B(\omega), A(\omega)).$$

Theorem 8 (Conjugate Symmetry for Real Signals). *Let $f(t) \leftrightarrow F(\omega)$. Then*

$$f(t) \in \mathbb{R} \text{ for all } t \iff F^*(\omega) = F(-\omega) \text{ for all } \omega.$$

Proof. Assume first that $f(t)$ is real. Then

$$\begin{aligned} F^*(\omega) &= \left(\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right)^* = \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) e^{-i(-\omega)t} dt = F(-\omega). \end{aligned}$$

Conversely, assume $F^*(\omega) = F(-\omega)$ for all ω . Take the inverse transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega.$$

Taking complex conjugates,

$$f^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F^*(\omega) e^{-i\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(-\omega) e^{-i\omega t} d\omega.$$

Change variables $\omega' = -\omega$; then

$$f^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega') e^{i\omega' t} d\omega' = f(t).$$

Hence $f(t)$ is real. □

Corollary 2. *If $f(t)$ is real, then*

- the real part $A(\omega)$ and the magnitude $|F(\omega)|$ are even,

- the imaginary part $B(\omega)$ is odd, and the phase $\Theta(\omega)$ can be chosen odd modulo 2π away from zeros of F .

Proof. From $F^*(\omega) = F(-\omega)$ and $F(\omega) = A(\omega) + iB(\omega)$ we obtain

$$A(\omega) - iB(\omega) = A(-\omega) + iB(-\omega).$$

Equating real and imaginary parts gives

$$A(\omega) = A(-\omega), \quad B(-\omega) = -B(\omega).$$

Thus A is even and B is odd. The magnitude satisfies $|F(-\omega)| = |F(\omega)|$, hence $|F|$ is even. The phase can be chosen so that $\Theta(-\omega) = -\Theta(\omega)$ wherever $F(\omega) \neq 0$, so Θ is odd modulo 2π . $\square$

Similar to the case of Fourier series, we have the following theorem and corollary for Fourier transform of a real signal:

Theorem 9 (Even/Odd Decomposition). *Suppose $f(t)$ is real and decomposed into even and odd parts*

$$f(t) = f_e(t) + f_o(t), \quad f_e(t) = \frac{f(t) + f(-t)}{2}, \quad f_o(t) = \frac{f(t) - f(-t)}{2}.$$

Let $f(t) \leftrightarrow F(\omega)$ with $F(\omega) = A(\omega) + iB(\omega)$. Then

$$f_e(t) \longleftrightarrow A(\omega), \quad f_o(t) \longleftrightarrow iB(\omega).$$

Corollary 3. *For real $f(t)$:*

1. $F(\omega)$ is real and even if and only if $f(t)$ is even.
2. $F(\omega)$ is purely imaginary and odd if and only if $f(t)$ is odd.

3.5 Parseval's Theorem for Pulses

Consider a voltage pulse $v(t)$ applied to an R -ohm resistor. The instantaneous power is

$$p(t) = \frac{|v(t)|^2}{R} \quad (\text{watts}).$$

For $R = 1$,

$$p(t) = |v(t)|^2.$$

The total energy delivered by the pulse is

$$E = \int_{-\infty}^{\infty} p(t) dt = \int_{-\infty}^{\infty} |v(t)|^2 dt \quad (\text{joules}).$$

The same expression holds for current pulses if one interprets $|f(t)|^2$ as instantaneous power in a 1-ohm load.

Definition 9 (Energy Pulse). *A signal $f(t)$ is called an energy pulse if*

$$\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty.$$

For such pulses, there is an energy relation in the transform domain.

Lemma 1 (Parseval's Lemma). *Let $f(t)$ and $g(t)$ be energy pulses with transforms $F(\omega)$ and $G(\omega)$, respectively. Then*

$$\int_{-\infty}^{\infty} f(t) g^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) G^*(\omega) d\omega.$$

Proof. Express $g^*(t)$ using the inverse transform:

$$g^*(t) = \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{i\omega t} d\omega \right)^* = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) e^{-i\omega t} d\omega.$$

Then

$$\begin{aligned} \int_{-\infty}^{\infty} f(t) g^*(t) dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right) G^*(\omega) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) G^*(\omega) d\omega. \end{aligned}$$

□

Theorem 10 (Parseval's Theorem for Pulses). *Let $f(t)$ be an energy pulse with Fourier transform $F(\omega)$. Then*

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega.$$

Proof. Apply Parseval's lemma with $g(t) = f(t)$ and $G(\omega) = F(\omega)$. Then

$$\int_{-\infty}^{\infty} f(t) f^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) F^*(\omega) d\omega.$$

This is the stated equality.

□

Definition 10 (Energy Spectrum). *The energy spectrum of a pulse with transform $F(\omega)$ is the function*

$$E(\omega) = \frac{1}{2\pi} |F(\omega)|^2 \quad (\text{joules/radian}).$$

The energy contained in the frequency band $[\omega_1, \omega_2]$ is then

$$\Delta E = \int_{\omega_1}^{\omega_2} E(\omega) d\omega = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} |F(\omega)|^2 d\omega.$$

3.6 Existence of the Fourier Integral

The derivation of (3.1)–(3.2) was heuristic. A rigorous treatment requires conditions on $f(t)$ under which the transform and inverse transform recover one another. No single set of necessary and sufficient conditions is both simple and completely general. For engineering use, the two practical tests below are usually enough: finite energy gives mean-square recovery, while Dirichlet-type assumptions give pointwise recovery away from jumps and half-value recovery at jumps.

Definition 11 (Square-Integrable Function). *A function $f(t)$ is square-integrable if*

$$\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty.$$

Theorem 11 (Square-Integrability Criterion). Suppose $f(t)$ is square-integrable. Define formally

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt,$$

and

$$\phi(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega.$$

Then:

1. The Fourier transform can be defined by energy-limit approximation even when the ordinary improper integral is not absolutely convergent.
2. The function $\phi(t)$ coincides with $f(t)$ almost everywhere and

$$\int_{-\infty}^{\infty} |\phi(t) - f(t)|^2 dt = 0.$$

In practical terms, the square-integrability criterion says that the transform pair is correct for energy calculations. It may not decide the value at an isolated jump, but isolated point values do not change signal energy.

A second class of sufficient conditions is based on a continuous-time analogue of the Dirichlet conditions.

Definition 12 (Dirichlet Conditions for Pulses). A function $f(t)$ is said to satisfy the Dirichlet conditions for pulses if:

1. $f(t)$ is bounded on $\mathbb{R}$,
2. in any finite interval it has at most a finite number of discontinuities and a finite number of local maxima and minima,
3. it is absolutely integrable,

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty.$$

Theorem 12 (Dirichlet–Jordan Criterion). Let $f(t)$ satisfy the Dirichlet conditions for pulses. Then:

1. The Fourier transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

exists for all ω .

2. The Fourier integral

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

converges to $f(t)$ at every point of continuity.

3. At a point of discontinuity t_0 , the Fourier integral converges to

$$\frac{1}{2} [f(t_0^+) + f(t_0^-)].$$

Piecewise-smooth pulses with finite area satisfy these Dirichlet conditions, and hence admit a well-defined Fourier transform with the usual inversion property. This covers the standard pulse models used throughout these notes.

3.7 Asymptotic Behavior of $F(\omega)$

The decay rate of the Fourier transform for large $|\omega|$ is related to the smoothness of the underlying time-domain signal.

Theorem 13 (Asymptotic Bound for $F(\omega)$). *Let $f(t)$ be a pulse that satisfies appropriate regularity conditions. Suppose $f(t)$ and its first m derivatives are continuous and vanish at infinity, and that the $(m+1)$ -st derivative is piecewise continuous and absolutely integrable. Then there exists a constant $K > 0$ such that, for sufficiently large $|\omega|$,*

$$|F(\omega)| \leq \frac{K}{|\omega|^{m+1}}.$$

Proof. Start from

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

Integrate by parts:

$$F(\omega) = \left[\frac{f(t)}{-i\omega} e^{-i\omega t} \right]_{-\infty}^{\infty} + \frac{1}{i\omega} \int_{-\infty}^{\infty} f'(t) e^{-i\omega t} dt.$$

The boundary term vanishes since $f(t) \rightarrow 0$ as $|t| \rightarrow \infty$. Repeating this procedure $m+1$ times gives

$$F(\omega) = \frac{1}{(i\omega)^{m+1}} \int_{-\infty}^{\infty} f^{(m+1)}(t) e^{-i\omega t} dt.$$

Absolute integrability of $f^{(m+1)}$ implies

$$\left| \int_{-\infty}^{\infty} f^{(m+1)}(t) e^{-i\omega t} dt \right| \leq \int_{-\infty}^{\infty} |f^{(m+1)}(t)| dt =: K_0.$$

Hence

$$|F(\omega)| \leq \frac{K_0}{|\omega|^{m+1}},$$

for all $\omega \neq 0$. Taking $K \geq K_0$ completes the proof. $\square$

Thus, as the number of continuous derivatives of $f(t)$ increases, the transform decays more rapidly in frequency. Since the energy spectrum is proportional to $|F(\omega)|^2$, smoother time-domain signals yield more rapidly decaying energy spectra.

4 Fourier Transforms of Some Important Functions

4.1 Introduction

The nineteenth century saw extensive work on placing the theories of Fourier and Laplace on a rigorous basis. For these notes, the main practical message is simpler: ordinary finite-energy pulses are handled by Fourier integrals, while ideal objects such as impulses, steps, constants, and eternal sinusoids are handled by consistent limiting rules.

Much of this development was driven by the need to clarify broad claims such as Fourier's assertion that every function $f(x)$ (or portion of a function) can be represented by a trigonometric series. Dirichlet provided a useful set of sufficient conditions, and counterexamples were later constructed to show that Fourier's statement is not universally valid.

A major extension relevant to engineering arose in the 1920s from quantum mechanics. In 1927, Paul A. M. Dirac introduced the *delta function* (now called the *Dirac delta* and denoted by $\delta(t - \tau)$) in his work on quantum theory. Although formulated for quantum mechanics, the Dirac delta soon became central in many areas of applied mathematics and electrical engineering.

We shall see that $\delta(t)$ does not satisfy square-integrability or Dirichlet-type conditions, carries infinite energy, and cannot be realized physically, yet still admits a well-defined Fourier transform. In fact, one can show that

$$\delta(t - \tau) \longleftrightarrow e^{-i\omega\tau},$$

so that the Dirac delta is closely tied to the complex exponentials that underlie Fourier analysis.

The appearance of such objects motivated the development of *distribution theory* (or *generalized function theory*). In these notes we use only the engineering-facing part of that theory: an impulse is not an ordinary function with a finite value at one point; it is an idealized object defined by what it does under an integral.

In this chapter we use a minimal amount of distribution theory, sufficient to derive the Fourier transforms of several functions of central importance:

- the Dirac delta (unit impulse),
- the unit step,
- the signum function,
- the eternal complex exponential,
- the constant (dc) signal,
- eternal sines and cosines,
- general periodic functions with a Fourier series representation.

All of these have infinite energy or infinite duration and lie outside the class handled by the square-integrability criterion. Nevertheless, each admits a consistent Fourier transform representation compatible with the transform of finite-energy pulses introduced earlier.

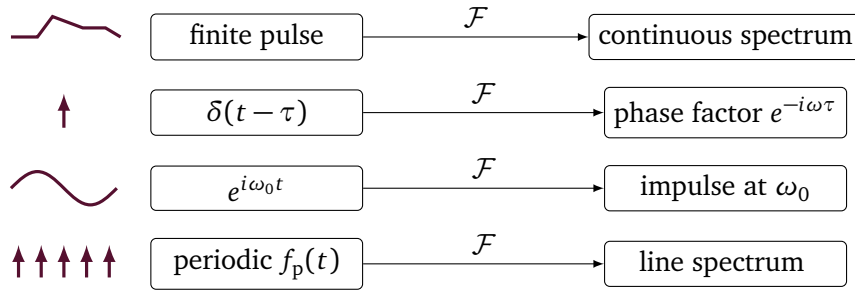


Figure 9: This chapter extends the ordinary pulse transform to idealized objects. The entries are visual reminders; the precise transform pairs are derived in the following sections.

4.2 The Rectangular Pulse $f(t) = \text{Rect}(t/T)$

We begin with the rectangular (gate) pulse of width T and unit height.

Definition 13 (Rectangular pulse). *The rectangular pulse of width $T > 0$ is defined by*

$$\text{Rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & |t| < \frac{T}{2}, \\ \frac{1}{2}, & |t| = \frac{T}{2}, \\ 0, & |t| > \frac{T}{2}. \end{cases}$$

The half-height endpoint value is immaterial for the integral, but it matches the inverse-transform value at the discontinuities.

The Fourier transform of $\text{Rect}(t/T)$ was obtained previously and is recalled here in our normalization

$$f(t) \leftrightarrow F(\omega) \quad \text{with} \quad F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega.$$

Proposition 1 (Fourier transform of a rectangular pulse). *Let $f(t) = \text{Rect}(t/T)$. Then*

$$f(t) = \text{Rect}\left(\frac{t}{T}\right) \longleftrightarrow F(\omega) = T \text{sinc}\left(\frac{\omega T}{2}\right), \quad (4.1)$$

where sinc is the unnormalized sinc function fixed in Notation Conventions.

Proof. From the definition of the transform,

$$F(\omega) = \int_{-\infty}^{\infty} \text{Rect}\left(\frac{t}{T}\right) e^{-i\omega t} dt = \int_{-T/2}^{T/2} e^{-i\omega t} dt.$$

Evaluating,

$$\begin{aligned} F(\omega) &= \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{t=-T/2}^{T/2} = \frac{e^{-i\omega T/2} - e^{i\omega T/2}}{-i\omega} \\ &= \frac{2 \sin(\omega T/2)}{\omega} = T \frac{\sin(\omega T/2)}{\omega T/2} = T \text{sinc}\left(\frac{\omega T}{2}\right), \end{aligned}$$

with the value at $\omega = 0$ given by continuity. □

A useful property of sinc that we shall need later is its total area.

Lemma 2 (Area under sinc). *The function $\text{sinc}(x)$ satisfies*

$$\int_{-\infty}^{\infty} \text{sinc}(x) dx = \pi. \quad (4.2)$$

Proof. From (4.1) and the inverse transform,

$$\text{Rect}\left(\frac{t}{T}\right) = \frac{1}{2\pi} \int_{-\infty}^{\infty} T \text{sinc}\left(\frac{\omega T}{2}\right) e^{i\omega t} d\omega.$$

Setting $t = 0$ and using $\text{Rect}(0/T) = 1$ gives

$$1 = \frac{T}{2\pi} \int_{-\infty}^{\infty} \text{sinc}\left(\frac{\omega T}{2}\right) d\omega.$$

With the change of variable $x = \omega T/2$, $d\omega = \frac{2}{T}dx$, this becomes

$$1 = \frac{T}{2\pi} \cdot \frac{2}{T} \int_{-\infty}^{\infty} \text{sinc}(x) dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \text{sinc}(x) dx,$$

which yields (4.2). □

4.3 The Single-Sided Decaying Exponential

Consider the one-sided decaying exponential

$$f(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{2}, & t = 0, \\ e^{-\beta t}, & t > 0, \end{cases} \quad \beta > 0. \quad (4.3)$$

It is convenient to write this using the unit step $u(t)$:

Definition 14 (Unit step). *The unit step (Heaviside step) is*

$$u(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{2}, & t = 0, \\ 1, & t > 0. \end{cases}$$

Then (4.3) becomes

$$f(t) = e^{-\beta t} u(t), \quad \beta > 0. \quad (4.4)$$

Proposition 2 (Fourier transform of a one-sided decaying exponential). *Let $f(t) = e^{-\beta t} u(t)$, $\beta > 0$. Then*

$$f(t) = e^{-\beta t} u(t) \longleftrightarrow F(\omega) = \frac{1}{\beta + i\omega}.$$

Proof. Using the definition of the transform,

$$F(\omega) = \int_{-\infty}^{\infty} e^{-\beta t} u(t) e^{-i\omega t} dt = \int_0^{\infty} e^{-(\beta + i\omega)t} dt.$$

Since $\beta > 0$, the integral converges and

$$F(\omega) = \left[-\frac{1}{\beta + i\omega} e^{-(\beta + i\omega)t} \right]_0^{\infty} = \frac{1}{\beta + i\omega}.$$

□

The real and imaginary parts of $F(\omega)$ are

$$F(\omega) = \frac{\beta}{\beta^2 + \omega^2} - i \frac{\omega}{\beta^2 + \omega^2}, \quad (4.5)$$

so that

$$|F(\omega)| = \frac{1}{\sqrt{\beta^2 + \omega^2}}, \quad \arg F(\omega) = -\operatorname{atan2}(\omega, \beta).$$

The magnitude is even in ω , and the phase is odd, consistent with the general properties for real signals.

The even and odd parts of $f(t)$ will be useful repeatedly.

For $f(t) = e^{-\beta t}u(t)$, one checks easily that

$$f_e(t) = \frac{1}{2}e^{-\beta|t|}, \quad (4.6)$$

$$f_o(t) = \frac{1}{2}(e^{-\beta t}u(t) - e^{\beta t}u(-t)).$$

Using the general results from previous chapters that the transform of the even part is the real part $A(\omega)$ and the transform of the odd part is $iB(\omega)$, we obtain:

Corollary 4 (Transforms of the even/odd parts). For $f(t) = e^{-\beta t}u(t)$,

$$f_e(t) = \frac{1}{2}e^{-\beta|t|} \longleftrightarrow F_e(\omega) = \frac{\beta}{\beta^2 + \omega^2}, \quad (4.7)$$

$$f_o(t) \longleftrightarrow F_o(\omega) = -i \frac{\omega}{\beta^2 + \omega^2}. \quad (4.8)$$

Proof. Writing $F(\omega) = A(\omega) + iB(\omega)$ as in (4.5), the general even/odd decomposition theorem gives

$$f_e(t) \longleftrightarrow A(\omega), \quad f_o(t) \longleftrightarrow iB(\omega).$$

From (4.5), $A(\omega) = \beta/(\beta^2 + \omega^2)$ and $iB(\omega) = -i\omega/(\beta^2 + \omega^2)$, which yields (4.7)–(4.8). $\square$

Thus a real, even time-domain function yields a real, even transform; a real, odd function yields a purely imaginary, odd transform.

4.4 The Double-Sided Decaying Exponential

The double-sided exponential is defined by

$$f(t) = e^{-\beta|t|}, \quad \beta > 0,$$

plotted symmetrically around the origin.

Proposition 3 (Fourier transform of a double-sided exponential). For $f(t) = e^{-\beta|t|}$ with $\beta > 0$,

$$f(t) = e^{-\beta|t|} \longleftrightarrow F(\omega) = \frac{2\beta}{\beta^2 + \omega^2}.$$

Proof. The function $e^{-\beta|t|}$ is twice the even part of the one-sided exponential (4.4):

$$e^{-\beta|t|} = 2f_e(t),$$

with $f_e(t)$ defined in (4.6). Hence, by (4.7),

$$F(\omega) = 2F_e(\omega) = 2 \cdot \frac{\beta}{\beta^2 + \omega^2} = \frac{2\beta}{\beta^2 + \omega^2}.$$

$\square$

The function is real and even in t , and its transform is real and even in ω , as expected. The transform decays as ω^{-2} for large $|\omega|$, consistent with the continuity and differentiability properties of $f(t)$.

4.5 The Signum Function

Definition 15 (Signum function). *The signum function is defined by*

$$\operatorname{sgn}(t) = \begin{cases} -1, & t < 0, \\ 0, & t = 0, \\ +1, & t > 0. \end{cases}$$

As an ordinary function, $\operatorname{sgn}(t)$ has infinite energy:

$$\int_{-\infty}^{\infty} |\operatorname{sgn}(t)|^2 dt = \int_{-\infty}^{\infty} 1 dt = \infty,$$

so the square-integrability criterion does not apply. Moreover, a direct application of the analysis integral,

$$F(\omega) = \int_{-\infty}^{\infty} \operatorname{sgn}(t) e^{-i\omega t} dt,$$

leads to divergent integrals if interpreted naively. To proceed, we treat $\operatorname{sgn}(t)$ as a *distribution*: compute with a family of ordinary decaying functions first, then take the limiting answer.

Definition 16 (Generalized function via sequence). *In this engineering use, a generalized function is represented by a limiting sequence $\{f_k(t)\}_{k>0}$ of ordinary functions, each having a well-defined Fourier transform. The limit is interpreted through integrals against smooth signals, not by comparing point values.*

For $\operatorname{sgn}(t)$, consider the sequence

$$f_k(t) = \begin{cases} -e^{kt}, & t < 0, \\ 0, & t = 0, \\ e^{-kt}, & t > 0, \end{cases} \quad k > 0.$$

For each fixed k , $f_k(t)$ is an ordinary function that tends pointwise to $\operatorname{sgn}(t)$ as $k \rightarrow 0^+$. Formally,

$$\operatorname{sgn}(t) = \lim_{k \rightarrow 0^+} f_k(t).$$

We first Fourier transform $f_k(t)$ for fixed k , then let $k \rightarrow 0^+$. For fixed k , since $f_k(t)$ is twice the odd part of $e^{-kt}u(t)$, with $\beta = k$. From (4.8) (with $\beta = k$), we obtain

$$\mathcal{F}\{f_k\}(\omega) = -i \frac{2\omega}{k^2 + \omega^2}.$$

Proposition 4 (Fourier transform of $\operatorname{sgn}(t)$). *As a distribution,*

$$\operatorname{sgn}(t) \longleftrightarrow F(\omega) = \frac{2}{i\omega}, \quad (4.9)$$

interpreted in the sense of principal value.

Sketch. For each $k > 0$,

$$F_k(\omega) = -i \frac{2\omega}{k^2 + \omega^2}.$$

As $k \rightarrow 0^+$, one can write

$$F(\omega) = -i \frac{2}{\omega},$$

in the distributional sense (principal-value interpretation at $\omega = 0$). $\square$

4.6 The Dirac Delta (Unit Impulse)

Definition 17 (Dirac delta). *The Dirac delta (unit impulse) at the origin is the distribution $\delta(t)$ characterized by*

$$\delta(t) = 0 \quad (t \neq 0), \quad \int_{-\infty}^{\infty} \delta(t) dt = 1.$$

This description is a working rule rather than an ordinary-function definition. The reliable rule is the sampling property below: under an integral, $\delta(t - \tau)$ extracts the value of the other factor at $t = \tau$. As with $\text{sgn}(t)$, one can make this precise by constructing $\delta(t)$ as the limit of a sequence of ordinary functions.

Definition 18 (Box sequence for δ). *For $k > 0$, define the box function*

$$B_k(t) = \begin{cases} \frac{1}{k}, & |t| < \frac{k}{2}, \\ 0, & |t| > \frac{k}{2}. \end{cases}$$

Then each B_k has unit area, $\int_{-\infty}^{\infty} B_k(t) dt = 1$, and

$$\delta(t) = \lim_{k \rightarrow 0^+} B_k(t)$$

in the distributional sense.

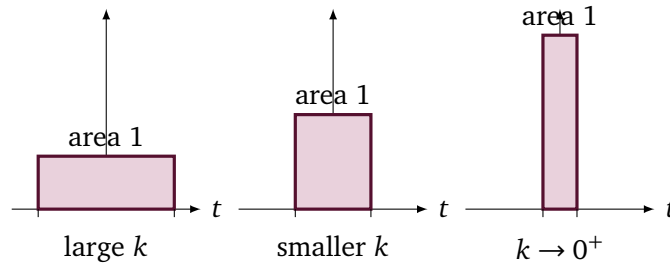


Figure 10: A unit-area box sequence becomes taller and narrower as it approaches the Dirac delta in the distributional sense.

Proposition 5 (Fourier transform of δ). *The Dirac delta satisfies*

$$\delta(t) \longleftrightarrow F(\omega) = 1.$$

Proof. For fixed k , the Fourier transform of B_k is

$$F_k(\omega) = \int_{-\infty}^{\infty} B_k(t) e^{-i\omega t} dt = \frac{1}{k} \int_{-k/2}^{k/2} e^{-i\omega t} dt = \frac{\sin(\omega k/2)}{\omega k/2} = \text{sinc}\left(\frac{\omega k}{2}\right).$$

As $k \rightarrow 0^+$, $\text{sinc}(\omega k/2) \rightarrow 1$ for each fixed ω , so the limit transform is $F(\omega) = 1$. $\square$

Thus the magnitude spectrum of $\delta(t)$ is identically one, and the phase is identically zero. The delta is even in time, and its transform is real and even.

The delta has infinite energy:

$$\int_{-\infty}^{\infty} |\delta(t)|^2 dt = \infty,$$

as also seen from Parseval's theorem applied formally to $F(\omega) \equiv 1$.

Definition 19 (Weight and shifted delta). *For any $\tau \in \mathbb{R}$, the shifted impulse $\delta(t - \tau)$ is the delta located at $t = \tau$. For any complex constant a , the impulse $a \delta(t - \tau)$ is said to have weight a . Its integral is*

$$\int_{-\infty}^{\infty} a \delta(t - \tau) dt = a.$$

The delta has the fundamental *sampling (sifting) property*.

Theorem 14 (Sampling property of the delta). *Let $\varphi(t)$ be continuous at $t = \tau$. Then*

$$\varphi(t) \delta(t - \tau) = \varphi(\tau) \delta(t - \tau)$$

as distributions, and

$$\int_{-\infty}^{\infty} \varphi(t) \delta(t - \tau) dt = \varphi(\tau).$$

The shifted delta has the transform

$$\delta(t - \tau) \longleftrightarrow e^{-i\omega\tau}, \quad (4.10)$$

obtained by direct integration:

$$\int_{-\infty}^{\infty} \delta(t - \tau) e^{-i\omega t} dt = e^{-i\omega\tau}.$$

As an important integral identity, one has

Theorem 15 (Integral representation of δ). *In the sense of distributions,*

$$\lim_{K \rightarrow \infty} \int_{-K}^K e^{i\omega t} d\omega = 2\pi \delta(t). \quad (4.11)$$

Proof. From Proposition 5, $\delta(t) \leftrightarrow 1$. The inverse transform formula gives

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 1 \cdot e^{i\omega t} d\omega.$$

Interpreting the improper integral as the distributional limit

$$\int_{-\infty}^{\infty} e^{i\omega t} d\omega = \lim_{K \rightarrow \infty} \int_{-K}^K e^{i\omega t} d\omega,$$

we obtain

$$\delta(t) = \frac{1}{2\pi} \lim_{K \rightarrow \infty} \int_{-K}^K e^{i\omega t} d\omega,$$

which is equivalent to (4.11). □

This can be viewed as a continuous superposition of complex exponentials yielding a sharply localized impulse in time.

4.7 The Eternal Constant

Consider the constant signal

$$f(t) = V_0, \quad t \in \mathbb{R}, \quad (4.12)$$

representing an ideal dc voltage or current. This is not an energy signal and must be treated as a distribution.

One may regard it as the limit of a rectangular pulse of height V_0 and width k as $k \rightarrow \infty$:

$$f(t) = \lim_{k \rightarrow \infty} V_0 \operatorname{Rect}\left(\frac{t}{k}\right).$$

For fixed k ,

$$F_k(\omega) = V_0 \int_{-\infty}^{\infty} \operatorname{Rect}\left(\frac{t}{k}\right) e^{-i\omega t} dt = V_0 k \operatorname{sinc}\left(\frac{\omega k}{2}\right).$$

Using Theorem 15 (with appropriate scaling), one finds that in the limit this converges to a single delta at the origin.

Proposition 6 (Fourier transform of a constant). *For the eternal constant $f(t) = V_0$,*

$$V_0 \longleftrightarrow F(\omega) = 2\pi V_0 \delta(\omega). \quad (4.13)$$

Proof. From (4.12), linearity of the transform and Proposition 5 applied in the frequency domain yield a direct argument. Consider

$$\delta(\omega) \longleftrightarrow \frac{1}{2\pi},$$

which follows from (4.10) with $\tau = 0$ and the inversion formula. By linearity, for any constant V_0 ,

$$2\pi V_0 \delta(\omega) \longleftrightarrow V_0,$$

which is (4.13). Equivalently, the limit of $V_0 k \operatorname{sinc}((\omega k)/2)$ as $k \rightarrow \infty$ is $2\pi V_0 \delta(\omega)$, by Theorem 15. $\square$

Thus the spectral content of an ideal dc signal is concentrated at $\omega = 0$ in the frequency domain. Since the ideal dc signal has infinite duration, this is a power-signal statement rather than a finite-energy statement.

4.8 The Unit Step as a Distribution

The unit step $u(t)$ can be expressed in terms of $\operatorname{sgn}(t)$ and a constant:

$$u(t) = \frac{1}{2} \operatorname{sgn}(t) + \frac{1}{2}. \quad (4.14)$$

Using (4.9) and (4.13), we obtain directly:

Proposition 7 (Fourier transform of the unit step). *As a distribution,*

$$u(t) \longleftrightarrow F(\omega) = \frac{1}{i\omega} + \pi \delta(\omega), \quad (4.15)$$

with the term $1/(i\omega)$ interpreted in the principal-value sense.

Proof. From (4.14),

$$u(t) = \frac{1}{2} \operatorname{sgn}(t) + \frac{1}{2}.$$

By linearity and (4.9)–(4.13),

$$\mathcal{F}\{u(t)\}(\omega) = \frac{1}{2} \cdot \frac{2}{i\omega} + \frac{1}{2} \cdot 2\pi\delta(\omega) = \frac{1}{i\omega} + \pi\delta(\omega),$$

with $1/(i\omega)$ interpreted as a principal value. This yields (4.15). $\square$

The odd part of $u(t)$ is $\frac{1}{2} \operatorname{sgn}(t)$, which transforms to the purely imaginary odd component $\frac{1}{i\omega}$, and the even part is the constant $\frac{1}{2}$, which transforms to $\pi\delta(\omega)$.

4.9 The Eternal Complex Exponential

Consider the complex exponential of single angular frequency ω_0 :

$$f(t) = e^{i\omega_0 t}, \quad \omega_0 \in \mathbb{R}.$$

Its energy integral diverges, so we again treat it as a distribution, for example as the limit

$$e^{i\omega_0 t} = \lim_{k \rightarrow \infty} e^{i\omega_0 t} \operatorname{Rect}\left(\frac{t}{k}\right).$$

For fixed k ,

$$F_k(\omega) = \int_{-\infty}^{\infty} e^{i\omega_0 t} \operatorname{Rect}\left(\frac{t}{k}\right) e^{-i\omega t} dt = \int_{-k/2}^{k/2} e^{-i(\omega - \omega_0)t} dt = k \operatorname{sinc}\left(\frac{(\omega - \omega_0)k}{2}\right).$$

As $k \rightarrow \infty$, this converges to a delta at $\omega = \omega_0$.

Proposition 8 (Fourier transform of an eternal complex exponential). *For $f(t) = e^{i\omega_0 t}$,*

$$e^{i\omega_0 t} \longleftrightarrow F(\omega) = 2\pi \delta(\omega - \omega_0). \quad (4.16)$$

Proof. Using the expression for $F_k(\omega)$, the limiting distribution is

$$\lim_{k \rightarrow \infty} k \operatorname{sinc}\left(\frac{(\omega - \omega_0)k}{2}\right) = 2\pi \delta(\omega - \omega_0),$$

which is the scaled and shifted version of Theorem 15. $\square$

Thus an ideal monochromatic complex exponential has its entire spectral content concentrated at a single frequency.

4.10 Eternal Cosine and Sine

The cosine and sine of angular frequency ω_0 can be expressed in terms of complex exponentials:

$$\cos(\omega_0 t) = \frac{1}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t}), \quad (4.17)$$

$$\sin(\omega_0 t) = \frac{1}{2i} (e^{i\omega_0 t} - e^{-i\omega_0 t}). \quad (4.18)$$

Using (4.16), we obtain:

Proposition 9 (Fourier transforms of cosine and sine). For $\omega_0 \in \mathbb{R}$,

$$\begin{aligned}\cos(\omega_0 t) &\longleftrightarrow F_{\cos}(\omega) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)], \\ \sin(\omega_0 t) &\longleftrightarrow F_{\sin}(\omega) = -i\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].\end{aligned}$$

Proof. From (4.17),

$$\cos(\omega_0 t) = \frac{1}{2}e^{i\omega_0 t} + \frac{1}{2}e^{-i\omega_0 t}.$$

Using linearity and (4.16),

$$\mathcal{F}\{\cos(\omega_0 t)\}(\omega) = \frac{1}{2} \cdot 2\pi\delta(\omega - \omega_0) + \frac{1}{2} \cdot 2\pi\delta(\omega + \omega_0) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)].$$

Similarly, from (4.18),

$$\sin(\omega_0 t) = \frac{1}{2i}e^{i\omega_0 t} - \frac{1}{2i}e^{-i\omega_0 t},$$

and hence

$$\mathcal{F}\{\sin(\omega_0 t)\}(\omega) = \frac{1}{2i} \cdot 2\pi\delta(\omega - \omega_0) - \frac{1}{2i} \cdot 2\pi\delta(\omega + \omega_0) = -i\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].$$

□

The cosine is real and even; its spectrum is real and even, consisting of two impulses of equal weight π at $\omega = \pm\omega_0$. The sine is real and odd; its spectrum is purely imaginary and odd, with impulses of weight $-i\pi$ at $\omega = \omega_0$ and $i\pi$ at $\omega = -\omega_0$.

4.11 Periodic Functions

Let $f_p(t)$ be a real-valued function periodic with period $T_p > 0$,

$$f_p(t + T_p) = f_p(t),$$

and assume that f_p has finite energy in each period. Then f_p admits a complex Fourier series

$$f_p(t) = \sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t}, \quad \omega_0 := \frac{2\pi}{T_p}, \quad (4.19)$$

where the Fourier series coefficients are

$$F(n) = \frac{1}{T_p} \int_{t_0}^{t_0+T_p} f_p(t) e^{-in\omega_0 t} dt,$$

for any $t_0 \in \mathbb{R}$.

Using (4.16) term-by-term, we can write the Fourier transform of f_p as a sum of impulses in frequency:

Theorem 16 (Fourier transform of a periodic function). Let $f_p(t)$ be periodic with period T_p and Fourier series coefficients $F(n)$. Then in the sense of distributions,

$$f_p(t) \longleftrightarrow F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F(n) \delta(\omega - n\omega_0), \quad \omega_0 = \frac{2\pi}{T_p}. \quad (4.20)$$

Proof. Starting from the Fourier series (4.19),

$$f_p(t) = \sum_{n=-\infty}^{\infty} F(n) e^{in\omega_0 t},$$

and using linearity of the transform together with (4.16) for each harmonic,

$$\mathcal{F}\{e^{in\omega_0 t}\}(\omega) = 2\pi \delta(\omega - n\omega_0),$$

we obtain

$$F(\omega) = \mathcal{F}\{f_p(t)\}(\omega) = \sum_{n=-\infty}^{\infty} F(n) \mathcal{F}\{e^{in\omega_0 t}\}(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F(n) \delta(\omega - n\omega_0),$$

with convergence understood in the distribution sense. This is (4.20). $\square$

Thus the Fourier transform of a periodic function consists of a discrete set of spectral lines (Dirac deltas) at integer multiples of the fundamental frequency ω_0 , with weights proportional to the Fourier series coefficients.

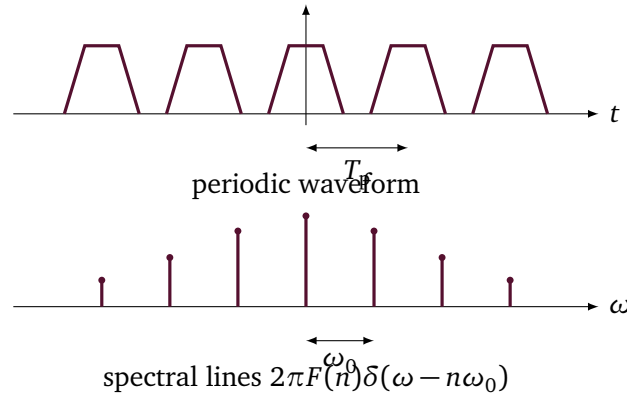


Figure 11: A periodic waveform has a Fourier transform made of impulses at harmonic frequencies. The impulse weights are proportional to the Fourier-series coefficients.

4.12 The Periodic Impulse Train (Dirac Comb)

An especially important periodic generalized function is the impulse train (Dirac comb) with period T_p :

Definition 20 (Periodic impulse train). For $T_p > 0$, the periodic impulse train in time is

$$\text{Sha}_{T_p}(t) := \sum_{k=-\infty}^{\infty} \delta(t - kT_p).$$

This signal is periodic with period T_p . We now compute its Fourier series coefficients and transform.

Over one period $(-T_p/2, T_p/2)$, only the impulse at the origin contributes. Using the Fourier series analysis formula,

$$F(n) = \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} \text{Sha}_{T_p}(t) e^{-in\omega_0 t} dt = \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} \delta(t) e^{-in\omega_0 t} dt = \frac{1}{T_p},$$

for all integers n . Thus

$$\text{Sha}_{T_p}(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T_p} e^{in\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_p},$$

in the distributional sense.

Applying Theorem 16, we obtain:

Theorem 17 (Fourier transform of the Dirac comb). *The Fourier transform of the periodic impulse train $\text{Sha}_{T_p}(t)$ is*

$$\text{Sha}_{T_p}(t) \longleftrightarrow F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} \frac{1}{T_p} \delta(\omega - n\omega_0), \quad \omega_0 = \frac{2\pi}{T_p}. \quad (4.21)$$

Equivalently, defining the frequency-domain comb

$$\text{Sha}_{\omega_0}(\omega) := \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0),$$

we can write

$$\text{Sha}_{T_p}(t) \longleftrightarrow F(\omega) = \frac{2\pi}{T_p} \text{Sha}_{\omega_0}(\omega), \quad \omega_0 = \frac{2\pi}{T_p}. \quad (4.22)$$

Proof. From the preceding computation, the Fourier series coefficients of $\text{Sha}_{T_p}(t)$ are $F(n) = 1/T_p$ for all integers n . Applying Theorem 16,

$$\mathcal{F}\{\text{Sha}_{T_p}(t)\}(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F(n) \delta(\omega - n\omega_0) = 2\pi \sum_{n=-\infty}^{\infty} \frac{1}{T_p} \delta(\omega - n\omega_0),$$

which is (4.21). Factoring out $2\pi/T_p$ and using the definition of Sha_{ω_0} gives (4.22). $\square$

Thus the Dirac comb is self-replicating under the Fourier transform: a comb in time transforms into a comb in frequency, with reciprocal spacing and appropriate scaling. This structure is central in the analysis of sampling and discrete-time representations, to be revisited later.

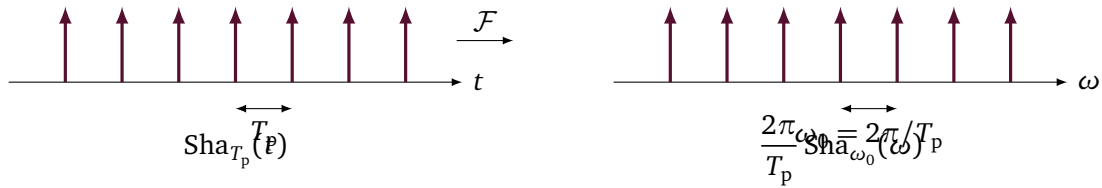


Figure 12: The Dirac comb transforms into another comb with reciprocal spacing and the scale factor required by the transform convention.

5 The Method of Successive Differentiation

5.1 The Differentiation Property

The Fourier transform is particularly useful because differentiation in time corresponds to multiplication in frequency. We first introduce the differential operator.

Definition 21 (Differential operator). *The operator D denotes differentiation with respect to time:*

$$Dx(t) = \frac{d}{dt}x(t), \quad D^2x(t) = \frac{d^2}{dt^2}x(t),$$

and, in general,

$$D^n x(t) = \frac{d^n}{dt^n}x(t).$$

We now state the basic relation between time-domain differentiation and the Fourier transform.

Theorem 18 (Time-domain differentiation). *Let $f(t) \longleftrightarrow F(\omega)$. Then*

$$Df(t) \longleftrightarrow i\omega F(\omega).$$

Proof. Using the synthesis formula,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega.$$

Differentiating with respect to t ,

$$\begin{aligned} Df(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \frac{d}{dt}(e^{i\omega t}) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) (i\omega) e^{i\omega t} d\omega, \end{aligned}$$

which is the synthesis formula for the Fourier transform $i\omega F(\omega)$. Hence $Df(t) \longleftrightarrow i\omega F(\omega)$. $\square$

Repeated differentiation follows immediately.

Corollary 5. *If $f(t) \longleftrightarrow F(\omega)$, then for any integer $n \geq 1$,*

$$D^n f(t) \longleftrightarrow (i\omega)^n F(\omega).$$

Polynomial differential operators act in an analogous way.

Corollary 6. *Let*

$$P(D) = a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0,$$

and let $f(t) \longleftrightarrow F(\omega)$. Then

$$P(D)f(t) \longleftrightarrow P(i\omega)F(\omega).$$

5.2 Differentiating Functions with Discontinuities

Differentiation of functions with jump discontinuities requires generalized functions. The key building block is the derivative of the unit step.

Recall that the Dirac delta $\delta(t)$ satisfies

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t),$$

where $u(t)$ is the unit step.

Proposition 10 (Derivative of the unit step).

$$\frac{d}{dt}u(t) = \delta(t).$$

By time shift, the same relation holds for shifted steps:

$$\frac{d}{dt}u(t - \tau) = \delta(t - \tau),$$

for any real τ .

Thus, for a function $x(t)$ with jump discontinuities, differentiation produces impulses at each jump.

Proposition 11 (Derivative of a piecewise-smooth function). *Let $x(t)$ be piecewise continuously differentiable and suppose that at $t = t_k$ it has a jump of height*

$$h_k = \lim_{t \rightarrow t_k^+} x(t) - \lim_{t \rightarrow t_k^-} x(t).$$

Then, in the sense of generalized functions,

$$Dx(t) = x'_{\text{cont}}(t) + \sum_k h_k \delta(t - t_k),$$

where $x'_{\text{cont}}(t)$ denotes the ordinary derivative in the intervals where x is continuous.

Combining Proposition 11 with Theorem 18 gives a systematic way to handle discontinuities in the time domain.

5.3 The Method of Successive Differentiation

For many signals of interest, especially piecewise-polynomial signals with finite support, repeated differentiation eventually yields a finite sum of impulses (and possibly derivatives of impulses). Once this derivative is known, the Fourier transform can be obtained by inspection in the frequency domain and then divided by an appropriate power of $i\omega$ to recover the original transform.

Definition 22 (Finite span). *A signal $x(t)$ is said to have finite span if there exists a finite interval $[a, b]$ such that $x(t) = 0$ for all $t \notin [a, b]$.*

For signals with finite span and piecewise-polynomial structure, the method proceeds as follows:

1. Differentiate $x(t)$ repeatedly until the result is a finite combination of impulses (and possibly impulse derivatives).

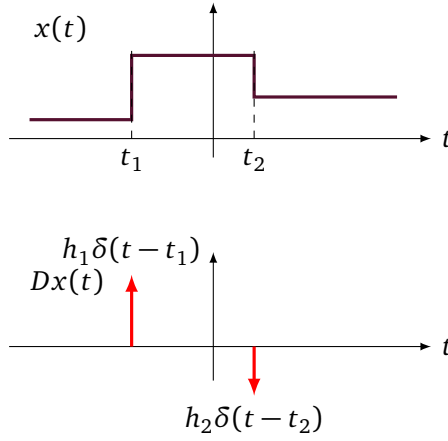


Figure 13: Ordinary derivatives handle the smooth pieces of a signal, while each jump contributes an impulse whose weight is the jump height.

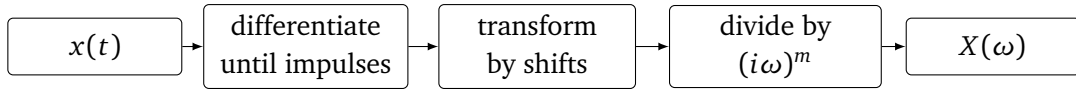


Figure 14: Successive differentiation avoids direct integration by converting piecewise-polynomial signals into weighted impulses, whose transforms are immediate.

2. Use known Fourier pairs to obtain the transform of the differentiated signal.
3. Divide by the corresponding power of $i\omega$ according to Corollary 5 to recover $X(\omega)$.

Example 3 (First derivative suffices). Let $x(t)$ be a piecewise-constant signal whose derivative is

$$Dx(t) = \delta(t) + \delta(t-1) - 2\delta(t-2).$$

Then, by Theorem 18 and the time-shift property,

$$i\omega X(\omega) = 1 + e^{-i\omega} - 2e^{-2i\omega},$$

so

$$X(\omega) = \frac{1 + e^{-i\omega} - 2e^{-2i\omega}}{i\omega}.$$

This expression is obtained without explicit use of the analysis integral.

Example 4 (Second derivative required). Consider a triangular pulse symmetric about the origin with support $[-1, 1]$. Its second derivative can be represented as

$$D^2x(t) = \delta(t+1) - 2\delta(t) + \delta(t-1).$$

Applying Theorem 18,

$$(i\omega)^2 X(\omega) = e^{i\omega} - 2 + e^{-i\omega},$$

so

$$X(\omega) = \frac{e^{i\omega} - 2 + e^{-i\omega}}{(i\omega)^2}.$$

This reproduces the known transform of a triangular pulse.

Impulses themselves can be differentiated.

Definition 23 (Doublet). *The derivative of the Dirac delta,*

$$\delta'(t) = D\delta(t),$$

is called a doublet.

From Theorem 18,

$$\delta'(t) \longleftrightarrow i\omega.$$

5.4 A Complication: Signals Without Finite Span

When the signal does not have finite span, repeated differentiation may remove constant components, which correspond to impulses at $\omega = 0$ in the frequency domain. These must be restored at the end of the procedure.

We distinguish two classes.

Definition 24 (Class I and Class II signals). Class I: *Signals that are piecewise polynomial and have finite span.* Class II: *Signals that are piecewise polynomial but do not have finite span.*

For Class I signals, the method of successive differentiation applies directly, and no additional correction at $\omega = 0$ is required.

For the Class II signals used in this section, the non-finite-span part is a constant tail or a step-like tail. A nonzero constant background corresponds to a $\delta(\omega)$ term in the frequency domain, and differentiation, which annihilates constants, removes that term. After using successive differentiation, one must add back the contribution corresponding to the average value. More general polynomial tails can produce derivatives of $\delta(\omega)$, so the simple dc-restoration rule below is meant only for the constant-background case.

Proposition 12 (Restoring the dc component). *Let $f(t)$ be a Class II signal whose only nondecaying background is a constant average value*

$$\bar{f} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f(t) dt.$$

Then the Fourier transform includes the term

$$2\pi\bar{f} \delta(\omega).$$

If successive differentiation has been used to determine the rest of the spectrum, the term $2\pi\bar{f} \delta(\omega)$ must be added to obtain the complete $F(\omega)$.

Example 5 (Rectangular pulse plus constant). *Let*

$$f(t) = \text{Rect}(t) + C,$$

where $\text{Rect}(t)$ has support on $[-1/2, 1/2]$. Then $f(t)$ is piecewise constant but does not have finite span because of the constant C . Differentiation gives

$$Df(t) = D\text{Rect}(t),$$

so the constant C is removed. Using the method of successive differentiation yields the transform of $\text{Rect}(t)$,

$$F_{\text{Rect}}(\omega) = \text{sinc}\left(\frac{\omega}{2}\right).$$

The average of $f(t)$ over $\mathbb{R}$ is $\bar{f} = C$. Hence the complete transform is

$$F(\omega) = \text{sinc}\left(\frac{\omega}{2}\right) + 2\pi C \delta(\omega).$$

Example 6 ($\text{sgn}(t)$). *The signum function*

$$\text{sgn}(t) = \begin{cases} -1, & t < 0, \\ 0, & t = 0, \\ +1, & t > 0, \end{cases}$$

is a Class II signal. Its derivative is

$$D \text{sgn}(t) = 2\delta(t),$$

so

$$i\omega F(\omega) = 2,$$

and

$$F(\omega) = \frac{2}{i\omega}.$$

The average of $\text{sgn}(t)$ over $\mathbb{R}$ is zero, so no $\delta(\omega)$ term is needed.

Example 7 (Unit step $u(t)$). *The unit step*

$$u(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{2}, & t = 0, \\ 1, & t > 0, \end{cases}$$

is also a Class II signal. Its derivative is

$$Du(t) = \delta(t),$$

so

$$i\omega F(\omega) = 1,$$

and

$$F(\omega) = \frac{1}{i\omega}.$$

The average of $u(t)$ over $\mathbb{R}$ is $1/2$, hence the complete transform is

$$u(t) \longleftrightarrow \frac{1}{i\omega} + \pi \delta(\omega).$$

5.5 Nonpolynomial Sections

The method of successive differentiation extends to many signals that are not piecewise polynomial, provided that differentiation produces expressions whose Fourier transforms are known or easily identified.

For example, the one-sided decaying exponential

$$f(t) = e^{-\alpha t} u(t), \quad \alpha > 0,$$

has the well-known transform

$$f(t) \longleftrightarrow \frac{1}{\alpha + i\omega},$$

which is consistent with the differentiation property and the unit-step transform. Similar reasoning applies to gated sinusoids such as $\cos(\omega_0 t) \text{Rect}(t/T)$ and related signals, where differentiation combined with time shifts and modulation can simplify the derivation of their Fourier transforms.

6 Frequency-Domain Analysis

6.1 Introduction

Perhaps the most important contribution that Fourier analysis makes to engineering, and especially to electrical engineering, is that it enables us to decompose input signals into their frequency components. Once this has been done, it becomes possible to understand how filters and other linear networks act on such signals, frequency by frequency, and to design them so that the resulting outputs have desired attributes.

In this chapter we consider three applications of Fourier analysis in the frequency domain:

1. The response of an electrical network to a pulse.
2. The response of a network to an ac input such as $\sin(\omega_0 t)$ or $\cos(\omega_0 t)$.
3. The response of a network to an arbitrary periodic signal.

The second application, usually referred to as ac linear circuit analysis, is one of the most frequently used analytical tools in electrical engineering (see, e.g., Edminster, 1972). Often it is taught in an algorithmic way at an early stage of study: students are shown that if they replace inductors and capacitors in a network by the quantities $i\omega L$ and $1/(i\omega C)$, respectively (sometimes termed “complex resistors”), and if they take as the input to the network the quantity $e^{i\omega_0 t}$ (a *phasor*), then the response may be computed directly using Kirchhoff’s voltage and current laws. Although this procedure may initially appear somewhat mysterious, its effectiveness leads to quick acceptance and the underlying justification is often not emphasized.

Historically, the method was first used by Oliver Heaviside in 1878 (Nahin, 1987). It can be derived via his operational calculus, but we shall avoid that route here. Instead, we show how it arises directly from Fourier analysis. In many introductory courses this connection is not made explicit because students typically lack the required background in Fourier transforms. Since we now have that background, one of the main objectives of this chapter is to show clearly how the techniques of ac circuit analysis are obtained from Fourier analysis and why they are valid. This provides a natural setting in which to apply the machinery developed in previous chapters.

Although the examples in this chapter are drawn from electrical networks, the methods are applicable to a wide range of systems composed of other types of components—mechanical, acoustical, hydraulic, and many others. More generally, the forthcoming results apply to any system whose behavior can be modeled by a constant-coefficient linear differential equation.

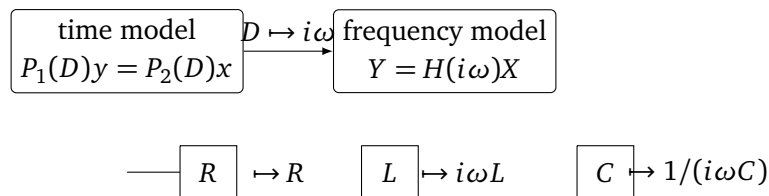


Figure 15: Frequency-domain circuit analysis replaces differentiation by multiplication by $i\omega$, turning constant-coefficient differential equations into algebraic transfer functions.

6.2 Response of a Linear, Time-Invariant System to a Pulse Function

We focus on systems composed of electrical components and, more specifically, on linear systems whose element values do not change with time. Such systems are called *linear, time-invariant*

(LTI) systems.

Definition 25 (LTI system and CCL DE). *An LTI system is one that can be characterized by a constant-coefficient linear differential equation (CCL DE). The LTI system and its associated CCL DE are mathematically equivalent, and we shall use these descriptions interchangeably.*

The effort required to derive the differential equation of an electrical network of nontrivial complexity can be substantial. One of the main advantages of the frequency-domain approach that we develop in this chapter is that, in practice, we will often not need to write down the differential equation explicitly. Instead, we will derive an operational method that allows us to obtain the response without first forming the differential equation. If desired, the differential equation can then be reconstructed directly from the frequency-domain characterization.

To begin, we examine how the Fourier transform may be used to solve CCL DEs and, in particular, to determine the response of an LTI network to various pulse inputs.

6.2.1 General form of a constant-coefficient linear differential equation

The general form of a CCL DE describing an LTI system is

$$P_1(D)y(t) = P_2(D)x(t), \quad (6.1)$$

where

- $P_1(D)$ and $P_2(D)$ are polynomials in the operator $D = \frac{d}{dt}$ with constant coefficients,
- $x(t)$ is the input or forcing function,
- $y(t)$ is the output (response, or solution).

An example is

$$(5D^2 + 6D + 1)y(t) = (3D + 2)x(t). \quad (6.2)$$

The coefficients of the two polynomials are constants and the equation is linear, so (6.2) is indeed a CCL DE.

Two points are important:

1. Equation (6.2) is assumed to hold for all t ; that is, the solution $y(t)$ must satisfy the equation for all t in $(-\infty, \infty)$.
2. No initial conditions are specified; that is, no constraints such as $y(0) = 1$ or $y'(5) = 2$ are imposed.

These two facts always apply, either explicitly or implicitly, when using the Fourier transform to solve CCL DEs.

From a system viewpoint, an LTI system/CCL DE can be represented by the block diagram in Figure 16.

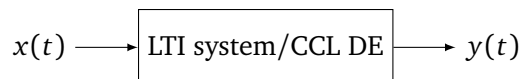


Figure 16: Input-output view of the LTI system represented by a constant-coefficient linear differential equation.

To *solve* the CCL DE means: given $x(t)$, find $y(t)$ so that (6.1) is satisfied for all t . Thus, solving the CCL DE associated with an LTI system is equivalent to finding the response of the system to the given input.

6.2.2 Fourier transform of a CCL DE

We now solve (6.2) using the differentiation property of the Fourier transform. Taking the Fourier transform of both sides of (6.2) yields

$$\int_{-\infty}^{\infty} (5D^2 + 6D + 1)y(t)e^{-i\omega t} dt = \int_{-\infty}^{\infty} (3D + 2)x(t)e^{-i\omega t} dt.$$

The integration over $(-\infty, \infty)$ is consistent with the assumption that the differential equation holds for all t .

Using the differentiation property of the Fourier transform, i.e. $D^k x(t) \leftrightarrow (i\omega)^k X(\omega)$, and denoting

$$x(t) \longleftrightarrow X(\omega), \quad y(t) \longleftrightarrow Y(\omega),$$

we obtain

$$[5(i\omega)^2 + 6(i\omega) + 1]Y(\omega) = [3(i\omega) + 2]X(\omega). \quad (6.3)$$

This is a purely algebraic equation in the frequency domain. More generally, we have the following result.

Theorem 19. *Fourier transformation of the time-domain CCL DE*

$$P_1(D)y(t) = P_2(D)x(t)$$

converts it to the frequency-domain algebraic equation

$$P_1(i\omega)Y(\omega) = P_2(i\omega)X(\omega),$$

where $x(t) \leftrightarrow X(\omega)$ and $y(t) \leftrightarrow Y(\omega)$.

Solving (6.3) for $Y(\omega)$ gives

$$Y(\omega) = \frac{3i\omega + 2}{5(i\omega)^2 + 6i\omega + 1}X(\omega). \quad (6.4)$$

Two observations are useful:

1. The right-hand side of (6.4) is the product of two factors: one depending only on the differential equation, and one depending only on the input $x(t)$.
2. The expression is valid for any $x(t)$ possessing a Fourier transform; once $X(\omega)$ is known, (6.4) yields $Y(\omega)$, and hence $y(t)$ by inversion.

6.2.3 Frequency response (transfer function)

Equation (6.4) suggests representing the LTI system in the frequency domain, where the system is characterized by a multiplicative function of frequency.

We define the frequency response (or frequency transfer function) as follows.

Definition 26 (Frequency response / frequency transfer function). *Let the CCL DE for an LTI system be*

$$P_1(D)y(t) = P_2(D)x(t).$$

The frequency response or frequency transfer function of the system is

$$H(i\omega) = \frac{P_2(i\omega)}{P_1(i\omega)}.$$

For the example (6.2), we have

$$H(i\omega) = \frac{3D + 2}{5D^2 + 6D + 1} \Big|_{D \rightarrow i\omega} = \frac{3i\omega + 2}{5(i\omega)^2 + 6i\omega + 1}.$$

The function $H(i\omega)$ encapsulates all information about the LTI system: the component values and their interconnection are reflected in its numerator and denominator. It is a frequency-domain quantity. In the next chapter, we will introduce a corresponding time-domain quantity, the *impulse response* $h(t)$, which also completely characterizes the system. As is well known, $H(i\omega)$ and $h(t)$ form a Fourier-transform pair.

With Definition 26, equation (6.4) may be written in the compact form

$$Y(\omega) = H(i\omega)X(\omega), \quad (6.5)$$

and, by the synthesis formula,

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(i\omega)X(\omega)e^{i\omega t} d\omega. \quad (6.6)$$

Equations (6.5) and (6.6) constitute a central result in the analysis of LTI systems.

Theorem 20. *Let the pulse $x(t) \leftrightarrow X(\omega)$ be the input to an LTI system whose frequency transfer function is $H(i\omega)$. Then:*

1. *In the frequency domain, the output transform is*

$$Y(\omega) = H(i\omega)X(\omega).$$

2. *In the time domain, the output is given by*

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(i\omega)X(\omega)e^{i\omega t} d\omega. \quad (6.7)$$

It is instructive to compare (6.7) with the Fourier integral representation of the input,

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{i\omega t} d\omega. \quad (6.8)$$

The output representation (6.7) differs from (6.8) only by the factor $H(i\omega)$ under the integral: the system acts in the frequency domain by pointwise multiplication of the input spectrum.

6.2.4 Magnitude and phase of the output spectrum

We now illustrate these ideas with two examples.

Example 8. *Assume that the frequency response of a network is*

$$H(i\omega) = \frac{1}{2 + i\omega}. \quad (6.9)$$

Find the magnitude and phase spectra of the output when the input is

$$x(t) = e^{-t}u(t),$$

where $u(t)$ is the unit step.

Solution. From (6.9),

$$H(i\omega) = \frac{1}{2 + i\omega} = \frac{1}{\sqrt{4 + \omega^2}} \exp(-i \operatorname{atan2}(\omega, 2)),$$

so the magnitude and phase spectra of the system are

$$|H(i\omega)| = \frac{1}{\sqrt{4 + \omega^2}}, \quad \phi_H(\omega) = -\operatorname{atan2}(\omega, 2).$$

From the standard transform pair

$$e^{-t}u(t) \longleftrightarrow \frac{1}{1 + i\omega},$$

we obtain

$$X(\omega) = \frac{1}{1 + i\omega} = \frac{1}{\sqrt{1 + \omega^2}} \exp(-i \operatorname{atan2}(\omega, 1)),$$

so

$$|X(\omega)| = \frac{1}{\sqrt{1 + \omega^2}}, \quad \phi_X(\omega) = -\operatorname{atan2}(\omega, 1).$$

Using Theorem 20, the output transform is

$$Y(\omega) = H(i\omega)X(\omega) = |H(i\omega)||X(\omega)| \exp(i[\phi_H(\omega) + \phi_X(\omega)]).$$

Therefore,

$$|Y(\omega)| = |H(i\omega)||X(\omega)| = \frac{1}{\sqrt{4 + \omega^2}} \frac{1}{\sqrt{1 + \omega^2}},$$

and

$$\phi_Y(\omega) \equiv \phi_H(\omega) + \phi_X(\omega) = -\operatorname{atan2}(\omega, 2) - \operatorname{atan2}(\omega, 1) \pmod{2\pi}.$$

6.3 AC Circuit Analysis from the Fourier Viewpoint

The familiar phasor method is now a direct consequence of the preceding theorem. If the input is the eternal complex exponential

$$x(t) = X_0 e^{i\omega_0 t},$$

then its spectrum is concentrated at the single frequency ω_0 . Since an LTI system multiplies each frequency component by $H(i\omega)$, the output must be

$$y(t) = H(i\omega_0)X_0 e^{i\omega_0 t}.$$

Thus $e^{i\omega_0 t}$ passes through the system unchanged in frequency; only its complex amplitude is multiplied by $H(i\omega_0)$. This is the engineering reason that phasors work.

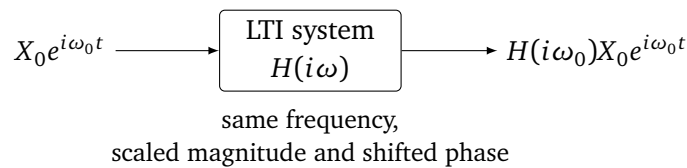


Figure 17: A complex exponential is an eigenfunction of an LTI system. The network only multiplies its complex amplitude by $H(i\omega_0)$.

For a circuit made of resistors, inductors, and capacitors, the differential relationships become algebraic at frequency ω_0 :

$$v_R = Ri, \quad v_L = L \frac{di}{dt} \mapsto i\omega_0 L I, \quad i_C = C \frac{dv}{dt} \mapsto i\omega_0 C V.$$

Equivalently, the frequency-domain impedances are

$$Z_R = R, \quad Z_L = i\omega_0 L, \quad Z_C = \frac{1}{i\omega_0 C}.$$

Kirchhoff's laws applied to these complex impedances give the same $H(i\omega_0)$ obtained from the differential equation, but with much less algebra. The Fourier-transform justification is that the calculation is being done one frequency component at a time.

6.4 Response to Periodic Inputs

If the input is periodic,

$$x_p(t) = \sum_{n=-\infty}^{\infty} X(n)e^{in\omega_0 t},$$

then every harmonic is again an eigenfunction of the LTI system. Applying the single-frequency result term by term gives

$$y_p(t) = \sum_{n=-\infty}^{\infty} H(in\omega_0)X(n)e^{in\omega_0 t}.$$

Thus the output Fourier-series coefficients are

$$Y(n) = H(in\omega_0)X(n).$$

This is the periodic counterpart of $Y(\omega) = H(i\omega)X(\omega)$. In practice, one analyzes a periodic input by finding its Fourier coefficients, multiplying each line by the transfer function at that harmonic, and synthesizing the result.

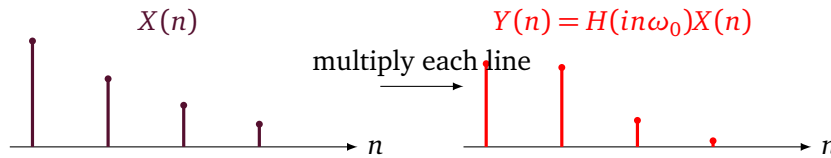


Figure 18: For a periodic input, the transfer function acts line by line on the Fourier-series spectrum.

6.5 Finding a Transfer Function from a Pulse Response

Sometimes the differential equation is not available, but a test input and output are. If the input pulse $x(t)$ and output pulse $y(t)$ can be measured or computed, and if $X(\omega) \neq 0$ on the frequency range of interest, then

$$H(i\omega) = \frac{Y(\omega)}{X(\omega)}.$$

This is the same relationship as before, solved for the unknown system. It is the frequency-domain version of system identification: apply a known signal, measure the response, transform both, and divide. Frequencies where $X(\omega)$ is zero or very small must be treated with care, since division there amplifies numerical error.

Observe that in Example 8:

- The magnitude spectra multiply,
- The phase spectra add modulo 2π .

This is a general property of LTI systems.

Theorem 21. Let the signal $x(t) \leftrightarrow X(\omega)$ be the input to an LTI network whose frequency response is $H(i\omega)$, and let $y(t) \leftrightarrow Y(\omega)$ be the output. Then

$$|Y(\omega)| = |H(i\omega)| |X(\omega)|,$$

and

$$\phi_Y(\omega) \equiv \phi_H(\omega) + \phi_X(\omega) \pmod{2\pi},$$

where ϕ_H , ϕ_X , and ϕ_Y denote the corresponding phase spectra.

In other words, an LTI network acts as a frequency-selective filter: it modifies the magnitude and phase of each frequency component of the input according to $H(i\omega)$.

Example 9. Consider the electrical network characterized by the CCL DE

$$12y''(t) + 7y'(t) + y(t) = 2x'(t) + x(t). \quad (6.10)$$

- (1) Find the frequency response $H(i\omega)$.
- (2) Find the magnitude and phase spectra of $y(t)$ when $x(t) = e^{-t}u(t)$.
- (3) Find the explicit time-domain expression for $y(t)$.
- (4) Verify that your result in (3) satisfies (6.10).

Solution. (1) Frequency response. Using the operator $D = \frac{d}{dt}$, (6.10) can be written as

$$(12D^2 + 7D + 1)y(t) = (2D + 1)x(t).$$

Hence, by Definition 26,

$$H(i\omega) = \frac{2D + 1}{12D^2 + 7D + 1} \Big|_{D \rightarrow i\omega} = \frac{2i\omega + 1}{12(i\omega)^2 + 7i\omega + 1}. \quad (6.11)$$

(2) Magnitude and phase spectra. We first rewrite the denominator of (6.11):

$$12(i\omega)^2 + 7i\omega + 1 = -12\omega^2 + 7i\omega + 1.$$

Thus

$$H(i\omega) = \frac{2i\omega + 1}{1 - 12\omega^2 + 7i\omega}.$$

The magnitude and phase are obtained as the ratio and difference of the corresponding quantities for numerator and denominator:

$$|H(i\omega)| = \frac{\sqrt{1 + 4\omega^2}}{\sqrt{(1 - 12\omega^2)^2 + 49\omega^2}},$$

$$\phi_H(\omega) = \text{atan2}(2\omega, 1) - \text{atan2}(7\omega, 1 - 12\omega^2).$$

The quadrant-aware atan2 is important here because the real part of the denominator changes sign.

For the input $x(t) = e^{-t}u(t)$, we have

$$x(t) = e^{-t}u(t) \longleftrightarrow X(\omega) = \frac{1}{1 + i\omega},$$

so

$$|X(\omega)| = \frac{1}{\sqrt{1 + \omega^2}}, \quad \phi_X(\omega) = -\text{atan2}(\omega, 1).$$

By Theorem 21, the output magnitude and phase spectra are

$$|Y(\omega)| = |H(i\omega)| |X(\omega)| = \frac{\sqrt{1 + 4\omega^2}}{\sqrt{(1 - 12\omega^2)^2 + 49\omega^2}} \frac{1}{\sqrt{1 + \omega^2}},$$

$$\phi_Y(\omega) \equiv \phi_H(\omega) + \phi_X(\omega) = \text{atan2}(2\omega, 1) - \text{atan2}(7\omega, 1 - 12\omega^2) - \text{atan2}(\omega, 1) \pmod{2\pi}.$$

(3) Time-domain expression for $y(t)$. The output transform is

$$Y(\omega) = H(i\omega)X(\omega) = \frac{2i\omega + 1}{12(i\omega)^2 + 7i\omega + 1} \frac{1}{1 + i\omega}. \quad (6.12)$$

We perform a partial fraction expansion of (6.12) in the form

$$Y(\omega) = \frac{A}{i\omega + \alpha} + \frac{B}{i\omega + \beta} + \frac{C}{i\omega + \gamma},$$

where α, β, γ are the poles of the denominator and A, B, C are constants determined algebraically.

Inversion term by term then yields

$$12(i\omega)^2 + 7i\omega + 1 = (3i\omega + 1)(4i\omega + 1),$$

so, with $s = i\omega$,

$$Y = \frac{2s + 1}{(s + 1)(3s + 1)(4s + 1)} = \frac{8}{3(4s + 1)} - \frac{3}{2(3s + 1)} - \frac{1}{6(s + 1)}.$$

Equivalently,

$$Y(\omega) = \frac{2/3}{i\omega + 1/4} - \frac{1/2}{i\omega + 1/3} - \frac{1/6}{i\omega + 1}.$$

Using $e^{-at}u(t) \leftrightarrow 1/(a + i\omega)$, this gives

$$y(t) = \left[(2/3)e^{-t/4} - (1/2)e^{-t/3} - (1/6)e^{-t} \right] u(t). \quad (6.13)$$

(4) Verification. For $t > 0$, ordinary differentiation gives

$$12y''(t) + 7y'(t) + y(t) = -e^{-t}.$$

Also $y(0^+) = 0$ and $y'(0^+) = 1/6$. Hence, in the generalized-function sense,

$$y''(t) = y''_{\text{cont}}(t) + \frac{1}{6}\delta(t),$$

so the left-hand side contains the impulse contribution $12(1/6)\delta(t) = 2\delta(t)$. Since

$$2x'(t) + x(t) = 2[\delta(t) - e^{-t}u(t)] + e^{-t}u(t) = 2\delta(t) - e^{-t}u(t),$$

we obtain

$$12y''(t) + 7y'(t) + y(t) = 2x'(t) + x(t)$$

for all t , with $x(t) = e^{-t}u(t)$, confirming that (6.13) is indeed a solution of the CCL DE.

7 Time-Domain Analysis

7.1 Introduction

Fourier analysis originated in the study of heat conduction and the solution of partial differential equations (PDEs). In its modern usage for electrical engineering, several shifts in emphasis have taken place:

- In Fourier's original work, a Fourier series was not required to represent a periodic function; it was viewed as an expansion of an initial temperature profile on a finite rod. In electrical engineering, the inherent periodicity of Fourier series makes them natural tools for representing periodic waveforms.
- The functions appearing in Fourier integrals were originally viewed as initial temperature distributions on an infinite rod, i.e. functions of the spatial variable x . In signal analysis, they are interpreted as one-time pulses $x(t)$, not necessarily of infinite duration.
- The inversion formula

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

did not play a central role in Fourier's own applications. In the analysis of linear, time-invariant (LTI) systems, however, such inversion formulas are fundamental: the application of Fourier analysis to LTI networks depended crucially on this relationship.

Historically, Fourier's methods were adapted to electrical engineering problems through the work of several authors, including Kelvin and Heaviside. Kelvin applied Fourier techniques to the telegraph equation, focusing on transmission lines rather than lumped networks. Heaviside, in particular, made systematic use of operational methods for electrical circuits and is credited with the development of the classical ac circuit analysis framework, in which inductors and capacitors are represented by $i\omega L$ and $1/(i\omega C)$, respectively. The function now commonly called the unit step is still referred to as the Heaviside function in some contexts.

A key observation is that Fourier series and Fourier integrals resolve signals into sums or superpositions of complex exponentials. For LTI systems, complex exponentials are eigenfunctions, and this makes Fourier methods natural tools for studying network responses.

7.2 The Impulse Response

In Chapter 6 the analysis of LTI systems was carried out primarily in the frequency domain. For an LTI system with input $x(t)$, output $y(t)$, and frequency response (or transfer function) $H(i\omega)$, the central result was

$$Y(\omega) = H(i\omega)X(\omega), \quad (7.1)$$

where

$$x(t) \longleftrightarrow X(\omega), \quad y(t) \longleftrightarrow Y(\omega).$$

The goal here is to interpret (7.1) purely in the time domain. The transform pairs for $x(t)$ and $y(t)$ are clear; it remains to identify the time-domain quantity corresponding to $H(i\omega)$.

Problem 1. What is the time-domain counterpart of $H(i\omega)$ in (7.1)?

Let the inverse Fourier transform of $H(i\omega)$ be denoted by $h(t)$. That is,

$$H(i\omega) \longleftrightarrow h(t), \quad (7.2)$$

in the sense that

$$h(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(i\omega) e^{i\omega t} d\omega.$$

Consider now an LTI system with frequency response $H(i\omega)$, and apply as input the Dirac delta

$$x(t) = \delta(t).$$

Since $\delta(t) \leftrightarrow 1$, we have $X(\omega) = 1$, and (7.1) reduces to

$$Y(\omega) = H(i\omega).$$

By (7.2), the time-domain counterpart of $Y(\omega)$ in this case is precisely $h(t)$:

$$y(t) = h(t).$$

Thus, when the input to the LTI system is a unit impulse, the output is the inverse Fourier transform of $H(i\omega)$.

Theorem 22 (Impulse response). *Let $H(i\omega)$ be the frequency response of an LTI system. Its inverse Fourier transform*

$$h(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(i\omega) e^{i\omega t} d\omega \quad (7.3)$$

is the output of the system when the input is the unit impulse $\delta(t)$.

Definition 27 (Impulse response). *The function $h(t)$ defined by (7.3) is called the impulse response of the LTI system.*

Example 10. *Consider two first-order LTI systems with frequency responses*

$$H_1(i\omega) = \frac{B}{B + i\omega}, \quad B > 0,$$

$$H_2(i\omega) = 1 - \frac{B}{B + i\omega} = \frac{i\omega}{B + i\omega}, \quad B > 0.$$

Find their impulse responses.

- For $H_1(i\omega)$, we recognize the standard transform pair

$$Be^{-Bt}u(t) \longleftrightarrow \frac{B}{B + i\omega}.$$

Hence

$$h_1(t) = Be^{-Bt}u(t).$$

- For $H_2(i\omega)$,

$$H_2(i\omega) = 1 - \frac{B}{B + i\omega}$$

and using linearity together with $\delta(t) \leftrightarrow 1$, we obtain

$$h_2(t) = \delta(t) - Be^{-Bt}u(t).$$

In both cases the time-domain impulse response follows directly by inverting $H(i\omega)$.

7.3 Convolution

Consider an LTI system with input $x(t)$, output $y(t)$, and impulse response $h(t)$, as indicated conceptually by

$$x(t) \longrightarrow \boxed{\text{LTI system}} \longrightarrow y(t),$$

with $h(t)$ defined as in Theorem 22. The goal is to obtain a time-domain formula expressing $y(t)$ in terms of $x(t)$ and $h(t)$.

Problem 2. *Given the input $x(t)$ and the impulse response $h(t)$ of an LTI system, what is the time-domain expression for the output $y(t)$?*

From frequency-domain analysis we already know that

$$Y(\omega) = H(i\omega)X(\omega), \quad (7.4)$$

with

$$H(i\omega) \longleftrightarrow h(t).$$

Thus $y(t)$ is the inverse Fourier transform of $H(i\omega)X(\omega)$:

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(i\omega)X(\omega)e^{i\omega t} d\omega.$$

The convolution theorem identifies a purely time-domain expression for this inverse transform.

Theorem 23 (Time-domain convolution theorem). *Let $x(t) \longleftrightarrow X(\omega)$ and $h(t) \longleftrightarrow H(i\omega)$. Then*

$$\int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau \longleftrightarrow X(\omega)H(i\omega).$$

Equivalently,

$$\mathcal{F}\{x * h\}(\omega) = X(\omega)H(i\omega),$$

where

$$(x * h)(t) := \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau.$$

Definition 28 (Convolution). *Given two functions $x(t)$ and $h(t)$, their convolution is the function*

$$(x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau.$$

The same operation is sometimes written $x(t) * h(t)$ for emphasis.

Combining Theorem 23 with (7.4), we obtain the time-domain input–output relation for LTI systems.

Theorem 24 (Impulse response representation of an LTI system). *Let an LTI system have impulse response $h(t)$, and let $x(t)$ be the input and $y(t)$ the output. Then*

$$\begin{aligned} Y(\omega) &= H(i\omega)X(\omega), \\ y(t) &= (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau. \end{aligned}$$

Thus, multiplication in the frequency domain corresponds to convolution in the time domain:

$$x * h \longleftrightarrow X(\omega)H(i\omega).$$

Convolution therefore provides a complete time-domain alternative to the frequency-domain techniques for computing the response of an LTI system to a pulse input.

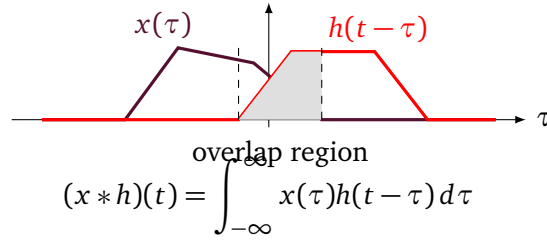


Figure 19: Convolution multiplies two functions after one has been reversed and shifted; the integral accumulates the area of their overlap.

7.4 Interpretation of the Convolution Product

The formula

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau \quad (7.5)$$

encapsulates the response of an LTI system to an arbitrary input $x(t)$ in terms of its impulse response $h(t)$. It is useful to interpret this expression in terms of superposition of scaled and shifted impulse responses.

Formally, any (sufficiently regular) pulse $x(t)$ can be represented as a superposition of weighted and shifted Dirac deltas:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau. \quad (7.6)$$

This identity follows from the sampling property of the delta function:

$$\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau = x(t).$$

Theorem 25 (Representation by impulses). *Any suitable pulse $x(t)$ can be written as a “continuous linear combination” of shifted Dirac deltas:*

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau,$$

where $x(\tau)$ acts as the coefficient function.

Now consider an LTI system with impulse response $h(t)$. By linearity and time invariance:

- The response to $\delta(t)$ is $h(t)$.
- The response to $\delta(t - \tau)$ is $h(t - \tau)$ (time shift).
- The response to $x(\tau) \delta(t - \tau)$ is $x(\tau)h(t - \tau)$ (scaling by $x(\tau)$).

Using the representation (7.6), the input can be viewed as a superposition of such weighted impulses. By superposition of responses,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau,$$

which is precisely the convolution expression (7.5).

Thus, convolution can be interpreted as a superposition of all shifted impulse responses, each weighted by the value of the input at the corresponding shift. Equivalently, the input $x(t)$ is a “train” of weighted impulses; the output $y(t)$ is the corresponding train of weighted, shifted impulse responses, summed over all shifts.

This viewpoint is consistent with the eigenfunction interpretation. There, the input was expanded in complex exponentials and the output was obtained as a superposition of eigen-responses. Here, the expansion is in terms of Dirac deltas rather than exponentials, but the linear superposition principle is the same.

7.5 Convolution Symmetry

The convolution product is symmetric in its arguments.

Theorem 26 (Convolution symmetry). *For suitable functions $x(t)$ and $h(t)$,*

$$\int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(t-\tau)h(\tau) d\tau.$$

Proof. Let $z = t - \tau$, so that $\tau = t - z$ and $d\tau = -dz$. Then

$$\begin{aligned} \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau &= \int_{\infty}^{-\infty} x(t-z)h(z)(-dz) \\ &= \int_{-\infty}^{\infty} x(t-z)h(z) dz \\ &= \int_{-\infty}^{\infty} x(t-\tau)h(\tau) d\tau. \end{aligned}$$

□

Because of this symmetry, one may write

$$(x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(t-\tau)h(\tau) d\tau,$$

and use whichever representation is more convenient in a given context, including graphical convolution constructions.

7.6 Convolution the Graphical Way

For hand calculation, the formula

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau$$

should be read as an instruction:

1. redraw the input as a function of the dummy variable τ ;
2. reverse the impulse response to obtain $h(-\tau)$;
3. shift it by t to obtain $h(t-\tau)$;
4. multiply $x(\tau)$ and $h(t-\tau)$;

5. integrate the overlap area.

As t moves, the overlap changes. Break points occur when an edge of one pulse meets an edge of the other pulse. Between consecutive break points, the limits of integration are fixed by inspection.

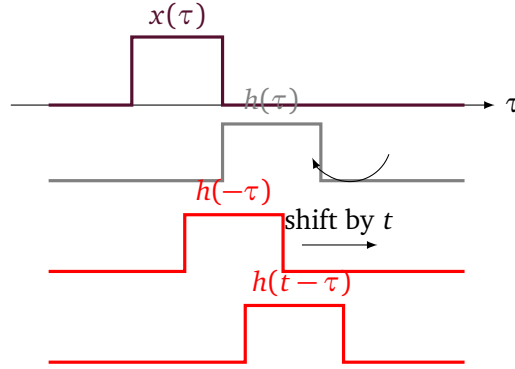


Figure 20: Graphical convolution: rewrite both functions in τ , reverse one, shift it, then integrate the overlap.

Example 11 (Two rectangular pulses). *Let*

$$x(t) = \begin{cases} 1, & 0 < t < a, \\ 0, & \text{otherwise,} \end{cases} \quad h(t) = \begin{cases} 1, & 0 < t < b, \\ 0, & \text{otherwise,} \end{cases}$$

with $0 < a \leq b$. Then $x(\tau)h(t - \tau)$ is equal to 1 exactly on the intersection of the intervals

$$0 < \tau < a, \quad t - b < \tau < t.$$

Therefore the convolution is the length of that intersection:

$$(x * h)(t) = \max\{0, \min(a, t) - \max(0, t - b)\}.$$

Writing this by intervals gives

$$(x * h)(t) = \begin{cases} 0, & t < 0, \\ t, & 0 < t < a, \\ a, & a < t < b, \\ a + b - t, & b < t < a + b, \\ 0, & t > a + b. \end{cases}$$

The output support has length $a + b$, which is the continuous counterpart of the span rule used later for FFT convolution.

7.7 Convolution in the Frequency Domain

The convolution theorem has a dual counterpart: whereas convolution in time corresponds to multiplication in frequency, multiplication in time corresponds to convolution in frequency.

Theorem 27 (Frequency-domain convolution theorem). Let $x(t) \leftrightarrow X(\omega)$ and $g(t) \leftrightarrow G(\omega)$. Then

$$x(t)g(t) \longleftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) G(\omega - \Omega) d\Omega.$$

Equivalently,

$$\mathcal{F}\{xg\}(\omega) = \frac{1}{2\pi} (X * G)(\omega),$$

where the convolution in frequency is defined by

$$(X * G)(\omega) = \int_{-\infty}^{\infty} X(\Omega) G(\omega - \Omega) d\Omega.$$

Definition 29 (Frequency-domain convolution). Given frequency-domain functions $X(\omega)$ and $G(\omega)$, their convolution is

$$(X * G)(\omega) = \int_{-\infty}^{\infty} X(\Omega) G(\omega - \Omega) d\Omega = \int_{-\infty}^{\infty} G(\Omega) X(\omega - \Omega) d\Omega.$$

Proof. Let

$$Y(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) G(\omega - \Omega) d\Omega.$$

Take its inverse Fourier transform:

$$\begin{aligned} y(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(\omega) e^{i\omega t} d\omega \\ &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} X(\Omega) G(\omega - \Omega) d\Omega \right] e^{i\omega t} d\omega \\ &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} X(\Omega) \left[\int_{-\infty}^{\infty} G(\omega - \Omega) e^{i\omega t} d\omega \right] d\Omega. \end{aligned}$$

Let $\sigma = \omega - \Omega$, so $\omega = \sigma + \Omega$ and $d\omega = d\sigma$. Then

$$\begin{aligned} \int_{-\infty}^{\infty} G(\omega - \Omega) e^{i\omega t} d\omega &= \int_{-\infty}^{\infty} G(\sigma) e^{i(\sigma + \Omega)t} d\sigma \\ &= e^{i\Omega t} \int_{-\infty}^{\infty} G(\sigma) e^{i\sigma t} d\sigma \\ &= 2\pi g(t) e^{i\Omega t}, \end{aligned}$$

by inversion of $G(\omega)$. Substituting back, we obtain

$$\begin{aligned} y(t) &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} X(\Omega) [2\pi g(t) e^{i\Omega t}] d\Omega \\ &= \frac{g(t)}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{i\Omega t} d\Omega \\ &= \frac{g(t)}{2\pi} \cdot 2\pi x(t) \\ &= x(t)g(t), \end{aligned}$$

as claimed. □

A basic and important application is modulation. If

$$y(t) = x(t) \cos(\omega_0 t),$$

then, using the transform pair

$$\cos(\omega_0 t) \longleftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)],$$

the convolution in frequency yields

$$Y(\omega) = \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0),$$

i.e. the baseband spectrum $X(\omega)$ is shifted to $\pm\omega_0$ and scaled. This underlies standard amplitude modulation schemes.

In summary:

- Convolution in time corresponds to multiplication in frequency:

$$x * h \longleftrightarrow X(\omega)H(i\omega).$$

- Multiplication in time corresponds to convolution in frequency:

$$xg \longleftrightarrow \frac{1}{2\pi} (X * G)(\omega).$$

These dual relationships are central to the use of Fourier analysis in time-domain and frequency-domain system analysis.

8 The Properties of the Fourier Transform

In this chapter we collect the standard properties of the continuous-time Fourier transform and provide proofs for each. Throughout, we use the following convention:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt,$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega.$$

We write

$$f(t) \longleftrightarrow F(\omega)$$

to denote such a Fourier-transform pair.

This chapter is best read as a toolbox. Each property describes a common engineering operation in one domain and tells us the corresponding operation in the other domain. The proofs are included so that the scale factors and signs are not mysterious, but the main practical value is the left-to-right reading:

operation on $f(t)$	effect on $F(\omega)$	engineering reading
scale t	reciprocal scale of ω	narrow pulses have wide spectra
shift in t	multiply by phase	delay changes phase, not magnitude
multiply by $e^{i\omega_0 t}$	shift in ω	modulation moves spectra
differentiate in t	multiply by $i\omega$	high frequencies are emphasized
convolve in t	multiply in ω	LTI filtering is spectral weighting

When using the table, keep the transform convention above in view. Most apparent mistakes in Fourier properties are really convention mistakes: a sign in the exponential, or a missing factor 2π .

8.1 Linearity

Theorem 28 (Linearity). Let $f_1(t) \longleftrightarrow F_1(\omega)$ and $f_2(t) \longleftrightarrow F_2(\omega)$. For constants $c_1, c_2 \in \mathbb{C}$, define

$$f(t) = c_1 f_1(t) + c_2 f_2(t).$$

Then

$$f(t) \longleftrightarrow F(\omega) = c_1 F_1(\omega) + c_2 F_2(\omega).$$

Proof. By definition,

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} (c_1 f_1(t) + c_2 f_2(t)) e^{-i\omega t} dt \\ &= c_1 \int_{-\infty}^{\infty} f_1(t) e^{-i\omega t} dt + c_2 \int_{-\infty}^{\infty} f_2(t) e^{-i\omega t} dt \\ &= c_1 F_1(\omega) + c_2 F_2(\omega), \end{aligned}$$

using linearity of the integral. □

8.2 Realness

Theorem 29 (Realness). Let $f(t) \leftrightarrow F(\omega)$ with $F(\omega)$ defined as above. Then

$$f(t) \text{ real-valued for all } t \iff F^*(\omega) = F(-\omega) \text{ for all } \omega.$$

Proof. Assume first that f is real-valued. Then

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt,$$

$$F^*(\omega) = \left(\int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt \right)^* = \int_{-\infty}^{\infty} f(t)e^{i\omega t} dt,$$

since $f(t)$ is real. On the other hand,

$$F(-\omega) = \int_{-\infty}^{\infty} f(t)e^{-i(-\omega)t} dt = \int_{-\infty}^{\infty} f(t)e^{i\omega t} dt.$$

Thus $F^*(\omega) = F(-\omega)$.

Conversely, suppose $F^*(\omega) = F(-\omega)$. Then

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega.$$

Taking the complex conjugate,

$$f^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F^*(\omega)e^{-i\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(-\omega)e^{-i\omega t} d\omega.$$

Make the change of variable $\nu = -\omega$:

$$f^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\nu)e^{i\nu t} d\nu = f(t).$$

Hence $f(t)$ is real-valued. □

8.3 Even–Odd and Symmetry Properties

Let $f(t)$ be real and

$$F(\omega) = A(\omega) + iB(\omega),$$

with A, B real-valued.

Theorem 30 (Symmetry of real transforms). If $f(t)$ is real, then

$$A(\omega) = \Re F(\omega) \text{ is even,}$$

$$B(\omega) = \Im F(\omega) \text{ is odd,}$$

$$|F(\omega)| = \sqrt{A^2(\omega) + B^2(\omega)} \text{ is even,}$$

$$\arg F(\omega) \text{ can be chosen odd modulo } 2\pi \text{ where } F(\omega) \neq 0.$$

Moreover,

$$F(\omega) \text{ real and even} \iff f(t) \text{ even,}$$

$$F(\omega) \text{ purely imaginary and odd} \iff f(t) \text{ odd.}$$

Proof. Since f is real, we have $F^*(\omega) = F(-\omega)$ by the realness theorem. Write $F(\omega) = A(\omega) + iB(\omega)$, so

$$F^*(\omega) = A(\omega) - iB(\omega).$$

On the other hand,

$$F(-\omega) = A(-\omega) + iB(-\omega).$$

Thus

$$A(-\omega) + iB(-\omega) = A(\omega) - iB(\omega),$$

which implies

$$\begin{aligned} A(-\omega) &= A(\omega) \quad (\text{even}), \\ B(-\omega) &= -B(\omega) \quad (\text{odd}). \end{aligned}$$

Then

$$|F(-\omega)| = \sqrt{A^2(-\omega) + B^2(-\omega)} = \sqrt{A^2(\omega) + B^2(\omega)} = |F(\omega)|,$$

so $|F(\omega)|$ is even. At frequencies where $F(\omega) \neq 0$, the phase satisfies

$$F(-\omega) = |F(\omega)|e^{-i \arg F(\omega)},$$

so $\arg F(-\omega) = -\arg F(\omega)$ modulo 2π . At zeros of F , the phase is undefined.

Next, if f is even,

$$f(t) = f(-t),$$

then

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(-t)e^{-i\omega t} dt.$$

Change variable $\tau = -t$:

$$F(\omega) = \int_{-\infty}^{\infty} f(\tau)e^{i\omega\tau} d\tau = \int_{-\infty}^{\infty} f(\tau)\cos(\omega\tau)d\tau - i \int_{-\infty}^{\infty} f(\tau)\sin(\omega\tau)d\tau.$$

Since f is even, $f(\tau)\cos(\omega\tau)$ is even and $f(\tau)\sin(\omega\tau)$ is odd, so the latter integral vanishes. Thus $F(\omega)$ is real and even. The converse follows by applying the inverse transform and similar symmetry arguments. The odd case is similar. $\square$

8.4 Area Property

Theorem 31 (Area). *If $f(t) \leftrightarrow F(\omega)$ and the quantities below exist, then*

$$\int_{-\infty}^{\infty} f(t) dt = F(0), \tag{8.1}$$

$$\int_{-\infty}^{\infty} F(\omega) d\omega = 2\pi f(0). \tag{8.2}$$

Proof. For (8.1), set $\omega = 0$ in the definition

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt.$$

Then

$$F(0) = \int_{-\infty}^{\infty} f(t)e^0 dt = \int_{-\infty}^{\infty} f(t) dt.$$

For (8.2), evaluate the inverse transform at $t = 0$:

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega \cdot 0} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) d\omega.$$

Rearranging gives

$$\int_{-\infty}^{\infty} F(\omega) d\omega = 2\pi f(0).$$

If the inverse transform converges to a half-value at a jump, then $f(0)$ in (8.2) should be read as that recovered value. $\square$

8.5 Duality

Theorem 32 (Duality). *If*

$$f(t) \longleftrightarrow F(\omega),$$

then

$$F(t) \longleftrightarrow 2\pi f(-\omega).$$

Proof. Start from the inverse transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega.$$

Interchange the roles of t and ω . Write

$$f(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(t)e^{i\omega t} dt.$$

Now compare this with the standard transform pair

$$g(t) \longleftrightarrow G(\omega) = \int_{-\infty}^{\infty} g(t)e^{-i\omega t} dt.$$

The expression for $f(\omega)$ can be written as

$$f(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(t)e^{-i(-\omega)t} dt.$$

Thus $f(\omega)$ is (up to the constant $1/(2\pi)$) the Fourier transform of $F(t)$ evaluated at $-\omega$. More precisely, if

$$F(t) \longleftrightarrow G(\omega),$$

then

$$G(-\omega) = \int_{-\infty}^{\infty} F(t)e^{i\omega t} dt.$$

Comparing,

$$f(\omega) = \frac{1}{2\pi} G(-\omega).$$

But by the original pairing $f \leftrightarrow F$, we have

$$G(\omega) = 2\pi f(-\omega).$$

Therefore

$$F(t) \longleftrightarrow 2\pi f(-\omega).$$

$\square$

8.6 Reciprocal Scaling

Theorem 33 (Scaling). *Let $a \in \mathbb{R} \setminus \{0\}$. If $f(t) \leftrightarrow F(\omega)$, then*

$$f(at) \longleftrightarrow \frac{1}{|a|} F\left(\frac{\omega}{a}\right).$$

Proof. Compute the transform of $f(at)$:

$$\mathcal{F}\{f(at)\}(\omega) = \int_{-\infty}^{\infty} f(at) e^{-i\omega t} dt.$$

Let $z = at$. Then $t = z/a$, $dt = dz/a$. For $a > 0$, the limits remain $-\infty$ to ∞ :

$$\begin{aligned} \mathcal{F}\{f(at)\}(\omega) &= \int_{-\infty}^{\infty} f(z) e^{-i\omega z/a} \frac{dz}{a} = \frac{1}{a} \int_{-\infty}^{\infty} f(z) e^{-i(\omega/a)z} dz \\ &= \frac{1}{a} F\left(\frac{\omega}{a}\right). \end{aligned}$$

For $a < 0$, one obtains the same expression, but the change of limits contributes a minus sign that is absorbed by $|a|$. Thus the general formula is

$$f(at) \longleftrightarrow \frac{1}{|a|} F\left(\frac{\omega}{a}\right).$$

□

As a useful special case, setting $a = -1$ gives

$$f(-t) \longleftrightarrow F(-\omega).$$

The scaling law is the formal version of the time–frequency tradeoff used throughout signal processing. If $a > 1$, then $f(at)$ is compressed in time by the factor a . Its spectrum becomes $F(\omega/a)/a$, which is stretched in frequency by the same factor and reduced in height so that the total area is preserved. Compression in one domain therefore produces expansion in the other.

8.7 Time Shift

Theorem 34 (Time Shift). *If $f(t) \leftrightarrow F(\omega)$, then for any real t_0 ,*

$$f(t - t_0) \longleftrightarrow e^{-i\omega t_0} F(\omega).$$

Proof. Compute the transform of $f(t - t_0)$:

$$\mathcal{F}\{f(t - t_0)\}(\omega) = \int_{-\infty}^{\infty} f(t - t_0) e^{-i\omega t} dt.$$

Let $\tau = t - t_0$. Then $t = \tau + t_0$, $dt = d\tau$, and the limits are unchanged:

$$\begin{aligned} \mathcal{F}\{f(t - t_0)\}(\omega) &= \int_{-\infty}^{\infty} f(\tau) e^{-i\omega(\tau + t_0)} d\tau \\ &= e^{-i\omega t_0} \int_{-\infty}^{\infty} f(\tau) e^{-i\omega \tau} d\tau \\ &= e^{-i\omega t_0} F(\omega). \end{aligned}$$

□

8.8 Frequency Shift (Modulation)

Theorem 35 (Frequency Shift). If $f(t) \leftrightarrow F(\omega)$, then for any real ω_0 ,

$$f(t)e^{i\omega_0 t} \longleftrightarrow F(\omega - \omega_0).$$

Proof. Compute

$$\begin{aligned} \mathcal{F}\{f(t)e^{i\omega_0 t}\}(\omega) &= \int_{-\infty}^{\infty} f(t)e^{i\omega_0 t} e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t)e^{-i(\omega - \omega_0)t} dt \\ &= F(\omega - \omega_0). \end{aligned}$$

□

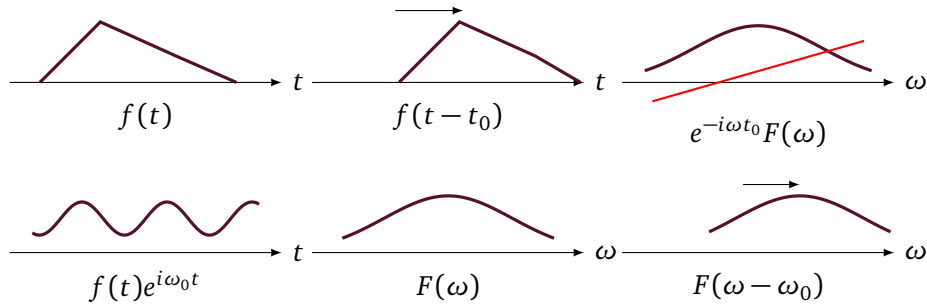


Figure 21: Time shifts leave the magnitude of the spectrum unchanged and add a linear phase factor; multiplication by a complex exponential shifts the spectrum in frequency.

There are two useful checks on these formulas. A time delay cannot change the amount of any frequency present, so it should not change $|F(\omega)|$; it only changes the phase by the straight-line factor $-\omega t_0$. By contrast, modulation by $e^{i\omega_0 t}$ really changes the frequency content, so the whole spectrum moves from its original location to a location centered at ω_0 .

8.9 Time-Domain Differentiation

Theorem 36 (Differentiation in Time). If $f(t) \leftrightarrow F(\omega)$ and $f'(t)$ exists and is absolutely integrable, then

$$\frac{d}{dt}f(t) \longleftrightarrow i\omega F(\omega).$$

More generally,

$$\frac{d^n}{dt^n}f(t) \longleftrightarrow (i\omega)^n F(\omega).$$

Proof. Consider $f'(t)$:

$$\mathcal{F}\{f'(t)\}(\omega) = \int_{-\infty}^{\infty} f'(t)e^{-i\omega t} dt.$$

Integrate by parts. Let

$$u = e^{-i\omega t}, \quad dv = f'(t)dt,$$

so

$$du = -i\omega e^{-i\omega t} dt, \quad v = f(t).$$

Then

$$\int_{-\infty}^{\infty} f'(t)e^{-i\omega t} dt = f(t)e^{-i\omega t} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(t)(-i\omega e^{-i\omega t}) dt.$$

Assuming $f(t)e^{-i\omega t} \rightarrow 0$ at $\pm\infty$, the boundary term vanishes and

$$\mathcal{F}\{f'(t)\}(\omega) = i\omega \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = i\omega F(\omega).$$

Iterating this argument yields $(i\omega)^n$ for the n th derivative. $\square$

For applications, this property is often more important as a diagnostic than as a proof technique. Multiplication by $|\omega|$ increases the weight of high frequencies, so differentiation makes sharp transitions more prominent. Repeated integration by parts gives the reverse lesson used in error estimates: every additional continuous derivative normally makes the transform decay one extra power of $1/|\omega|$.

8.10 Frequency-Domain Differentiation

Theorem 37 (Differentiation in Frequency). *If $f(t) \leftrightarrow F(\omega)$ and $tf(t)$ is integrable, then*

$$tf(t) \longleftrightarrow i \frac{d}{d\omega} F(\omega),$$

and, more generally,

$$t^n f(t) \longleftrightarrow i^n \frac{d^n}{d\omega^n} F(\omega).$$

Proof. Differentiate $F(\omega)$ with respect to ω :

$$\begin{aligned} \frac{d}{d\omega} F(\omega) &= \frac{d}{d\omega} \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(t) \frac{d}{d\omega} e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t)(-it)e^{-i\omega t} dt = -i \int_{-\infty}^{\infty} tf(t)e^{-i\omega t} dt. \end{aligned}$$

Thus

$$\int_{-\infty}^{\infty} tf(t)e^{-i\omega t} dt = i \frac{d}{d\omega} F(\omega),$$

which implies

$$tf(t) \longleftrightarrow iF'(\omega).$$

Repeated differentiation yields the general formula by induction. $\square$

Thus multiplication by t in the time domain measures how fast the spectrum is changing in frequency. This is the dual of the previous result: differentiation in one domain corresponds to multiplication by the independent variable in the other.

8.11 Time-Domain Convolution

Theorem 38 (Convolution in Time). *Define*

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau.$$

If $f(t) \leftrightarrow F(\omega)$ and $g(t) \leftrightarrow G(\omega)$, then

$$(f * g)(t) \longleftrightarrow F(\omega)G(\omega).$$

Proof. Compute the transform of $(f * g)(t)$:

$$\mathcal{F}\{f * g\}(\omega) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \right) e^{-i\omega t} dt.$$

Assume absolute integrability so that Fubini's theorem permits interchange of integrals:

$$\mathcal{F}\{f * g\}(\omega) = \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(t - \tau) e^{-i\omega t} dt \right] d\tau.$$

Set $u = t - \tau$. Then $t = u + \tau$, $dt = du$:

$$\begin{aligned} \int_{-\infty}^{\infty} g(t - \tau) e^{-i\omega t} dt &= \int_{-\infty}^{\infty} g(u) e^{-i\omega(u+\tau)} du = e^{-i\omega\tau} \int_{-\infty}^{\infty} g(u) e^{-i\omega u} du \\ &= e^{-i\omega\tau} G(\omega). \end{aligned}$$

Substitute back:

$$\begin{aligned} \mathcal{F}\{f * g\}(\omega) &= \int_{-\infty}^{\infty} f(\tau) e^{-i\omega\tau} G(\omega) d\tau = G(\omega) \int_{-\infty}^{\infty} f(\tau) e^{-i\omega\tau} d\tau \\ &= G(\omega) F(\omega). \end{aligned}$$

□

The constant factors are worth noticing. With the convention used here, convolution in time has no 2π factor:

$$f * g \longleftrightarrow FG.$$

The reciprocal factor appears in the dual statement below. Remembering where the 2π lives prevents many scaling errors in sampling and FFT chapters.

8.12 Frequency-Domain Convolution

Theorem 39 (Convolution in Frequency). *If $f(t) \leftrightarrow F(\omega)$ and $g(t) \leftrightarrow G(\omega)$, then*

$$f(t)g(t) \longleftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\theta) G(\omega - \theta) d\theta = \frac{1}{2\pi} (F * G)(\omega).$$

Proof. Start from

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\theta) e^{i\theta t} d\theta, \quad g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\nu) e^{i\nu t} d\nu.$$

Then

$$\begin{aligned} f(t)g(t) &= \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\theta) e^{i\theta t} d\theta \right] \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} G(\nu) e^{i\nu t} d\nu \right] \\ &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\theta) G(\nu) e^{i(\theta+\nu)t} d\theta d\nu. \end{aligned}$$

Now take the Fourier transform:

$$\begin{aligned} \mathcal{F}\{fg\}(\omega) &= \int_{-\infty}^{\infty} f(t)g(t) e^{-i\omega t} dt \\ &= \frac{1}{(2\pi)^2} \int \int F(\theta) G(\nu) \left[\int_{-\infty}^{\infty} e^{i(\theta+\nu-\omega)t} dt \right] d\theta d\nu. \end{aligned}$$

The inner integral is

$$\int_{-\infty}^{\infty} e^{i(\theta+\nu-\omega)t} dt = 2\pi \delta(\theta + \nu - \omega).$$

Hence

$$\begin{aligned} \mathcal{F}\{fg\}(\omega) &= \frac{1}{(2\pi)^2} \int \int F(\theta) G(\nu) 2\pi \delta(\theta + \nu - \omega) d\theta d\nu \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\theta) G(\omega - \theta) d\theta, \end{aligned}$$

using the sifting property of the delta. This is the desired result. $\square$

8.13 Dirac Delta: Sampling and Convolution

The Dirac delta $\delta(t)$ is defined by the properties

$$\delta(t) = 0 \text{ for } t \neq 0, \quad \int_{-\infty}^{\infty} \delta(t) dt = 1,$$

and the sampling property

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0).$$

Theorem 40 (Sampling). *For any suitable f and G ,*

$$\begin{aligned} f(t) \delta(t - t_0) &= f(t_0) \delta(t - t_0), \\ G(\omega) \delta(\omega - \omega_0) &= G(\omega_0) \delta(\omega - \omega_0). \end{aligned}$$

Proof. Consider

$$f(t) \delta(t - t_0).$$

Viewed as a distribution, for any test function $\varphi(t)$,

$$\begin{aligned} \int_{-\infty}^{\infty} f(t) \delta(t - t_0) \varphi(t) dt &= \int_{-\infty}^{\infty} \delta(t - t_0) (f(t) \varphi(t)) dt = f(t_0) \varphi(t_0) \\ &= \int_{-\infty}^{\infty} f(t_0) \delta(t - t_0) \varphi(t) dt. \end{aligned}$$

Hence $f(t) \delta(t - t_0) = f(t_0) \delta(t - t_0)$ as distributions. The frequency-domain statement is identical in form, with t, t_0 replaced by ω, ω_0 . $\square$

Theorem 41 (Convolution with a Delta).

$$f(t) * \delta(t - t_0) = f(t - t_0),$$

$$G(\omega) * \delta(\omega - \omega_0) = G(\omega - \omega_0).$$

Proof. By definition,

$$(f * \delta(\cdot - t_0))(t) = \int_{-\infty}^{\infty} f(\tau) \delta(t - \tau - t_0) d\tau.$$

Use the sampling property in τ with $g(\tau) = f(\tau)$ and shift $t - \tau - t_0 = 0$ giving $\tau = t - t_0$:

$$(f * \delta(\cdot - t_0))(t) = f(t - t_0).$$

The frequency-domain version is analogous. □

8.14 Integration Property

Theorem 42 (Integration). Let $f(t) \leftrightarrow F(\omega)$ and define

$$h(t) = \int_{-\infty}^t f(\tau) d\tau.$$

Then

$$h(t) \longleftrightarrow H(\omega) = \frac{1}{i\omega} F(\omega) + \pi F(0) \delta(\omega),$$

in the sense of distributions.

Proof. Write

$$h(t) = \int_{-\infty}^t f(\tau) d\tau = (f * u)(t),$$

where $u(t)$ is the unit step. Using

$$u(t) \longleftrightarrow \text{PV}\left(\frac{1}{i\omega}\right) + \pi\delta(\omega),$$

and the time-convolution theorem, we get

$$\begin{aligned} H(\omega) &= F(\omega) \left[\text{PV}\left(\frac{1}{i\omega}\right) + \pi\delta(\omega) \right] \\ &= \text{PV}\left(\frac{F(\omega)}{i\omega}\right) + \pi F(0)\delta(\omega). \end{aligned}$$

When F is regular at $\omega = 0$, this is commonly written as

$$H(\omega) = \frac{1}{i\omega} F(\omega) + \pi F(0)\delta(\omega),$$

with $1/\omega$ understood in the principal-value sense. □

9 The Sampling Theorems

In this chapter we study sampling in the time and frequency domains. We adopt the same Fourier-transform convention as before:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt, \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega.$$

We write

$$x(t) \longleftrightarrow X(\omega)$$

to denote a Fourier-transform pair.

Sampling refers to evaluating a continuous function at discrete points. For a continuous-time signal $x(t)$, sampling with period T_s gives the sequence

$$\{x(nT_s)\}_{n \in \mathbb{Z}}.$$

In the frequency domain, sampling a spectrum $X(\omega)$ at $\omega = n\omega_0$ produces

$$\{X(n\omega_0)\}_{n \in \mathbb{Z}}.$$

We analyze these operations using impulse (Dirac-delta) models, then generalize to non-impulsive sampling pulses.

The central idea is reciprocal: making one domain discrete makes the other domain periodic. If the samples in time are close together, the spectral copies are far apart; if samples in frequency are close together, the repeated time copies are far apart. Aliasing is simply the failure of those repeated copies to fit without overlap.

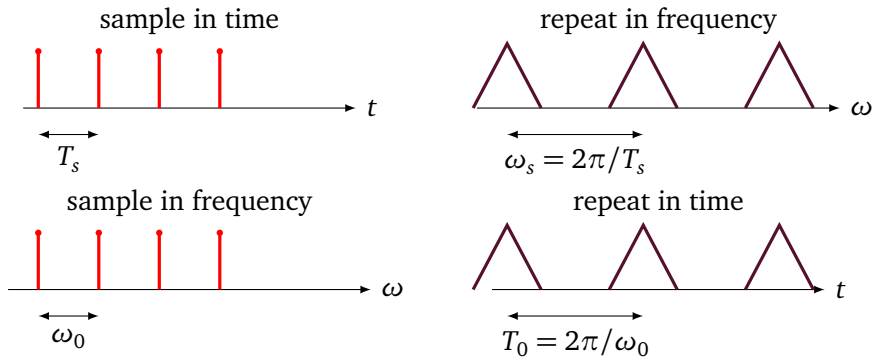


Figure 22: Sampling and repetition are dual operations. Smaller spacing in one domain gives larger spacing between copies in the other domain.

9.1 Time-Domain Impulse Sampling

Let $x(t)$ be a continuous-time signal. The idealized impulse sampler multiplies $x(t)$ by the Dirac-comb

$$\Delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s),$$

where T_s is the sampling interval and

$$\omega_s = \frac{2\pi}{T_s}$$

is the sampling angular frequency.

The sampled signal is

$$x_s(t) = x(t) \Delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s).$$

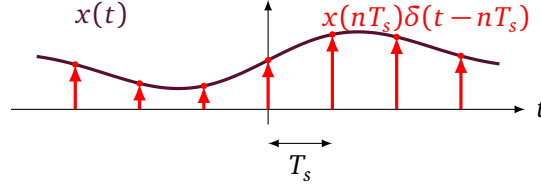


Figure 23: Ideal impulse sampling keeps the sample values as impulse weights located at integer multiples of T_s .

We now derive the spectrum $X_s(\omega)$ of $x_s(t)$ in two ways.

Method 1: Directly from the definition

Using the sampling property

$$\delta(t - nT_s) \longleftrightarrow e^{-i\omega nT_s},$$

we have

$$\begin{aligned} X_s(\omega) &= \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) \right] e^{-i\omega t} dt \\ &= \sum_{n=-\infty}^{\infty} x(nT_s) \int_{-\infty}^{\infty} \delta(t - nT_s) e^{-i\omega t} dt \\ &= \sum_{n=-\infty}^{\infty} x(nT_s) e^{-i\omega nT_s}. \end{aligned}$$

This is a continuous function of ω , but it is built from the discrete sample values $x(nT_s)$. The expression is periodic in ω with period $\omega_s = 2\pi/T_s$, because replacing ω by $\omega + \omega_s$ adds the factor $e^{-i2\pi n} = 1$ to each term.

Method 2: Using convolution in the frequency domain

We first compute the Fourier transform of $\Delta_{T_s}(t)$.

Lemma 3 (Spectrum of the Dirac comb). *With the above transform convention,*

$$\Delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow \Delta_{\omega_s}(\omega) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s).$$

Proof. Consider the periodic train $\Delta_{T_s}(t)$ with period T_s . Its Fourier series representation is

$$\Delta_{T_s}(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_s t}, \quad \omega_s = \frac{2\pi}{T_s},$$

with coefficients

$$c_k = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \Delta_{T_s}(t) e^{-ik\omega_s t} dt = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-ik\omega_s t} dt = \frac{1}{T_s}.$$

The symmetric interval is used only to include exactly one impulse of the comb. Hence

$$\Delta_{T_s}(t) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{ik\omega_s t}.$$

Taking the Fourier transform term-wise,

$$\mathcal{F}\{e^{ik\omega_s t}\}(\omega) = 2\pi \delta(\omega - k\omega_s),$$

so

$$\Delta_{\omega_s}(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - k\omega_s) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s).$$

□

Now

$$x_s(t) = x(t)\Delta_{T_s}(t) \iff X_s(\omega) = \frac{1}{2\pi}(X * \Delta_{\omega_s})(\omega).$$

Thus

$$\begin{aligned} X_s(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) \Delta_{\omega_s}(\omega - \Omega) d\Omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - \Omega - k\omega_s) d\Omega \\ &= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} X(\Omega) \delta(\omega - \Omega - k\omega_s) d\Omega \\ &= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s). \end{aligned}$$

We obtain the fundamental replication formula:

$$\boxed{X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).} \quad (9.1)$$

Thus time-domain sampling replicates the original spectrum periodically with period ω_s , scaled by $1/T_s$.

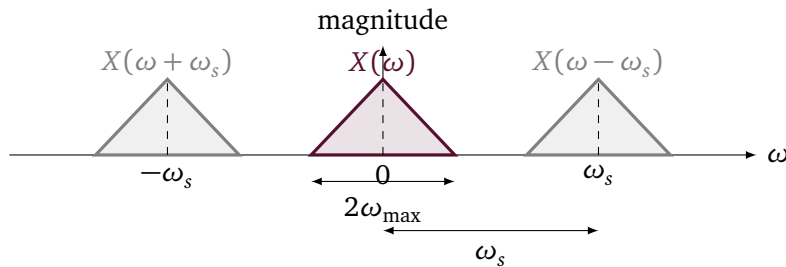


Figure 24: Impulse sampling creates shifted copies of the spectrum. If $\omega_s > 2\omega_{\max}$, adjacent copies do not overlap and ideal low-pass reconstruction can isolate the center copy.

9.2 The Time-Domain Sampling Theorem

Assume that $x(t)$ is strictly band-limited.

Definition 30 (Strict band limitation). We say that $x(t)$ is strictly band-limited to $\omega_{\max} > 0$ if

$$X(\omega) = 0 \quad \text{for } |\omega| > \omega_{\max}.$$

From (9.1), $X_s(\omega)$ is obtained by shifting copies of $X(\omega)$ by integer multiples of ω_s . Thus the theorem below is a fitting condition: the shifted copies must fit side by side on the frequency axis. If they overlap, the overlapped portion is a sum of several unknown pieces of X , and an ideal recovery filter cannot know how to separate them.

Theorem 43 (Time-domain sampling theorem). Let $x(t) \leftrightarrow X(\omega)$ be strictly band-limited to $|\omega| \leq \omega_{\max}$. Let the sampling frequency be $\omega_s = 2\pi/T_s$. Then exact reconstruction from samples $\{x(nT_s)\}_{n \in \mathbb{Z}}$ is guaranteed whenever

$$\omega_s > 2\omega_{\max}.$$

If $\omega_s < 2\omega_{\max}$, exact recovery is impossible in general. The equality case $\omega_s = 2\omega_{\max}$ is the critical edge case where replicas just touch and must be handled separately. Equivalently, in hertz,

$$f_s > 2f_{\max}, \quad f_s = \frac{\omega_s}{2\pi}, \quad f_{\max} = \frac{\omega_{\max}}{2\pi}.$$

The critical sampling frequency

$$\omega_N = 2\omega_{\max}, \quad f_N = 2f_{\max}$$

is called the Nyquist angular (resp. ordinary) sampling frequency (Nyquist rate).

Proof. From (9.1),

$$X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).$$

Each term $X(\omega - k\omega_s)$ is supported in $|\omega - k\omega_s| \leq \omega_{\max}$, i.e.

$$k\omega_s - \omega_{\max} \leq \omega \leq k\omega_s + \omega_{\max}.$$

(No-overlap condition). For $k = 0$ and $k = 1$, the supports

$$[-\omega_{\max}, \omega_{\max}] \quad \text{and} \quad [\omega_s - \omega_{\max}, \omega_s + \omega_{\max}]$$

do not overlap if and only if

$$\omega_s - \omega_{\max} > \omega_{\max} \quad \Longleftrightarrow \quad \omega_s > 2\omega_{\max}.$$

Similarly for $k = -1$, compare

$$[-\omega_s - \omega_{\max}, -\omega_s + \omega_{\max}] \quad \text{and} \quad [-\omega_{\max}, \omega_{\max}].$$

Under the same condition $\omega_s > 2\omega_{\max}$, these intervals do not overlap either. Hence all shifted copies of X are nonoverlapping.

(Reconstruction). Define the ideal low-pass recovery filter with frequency response

$$G(\omega) = \begin{cases} T_s, & |\omega| \leq \omega_{\max}, \\ 0, & \text{otherwise.} \end{cases}$$

Let $Y(\omega) = G(\omega)X_s(\omega)$ be the spectrum at the filter output. For $|\omega| \leq \omega_{\max}$, only the $k = 0$ term survives in the sum:

$$X_s(\omega) = \frac{1}{T_s}X(\omega), \quad |\omega| \leq \omega_{\max},$$

since all $X(\omega - k\omega_s)$ with $k \neq 0$ are zero there by non-overlap. Hence

$$Y(\omega) = G(\omega)X_s(\omega) = T_s \cdot \frac{1}{T_s}X(\omega) = X(\omega) \quad (|\omega| \leq \omega_{\max}),$$

and $Y(\omega) = 0$ outside this band. Thus $Y(\omega) = X(\omega)$ for all ω , so

$$y(t) = x(t).$$

Therefore exact reconstruction is possible when $\omega_s > 2\omega_{\max}$.

(*Necessity for undersampling*). If $\omega_s < 2\omega_{\max}$, the supports of at least two distinct shifts $X(\omega - k\omega_s)$ overlap on intervals of nonzero width. Then in the overlap region $X_s(\omega)$ is a sum of two or more unknown contributions of X , and no linear frequency-domain filter can uniquely separate these contributions. Hence exact recovery is impossible in general under undersampling.

At $\omega_s = 2\omega_{\max}$, replicas meet at band edges (critical Nyquist case).

□

The condition $f_s > 2f_{\max}$ is the strict Nyquist sampling criterion. It is often summarized as “at least two samples per period of the highest frequency component,” but the exact equality case is delicate because the shifted spectra touch at the band edge. For engineering use, the safest reading is: sampling above the Nyquist rate gives room to isolate the center copy of the spectrum; sampling below it folds neighboring copies into the same frequency interval. The word *folds* is literal in plots: spectral content beyond the allowed band appears back inside the displayed band as aliasing.

9.3 Time-Domain Reconstruction Formula (Interpolation)

Under the assumptions of Theorem 43, choose an ideal reconstruction filter that passes the baseband copy and rejects all shifted copies:

$$G(\omega) = T_s \text{Rect}\left(\frac{\omega}{\omega_s}\right).$$

Its passband is $|\omega| < \omega_s/2$. Since $X(\omega) = 0$ for $|\omega| > \omega_{\max}$ and $\omega_s > 2\omega_{\max}$, this passband contains the whole original spectrum and lies before the first replica. The impulse response is

$$\begin{aligned} g(t) &= \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} T_s e^{i\omega t} d\omega = \frac{T_s}{2\pi} \cdot \frac{2 \sin((\omega_s/2)t)}{t} \\ &= \frac{T_s}{\pi} \frac{\sin((\omega_s/2)t)}{t}. \end{aligned}$$

Because $\omega_s = 2\pi/T_s$, this becomes

$$g(t) = \frac{T_s}{\pi} \frac{\sin((\pi/T_s)t)}{t} = \text{sinc}\left(\frac{\omega_s}{2}t\right),$$

using the sinc convention from Notation Conventions.

Because the reconstruction system is linear and shift-invariant, feeding $\delta(t - nT_s)$ produces $g(t - nT_s)$ at the output. Since the sampled signal can be written as

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s),$$

the output is

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) g(t - nT_s).$$

For the ideal brick-wall reconstruction filter with cutoff $\omega_s/2$, this yields the interpolation formula

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{\omega_s}{2}(t - nT_s)\right).$$

Each sample $x(nT_s)$ multiplies a shifted sinc pulse that vanishes at all other sampling instants.

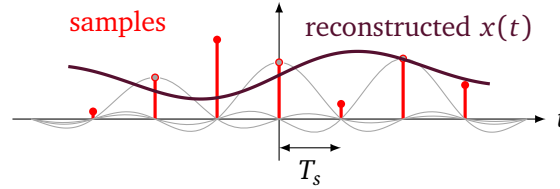


Figure 25: In ideal band-limited reconstruction, each sample scales a shifted sinc pulse; the sinc pulses vanish at all other sampling instants.

9.4 Sampling with General Periodic Pulses

So far we used the Dirac comb as the sampling waveform. More generally, let $p_s(t)$ be any T_s -periodic sampling waveform with Fourier series

$$p_s(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_s t}, \quad \omega_s = \frac{2\pi}{T_s}.$$

Sampling $x(t)$ with $p_s(t)$ corresponds to the product

$$x_s(t) = x(t)p_s(t).$$

Proposition 13 (Spectrum under multiplication by a periodic pulse). *Let $x(t) \leftrightarrow X(\omega)$ and $p_s(t)$ be as above. Then*

$$x_s(t) = x(t)p_s(t) \longleftrightarrow X_s(\omega) = \sum_{n=-\infty}^{\infty} c_n X(\omega - n\omega_s).$$

Proof. Using linearity and the frequency-shift property:

$$x_s(t) = x(t) \sum_n c_n e^{in\omega_s t} = \sum_n c_n [x(t)e^{in\omega_s t}].$$

Each term satisfies

$$x(t)e^{in\omega_s t} \longleftrightarrow X(\omega - n\omega_s).$$

Hence

$$X_s(\omega) = \sum_{n=-\infty}^{\infty} c_n X(\omega - n\omega_s).$$

□

Thus the effect of using a non-impulsive sampling pulse is to replicate the spectrum as before, but scale each replica by the corresponding Fourier-series coefficient c_n .

For a Dirac comb with period T_s , the coefficients satisfy $c_n = 1/T_s$, and we recover (9.1).

Under the Nyquist condition (and strict band limitation), one can still select an ideal low-pass filter to pass only a single, scaled copy of $X(\omega)$, so the sampling theorem and Nyquist criterion remain valid. In practice, sampling pulses are chosen narrow relative to T_s (e.g. narrow rectangular pulses) so that $x(t)$ does not change significantly during each pulse.

The impulse sampler is therefore an ideal reference model. Real sampling pulses change the weights of the replicas through the coefficients c_n , but they do not change the basic geometry of aliasing: copies still appear every ω_s , and recovery still depends on keeping the desired copy separated from its neighbors.

9.5 Sampling in the Frequency Domain

We now consider the dual situation: sampling in the frequency domain.

Let $x(t) \leftrightarrow X(\omega)$ be given, and sample $X(\omega)$ at points $\omega = n\omega_0$ using an impulse train in frequency:

$$\Delta_{\omega_0}(\omega) = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0).$$

Define

$$X_s(\omega) = X(\omega) \Delta_{\omega_0}(\omega) = \omega_0 \sum_{n=-\infty}^{\infty} X(n\omega_0) \delta(\omega - n\omega_0).$$

This is a discrete-line spectrum in frequency.

To find the corresponding time-domain signal $x_s(t)$, invert the transform.

Method 1: Convolution in time

By duality, $\Delta_{\omega_0}(\omega)$ is the transform of a time-domain impulse train with period

$$T_0 = \frac{2\pi}{\omega_0}.$$

In particular,

$$\Delta_{\omega_0}(\omega) \longleftrightarrow \sum_{k=-\infty}^{\infty} \delta(t - kT_0).$$

Multiplication in frequency corresponds to convolution in time:

$$\begin{aligned} x_s(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_s(\omega) e^{i\omega t} d\omega \\ &= (x * \Delta_{T_0})(t) = \int_{-\infty}^{\infty} x(\tau) \sum_{k=-\infty}^{\infty} \delta(t - \tau - kT_0) d\tau \\ &= \sum_{k=-\infty}^{\infty} x(t - kT_0). \end{aligned}$$

Hence frequency-domain impulse sampling produces a periodic repetition of $x(t)$.

$$x_s(t) = \sum_{k=-\infty}^{\infty} x(t - kT_0), \quad T_0 = \frac{2\pi}{\omega_0}.$$

Aliasing and the frequency-domain sampling theorem

Assume that $x(t)$ is strictly time-limited.

Definition 31 (Strict time limitation). We say that $x(t)$ is strictly time-limited to $|t| \leq T_{\max}$ if

$$x(t) = 0 \quad \text{for } |t| > T_{\max}.$$

Then each shifted copy $x(t - kT_0)$ is supported in

$$kT_0 - T_{\max} \leq t \leq kT_0 + T_{\max}.$$

These supports do not overlap if and only if

$$T_0 > 2T_{\max}.$$

Theorem 44 (Frequency-domain sampling theorem). Let $x(t) \leftrightarrow X(\omega)$ be strictly time-limited to $|t| \leq T_{\max}$. Sample $X(\omega)$ at $\omega = n\omega_0$ using an impulse train $\Delta_{\omega_0}(\omega)$ as above. Then the inverse transform $x_s(t)$ is a periodic repetition of $x(t)$ with period $T_0 = 2\pi/\omega_0$. It is guaranteed to be free of time-domain aliasing whenever

$$T_0 > 2T_{\max}, \quad \text{equivalently} \quad \omega_0 < \frac{\pi}{T_{\max}}.$$

If $T_0 < 2T_{\max}$, aliasing is unavoidable in general; $T_0 = 2T_{\max}$ is the critical touching case.

Proof. We have shown that

$$x_s(t) = \sum_{k=-\infty}^{\infty} x(t - kT_0).$$

Each copy is supported in $[kT_0 - T_{\max}, kT_0 + T_{\max}]$. For the copies not to overlap for any k , we require that

$$kT_0 + T_{\max} < (k+1)T_0 - T_{\max} \quad \text{for all } k,$$

which simplifies to

$$2T_{\max} < T_0.$$

This is both necessary and sufficient for non-overlap; if the condition fails, the shifted supports intersect and aliasing occurs in the sense that $x_s(t)$ is a sum of overlapping copies of $x(t)$. $\square$

9.6 Poisson Summation Formula

The previous derivation leads directly to a useful identity relating sums in time to sums in frequency.

Theorem 45 (Poisson summation). Let $x(t) \leftrightarrow X(\omega)$ be such that both are integrable and the series below converge. Let $T_0 > 0$ and $\omega_0 = 2\pi/T_0$. Then

$$\sum_{k=-\infty}^{\infty} x(t - kT_0) = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} X(n\omega_0) e^{in\omega_0 t}.$$

Proof. Define

$$x_s(t) = \sum_{k=-\infty}^{\infty} x(t - kT_0).$$

We have already shown that $x_s(t)$ is periodic with period T_0 and is the inverse Fourier transform of the frequency-sampled spectrum

$$X_s(\omega) = \omega_0 \sum_{n=-\infty}^{\infty} X(n\omega_0) \delta(\omega - n\omega_0).$$

On the other hand, any T_0 -periodic function admits a Fourier series

$$x_s(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}.$$

Compute c_n from the inverse transform of $X_s(\omega)$:

$$\begin{aligned} x_s(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_s(\omega) e^{i\omega t} d\omega \\ &= \frac{1}{2\pi} \omega_0 \sum_{n=-\infty}^{\infty} X(n\omega_0) \int_{-\infty}^{\infty} \delta(\omega - n\omega_0) e^{i\omega t} d\omega \\ &= \frac{\omega_0}{2\pi} \sum_{n=-\infty}^{\infty} X(n\omega_0) e^{in\omega_0 t}. \end{aligned}$$

Hence

$$c_n = \frac{\omega_0}{2\pi} X(n\omega_0) = \frac{1}{T_0} X(n\omega_0).$$

Thus

$$x_s(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 t} = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} X(n\omega_0) e^{in\omega_0 t},$$

which yields the Poisson summation formula. $\square$

As a direct consequence, if $x(t)$ is strictly time-limited to $|t| \leq T_0/2$, then $x_s(t)$ in Theorem 45 is obtained by tiling $\mathbb{R}$ with non-overlapping copies of $x(t)$.

The formula also admits an inverse corollary:

Corollary 7. Let $x_p(t)$ be T_0 -periodic with Fourier series

$$x_p(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}.$$

If $x(t)$ is defined by restricting $x_p(t)$ to one period and setting it to zero outside $(-T_0/2, T_0/2)$, then the Fourier transform $X(\omega)$ of $x(t)$ satisfies

$$X(n\omega_0) = T_0 C_n,$$

so the Fourier transform of one isolated period agrees with the Fourier-series coefficients at the harmonic frequencies after the scale factor T_0 . Values of $X(\omega)$ away from the grid are obtained from the defining integral for the isolated pulse, not from the samples alone.

9.7 Discrete vs. Periodic in Time and Frequency

The transform properties can be summarized as follows.

Theorem 46.

1. $x(t)$ is discrete in time (i.e. a sum of impulses spaced by a fixed period) if and only if $X(\omega)$ is periodic in frequency.
2. $x(t)$ is periodic in time if and only if $X(\omega)$ is discrete in frequency (i.e. a sum of impulses spaced by a fixed frequency).

Proof. (1) If $x(t)$ is discrete with period T_s :

$$x(t) = \sum_{n=-\infty}^{\infty} a_n \delta(t - nT_s),$$

then

$$X(\omega) = \sum_{n=-\infty}^{\infty} a_n e^{-i\omega nT_s}.$$

This is a Fourier series in ω with fundamental frequency $2\pi/T_s$, hence periodic in ω .

Conversely, if $X(\omega)$ is periodic with period ω_s , it admits a Fourier series

$$X(\omega) = \sum_{k=-\infty}^{\infty} b_k e^{-ik\omega T_0}, \quad T_0 = \frac{2\pi}{\omega_s}.$$

Inverting term-wise shows that $x(t)$ is a discrete train of impulses spaced by T_0 .

(2) If $x(t)$ is periodic with period T_0 , it has a Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}.$$

Using linearity and the transform of exponentials gives

$$X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} C_n \delta(\omega - n\omega_0),$$

which is discrete in frequency.

Conversely, if $X(\omega)$ is discrete, then inverting yields a periodic sum of exponentials in time, so $x(t)$ is periodic. $\square$

An immediate consequence is:

Corollary 8. *A function that is discrete in both time and frequency must be periodic in both time and frequency.*

This structure is characteristic of the discrete Fourier transform (DFT), which is simultaneously discrete and periodic in both domains.

10 The Discrete Fourier Transform

10.1 Introduction

The discrete Fourier transform (DFT) is the standard tool for frequency–domain analysis of discrete signals. When implemented directly, evaluation of the DFT requires on the order of N^2 complex multiplications. The fast Fourier transform (FFT) is an algorithmic implementation that reduces this to $\mathcal{O}(N \log N)$ when N has suitable factorizations (in particular, when N is a power of 2), which explains its central role in modern signal processing and numerical computation.

This chapter develops the DFT as a discrete analysis–synthesis system based on the orthogonality of discrete complex exponentials, and establishes its basic properties and symmetry relations. In later chapters the connections between the DFT, Fourier series, and the continuous Fourier transform are examined in more detail, and the structure of FFT algorithms is outlined.

The DFT should be read as the finite-dimensional member of the same family:

Fourier series: periodic in time, discrete in frequency,

Fourier transform: continuous in time, continuous in frequency,

DFT: discrete and periodic in both indices.

The last phrase is the key. A length- N DFT does not know about an infinite line of samples unless we add that interpretation ourselves. Algebraically, the DFT treats the N samples as one period of a periodic discrete sequence.

10.2 The Discrete Complex Exponentials

In continuous Fourier analysis, two families of complex exponentials are used:

$$\mathcal{S}_{\text{FS}} = \{e^{in\omega_0 t} \mid n \in \mathbb{Z}\}, \quad \mathcal{S}_{\text{FT}} = \{e^{i\omega t} \mid \omega \in \mathbb{R}\},$$

which serve as orthogonal bases for Fourier series and Fourier transforms, respectively. Their orthogonality relations are

$$\int_{-T/2}^{T/2} e^{in\omega_0 t} e^{-im\omega_0 t} dt = T \delta_{nm},$$

$$\int_{-\infty}^{\infty} e^{i\omega t} e^{-i\omega' t} dt = 2\pi \delta(\omega - \omega').$$

In the discrete setting we introduce a third family, the *discrete complex exponentials of order N* :

$$\mathcal{S}_N = \{e^{i2\pi mk/N} \mid m = 0, 1, \dots, N-1\},$$

where k is an integer index. These exponentials will serve as a discrete orthogonal basis.

10.2.1 Modulo- N arithmetic and circularity

Two integers a, b are said to be equal modulo N if they differ by an integer multiple of N :

$$a \equiv b \pmod{N} \iff a - b = \ell N \text{ for some } \ell \in \mathbb{Z}.$$

We sometimes write $[a]_N$ for the unique representative in $\{0, 1, \dots, N-1\}$ that is congruent to a modulo N .

The discrete exponentials are N -periodic in their integer argument:

$$e^{i2\pi(m+N)/N} = e^{i2\pi m/N} e^{i2\pi} = e^{i2\pi m/N},$$

and more generally

$$e^{i2\pi m/N} = e^{i2\pi [m]_N/N}.$$

This will be referred to as the *circularity property*.

This circularity is not just notation. It is why the DFT naturally produces circular shifts, circular convolution, and periodic spectra. If an index leaves the written range $0, \dots, N-1$, it re-enters at the other end, exactly as a point moving around a circle.

10.2.2 Notation for the primitive root of unity

It is convenient to introduce

$$W = e^{-i2\pi/N}.$$

Then

$$e^{i2\pi mk/N} = W^{-mk}.$$

Powers of W are N th roots of unity with

$$W^N = e^{-i2\pi} = 1.$$

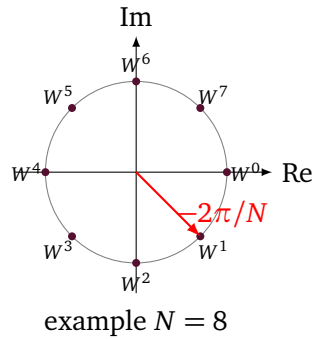


Figure 26: The DFT samples powers of a primitive root of unity. Because $W^N = 1$, the powers wrap around the unit circle and discrete frequency indices are circular modulo N .

10.2.3 Inner product and orthogonality

For two complex N -vectors

$$a = (a_0, a_1, \dots, a_{N-1}), \quad b = (b_0, b_1, \dots, b_{N-1}),$$

define their inner product

$$(a, b) = \sum_{k=0}^{N-1} a_k b_k^*.$$

They are said to be *orthogonal* if $(a, b) = 0$.

We now state the orthogonality property of the discrete exponentials.

Theorem 47 (Orthogonality of the discrete complex exponentials). Let $q, r \in \{0, 1, \dots, N-1\}$. Then

$$\sum_{k=0}^{N-1} W^{-(q-r)k} = \begin{cases} N, & q = r, \\ 0, & q \neq r. \end{cases} \quad (10.1)$$

Equivalently, the vectors

$$(1, W^{-q}, W^{-2q}, \dots, W^{-(N-1)q}), \quad q = 0, 1, \dots, N-1,$$

are mutually orthogonal, with norm $\sqrt{N}$.

Proof. Let $p = q - r$, so that p is an integer with $-(N-1) \leq p \leq N-1$. Define

$$S = \sum_{k=0}^{N-1} W^{-pk}.$$

If $p = 0$, then $W^0 = 1$ and

$$S = \sum_{k=0}^{N-1} 1 = N.$$

Assume $p \neq 0$. Then $W^{-p} \neq 1$, and the sum is a geometric series:

$$S = \sum_{k=0}^{N-1} (W^{-p})^k = \frac{1 - (W^{-p})^N}{1 - W^{-p}}.$$

Since $W^N = 1$, we have $(W^{-p})^N = W^{-pN} = (W^N)^{-p} = 1$. Thus the numerator $1 - (W^{-p})^N$ is zero, while the denominator is nonzero. Hence $S = 0$, which is (10.1). $\square$

10.3 The Discrete Fourier Transform

Let f_k , $k = 0, 1, \dots, N-1$, be a discrete sequence (possibly complex-valued). We wish to expand f_k as a linear combination of the N discrete exponentials

$$1, W^{-k}, W^{-2k}, \dots, W^{-(N-1)k}.$$

Following the continuous case, we postulate a *synthesis representation*

$$f_k = \frac{1}{N} \sum_{n=0}^{N-1} F_n W^{-nk}, \quad 0 \leq k \leq N-1, \quad (10.2)$$

for some complex coefficients F_n that remain to be determined. These coefficients will be the DFT of f_k .

To solve for F_n , multiply (10.2) by W^{mk} and sum over $k = 0, 1, \dots, N-1$:

$$\begin{aligned} \sum_{k=0}^{N-1} f_k W^{mk} &= \frac{1}{N} \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} F_n W^{-nk} W^{mk} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} F_n \sum_{k=0}^{N-1} W^{-(n-m)k}. \end{aligned} \quad (10.3)$$

By Theorem 47,

$$\sum_{k=0}^{N-1} W^{-(n-m)k} = \begin{cases} N, & n = m, \\ 0, & n \neq m, \end{cases}$$

so only the term with $n = m$ survives in (10.3) and we obtain

$$\sum_{k=0}^{N-1} f_k W^{mk} = F_m.$$

Renaming m as n yields the *analysis equation*

$$F_n = \sum_{k=0}^{N-1} f_k W^{nk}, \quad 0 \leq n \leq N-1.$$

We summarize the result.

Theorem 48 (Discrete Fourier transform). *Let f_k , $k = 0, 1, \dots, N-1$, be a complex sequence. Define the numbers F_n , $n = 0, 1, \dots, N-1$, by*

$$F_n = \sum_{k=0}^{N-1} f_k W^{nk}, \quad W = e^{-i2\pi/N}. \quad (10.4)$$

Then f_k can be reconstructed from F_n via

$$f_k = \frac{1}{N} \sum_{n=0}^{N-1} F_n W^{-nk}, \quad (10.5)$$

for $k = 0, 1, \dots, N-1$. Equations (10.4) and (10.5) form an analysis–synthesis pair known as the discrete Fourier transform (DFT) and inverse DFT (IDFT).

There are three scale facts to keep straight:

- The analysis formula has no $1/N$ factor.
- The synthesis formula has the $1/N$ factor.
- With this convention, a constant sequence $f_k = A$ has $F_0 = NA$ and all other bins zero.

This convention matches the FFT chapters that follow.

It is convenient to use the shorthand notation

$$\{f_k\} \longleftrightarrow \{F_n\}$$

to indicate that f_k and F_n are related by the DFT pair (10.4)–(10.5).

10.3.1 The DFT as a pair of inverse transforms

Corollary 9. *The transformations (10.4) and (10.5) are mutual inverses: if F_n is defined from f_k by (10.4), then the synthesis formula (10.5) reproduces f_k exactly.*

Proof. Substitute (10.4) into (10.5):

$$\begin{aligned} f_k &= \frac{1}{N} \sum_{n=0}^{N-1} \left(\sum_{r=0}^{N-1} f_r W^{nr} \right) W^{-nk} \\ &= \frac{1}{N} \sum_{r=0}^{N-1} f_r \sum_{n=0}^{N-1} W^{n(r-k)}. \end{aligned}$$

By orthogonality,

$$\sum_{n=0}^{N-1} W^{n(r-k)} = \begin{cases} N, & r = k, \\ 0, & r \neq k, \end{cases}$$

so only $r = k$ survives and we obtain $f_k = f_k$. □

10.3.2 Periodicity of the DFT indices

Because $W^N = 1$, the exponentials are N -periodic in both indices:

$$W^{(n+N)k} = W^{nk}, \quad W^{n(k+N)} = W^{nk}.$$

Consequently, the DFT coefficients and the reconstructed sequence are both periodic in their indices.

Corollary 10 (Periodicity). *Let $\{f_k\}$ and $\{F_n\}$ be related by the DFT pair (10.4)–(10.5). Extend f_k and F_n to all integer indices using these formulas. Then*

$$F_{n+N} = F_n, \quad f_{k+N} = f_k, \quad \forall n, k \in \mathbb{Z}.$$

Proof. For F_n :

$$F_{n+N} = \sum_{k=0}^{N-1} f_k W^{(n+N)k} = \sum_{k=0}^{N-1} f_k W^{nk} (W^N)^k = \sum_{k=0}^{N-1} f_k W^{nk} = F_n.$$

Similarly, in (10.5),

$$f_{k+N} = \frac{1}{N} \sum_{n=0}^{N-1} F_n W^{-n(k+N)} = \frac{1}{N} \sum_{n=0}^{N-1} F_n W^{-nk} (W^{-N})^n = f_k.$$

□

Thus both the input sequence and its DFT may be regarded as periodic sequences with period N , and the indices k and n are often interpreted modulo N .

10.3.3 Example: a discrete ramp

Consider the discrete sequence

$$f_k = \begin{cases} k, & k = 0, 1, 2, 3, \\ 0, & \text{otherwise,} \end{cases} \quad (N = 4).$$

Within one period $0 \leq k \leq 3$ we have the vector

$$f = (f_0, f_1, f_2, f_3) = (0, 1, 2, 3).$$

For $N = 4$ we have

$$W = e^{-i2\pi/4} = e^{-i\pi/2} = -i.$$

The DFT analysis equations (10.4) become

$$F_0 = f_0 W^{0 \cdot 0} + f_1 W^{0 \cdot 1} + f_2 W^{0 \cdot 2} + f_3 W^{0 \cdot 3} = 0 + 1 + 2 + 3 = 6,$$

$$F_1 = f_0 W^{1 \cdot 0} + f_1 W^{1 \cdot 1} + f_2 W^{1 \cdot 2} + f_3 W^{1 \cdot 3} = 0 + 1(-i) + 2(-1) + 3(i) = -2 + 2i,$$

$$F_2 = f_0 W^{2 \cdot 0} + f_1 W^{2 \cdot 1} + f_2 W^{2 \cdot 2} + f_3 W^{2 \cdot 3} = 0 + 1(-1) + 2(1) + 3(-1) = -2,$$

$$F_3 = f_0 W^{3 \cdot 0} + f_1 W^{3 \cdot 1} + f_2 W^{3 \cdot 2} + f_3 W^{3 \cdot 3} = 0 + 1(i) + 2(-1) + 3(-i) = -2 - 2i.$$

Thus

$$F = (F_0, F_1, F_2, F_3) = (6, -2 + 2i, -2, -2 - 2i).$$

Now apply the synthesis formula (10.5):

$$f_0 = \frac{1}{4}(F_0 W^0 + F_1 W^0 + F_2 W^0 + F_3 W^0) = \frac{1}{4}(6 - 2 + 2i - 2 - 2 - 2i) = 0,$$

$$f_1 = \frac{1}{4}(F_0 W^0 + F_1 W^{-1} + F_2 W^{-2} + F_3 W^{-3}) = 1,$$

$$f_2 = \frac{1}{4}(F_0 W^0 + F_1 W^{-2} + F_2 W^{-4} + F_3 W^{-6}) = 2,$$

$$f_3 = \frac{1}{4}(F_0 W^0 + F_1 W^{-3} + F_2 W^{-6} + F_3 W^{-9}) = 3.$$

Thus the IDFT recovers the original sequence $(0, 1, 2, 3)$ exactly, in accordance with Corollary 9.

It is instructive to view the synthesis step as a linear combination of orthogonal vectors. For $N = 4$, the vectors

$$v_n = (1, W^{-n}, W^{-2n}, W^{-3n}), \quad n = 0, 1, 2, 3,$$

form an orthogonal set. Then the synthesis formula can be written symbolically as

$$f = \frac{1}{4} \sum_{n=0}^3 F_n v_n,$$

i.e., f is a linear combination of the discrete complex exponentials with weights F_n .

10.4 Properties of the DFT

10.4.1 Cartesian and polar forms of F_n

For each n we write

$$F_n = A_n + iB_n,$$

where $A_n = \operatorname{Re} F_n$ and $B_n = \operatorname{Im} F_n$. In polar form,

$$F_n = |F_n| e^{i\theta_n},$$

where

$$|F_n| = \sqrt{A_n^2 + B_n^2}, \quad \theta_n = \arg(F_n).$$

When f_k is real-valued, F_n satisfies symmetry relations analogous to those for the continuous Fourier transform.

10.4.2 Symmetry for real data

Assume $f_k \in \mathbb{R}$ for $k = 0, 1, \dots, N-1$. Using the periodic extension in n (Corollary 10), it is convenient to regard F_n as defined for all $n \in \mathbb{Z}$ with period N .

Theorem 49 (Conjugate symmetry for real sequences). *If f_k is real, then*

$$F_{-n} = F_n^*, \quad \forall n \in \mathbb{Z}, \quad (10.6)$$

where indices are understood modulo N . Equivalently,

$$A_{-n} = A_n, \quad B_{-n} = -B_n,$$

so the real part A_n and the magnitude $|F_n|$ are even in n , while the imaginary part B_n and the phase θ_n are odd (up to 2π -periodic ambiguities in the phase).

Proof. Starting from the definition

$$F_n = \sum_{k=0}^{N-1} f_k W^{nk},$$

take the complex conjugate:

$$F_n^* = \sum_{k=0}^{N-1} f_k^* (W^{nk})^* = \sum_{k=0}^{N-1} f_k W^{-nk},$$

since f_k is real and $W^* = W^{-1}$. But

$$F_{-n} = \sum_{k=0}^{N-1} f_k W^{-nk},$$

so $F_{-n} = F_n^*$, which is (10.6). Writing $F_n = A_n + iB_n$ and $F_{-n} = A_{-n} + iB_{-n}$, the relation $F_{-n} = F_n^*$ implies

$$A_{-n} + iB_{-n} = A_n - iB_n,$$

so $A_{-n} = A_n$ and $B_{-n} = -B_n$. The statements about evenness and oddness follow. $\square$

Thus, for a real input sequence, the DFT coefficients satisfy a discrete conjugate symmetry: values at negative indices (modulo N) are complex conjugates of those at positive indices.

10.4.3 Indexing over a symmetric interval

Because F_n is periodic with period N , it is often useful to label one period symmetrically about the origin. For even N , a convenient choice is

$$n = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, -1, 0, 1, \dots, \frac{N}{2} - 1.$$

On this symmetric index set, Theorem 49 states that pairs F_n and F_{-n} are conjugates, so that

$$A_n = A_{-n}, \quad B_n = -B_{-n},$$

and the center index $n = 0$ (and, for even N , also $n = \pm N/2$) is real.

Viewed modulo N , the same symmetry can be described within the primary interval $n = 0, 1, \dots, N-1$: for n in this range,

$$F_n = F_{(N-n) \bmod N}^*,$$

which is the form typically exploited in practical implementations.

Notes and Comments

A lemma on sums of discrete exponentials

Lemma 4. For any integer $N > 1$,

$$\sum_{n=0}^{N-1} e^{i2\pi n/N} = 0.$$

Proof. The sum is a geometric series with ratio

$$\alpha = e^{i2\pi/N} \neq 1.$$

Hence

$$\sum_{n=0}^{N-1} \alpha^n = \frac{1 - \alpha^N}{1 - \alpha} = \frac{1 - e^{i2\pi}}{1 - \alpha} = 0,$$

since $e^{i2\pi} = 1$. □

The same argument applies with $\alpha = W^p = e^{-i2\pi p/N}$ and $p \not\equiv 0 \pmod{N}$, yielding the orthogonality relation of Theorem 47.

Remarks on the argument of a complex number

For a complex number $z = x + iy$, the principal value of the argument is often taken as

$$\arg(z) = \text{atan2}(y, x),$$

which returns a value in $(-\pi, \pi]$. Two special cases are worth noting:

- If $z = 0$ (i.e. $x = 0$ and $y = 0$), the argument is undefined. In practice one often assigns $\arg(0) = 0$ for convenience when plotting phase.
- If $y = 0$ and $x < 0$, then z lies on the negative real axis. The points $z = -1$ and its periodic copies in the DFT correspond to arguments π or $-\pi$, which differ by 2π but represent the same direction in the complex plane. When imposing both periodicity and odd symmetry for phase plots, this leads to a choice at such points. In practice, one typically chooses the branch that yields a visually continuous phase plot, and may mark ambiguous points explicitly when necessary.

These subtleties do not affect the algebraic properties of the DFT, but are relevant when interpreting and plotting the phase $\theta_n = \arg(F_n)$.

11 Inside the Fast Fourier Transform

11.1 Introduction

The radix-2 fast Fourier transform (FFT) is an efficient algorithm for evaluating the discrete Fourier transform (DFT) when the data length

$$N = 2^v, \quad v \in \mathbb{N}.$$

Instead of evaluating each DFT coefficient directly from its definition, the FFT factors the computation into v stages of simple two-point operations (*butterflies*), arranged in a structured way. This reduces the operation count from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log_2 N)$.

In this section the radix-2 structure is first illustrated for small N , then generalized using flow graphs and simple indexing rules. A derivation of the radix-2 algorithm following Cooley and Tukey is sketched for $N = 8$.

The basic idea is reuse. Direct DFT evaluation recomputes many nearly identical partial sums. The FFT splits the data into smaller groups, computes those partial sums once, and recombines them with simple sum-and-difference operations. A butterfly is exactly this reuse pattern: one twiddle multiplication feeds two outputs.

11.2 DFT in index notation and radix-2 setting

The DFT analysis equation can be written as

$$F_n = \sum_{k=0}^{N-1} f_0(k) W^{nk}, \quad n = 0, 1, \dots, N-1,$$

where

$$W = e^{-i2\pi/N}.$$

Here:

- F_n denotes the n th DFT spectral value.
- $f_0(k)$ denotes the k th element of the input (stage-0) data vector.
- The subscript in $f_\sigma(k)$ will be used for stage index $\sigma = 0, 1, \dots, v$.

Thus F_n is the final DFT coefficient notation from Chapter 10, while $f_\sigma(k)$ is temporary FFT stage notation used only inside the algorithm. The exponential can be written compactly as

$$e^{-i2\pi nk/N} = W^{nk}.$$

The minus sign in the exponent is fixed throughout this chapter.

The FFT is obtained by factoring the matrix representation of this DFT sum into a product of v sparse matrices, each corresponding to one stage of butterflies.

11.3 Small examples

Case $N = 2$

For $N = 2$,

$$W = e^{-i2\pi/2} = -1.$$

The DFT is

$$\begin{aligned} F_0 &= f_0(0)W^{0 \cdot 0} + f_0(1)W^{0 \cdot 1} = f_0(0) + f_0(1), \\ F_1 &= f_0(0)W^{1 \cdot 0} + f_0(1)W^{1 \cdot 1} = f_0(0) - f_0(1). \end{aligned}$$

So for $N = 2$ the DFT consists of one sum and one difference.

Case $N = 4$

For $N = 4$,

$$W = e^{-i2\pi/4} = -i, \quad W^2 = -1, \quad W^4 = 1.$$

The DFT in matrix form is

$$\begin{bmatrix} F_0 \\ F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} W^{0 \cdot 0} & W^{0 \cdot 1} & W^{0 \cdot 2} & W^{0 \cdot 3} \\ W^{1 \cdot 0} & W^{1 \cdot 1} & W^{1 \cdot 2} & W^{1 \cdot 3} \\ W^{2 \cdot 0} & W^{2 \cdot 1} & W^{2 \cdot 2} & W^{2 \cdot 3} \\ W^{3 \cdot 0} & W^{3 \cdot 1} & W^{3 \cdot 2} & W^{3 \cdot 3} \end{bmatrix} \begin{bmatrix} f_0(0) \\ f_0(1) \\ f_0(2) \\ f_0(3) \end{bmatrix}.$$

Using $W^4 = 1$ and $W^2 = -1$, the matrix simplifies so that only W^0 and W^1 appear explicitly, and the rows can be written in a form where only W^0 and W^1 are required, together with their negations.

This matrix can be factorized into a product of two 4×4 matrices, corresponding to two stages of butterflies. In stage 1, inputs are combined in pairs (0, 2) and (1, 3):

$$\begin{aligned} f_1(0) &= f_0(0) + W^0 f_0(2), \\ f_1(2) &= f_0(0) - W^0 f_0(2), \\ f_1(1) &= f_0(1) + W^0 f_0(3), \\ f_1(3) &= f_0(1) - W^0 f_0(3). \end{aligned}$$

These are two independent two-point DFTs: one built from the even-indexed samples $f_0(0), f_0(2)$, and one built from the odd-indexed samples $f_0(1), f_0(3)$. The first stage therefore makes the partial sums that will be reused by both the low and high frequency bins.

In stage 2, pairs (0, 1) and (2, 3) are combined:

$$\begin{aligned} f_2(0) &= f_1(0) + W^0 f_1(1), \\ f_2(1) &= f_1(0) - W^0 f_1(1), \\ f_2(2) &= f_1(2) + W^1 f_1(3), \\ f_2(3) &= f_1(2) - W^1 f_1(3). \end{aligned}$$

Substituting the stage 1 values into stage 2 gives

$$\begin{aligned} f_2(0) &= f_0(0) + f_0(1) + f_0(2) + f_0(3) = F_0, \\ f_2(1) &= f_0(0) - f_0(1) + f_0(2) - f_0(3) = F_2, \\ f_2(2) &= f_0(0) - if_0(1) - f_0(2) + if_0(3) = F_1, \\ f_2(3) &= f_0(0) + if_0(1) - f_0(2) - if_0(3) = F_3. \end{aligned}$$

Thus the stage output is

$$(f_2(0), f_2(1), f_2(2), f_2(3)) = (F_0, F_2, F_1, F_3).$$

The final outputs F_n are therefore obtained from $f_2(k)$ by a permutation (bit reversal) of the indices.

This output order is not an error in the arithmetic. It is a bookkeeping effect of the order in which the even/odd splits were performed. The bit-reversal step renames the final stage entries so that the bins are again listed as $F_0, F_1, \dots, F_{N-1}$.

Each pair of equations corresponds to a two-point butterfly. In each butterfly only one complex multiplication $W^p \cdot (\cdot)$ is required; the second result is formed by a subtraction using the same product.

For $N = 4$:

- Direct DFT (no precomputed twiddles): $4 \times 4 = 16$ complex multiplications, $4(4-1) = 12$ complex additions.
- FFT: 4 complex multiplications, 4 complex additions and 4 complex subtractions.

11.4 Operation count for radix-2 FFT

For $N = 2^v$, the radix-2 FFT has v stages. At each stage, $N/2$ butterflies are computed, each requiring one complex multiplication and two complex add/subtract operations. Thus

$$\begin{aligned}\text{complex multiplications} &= \frac{N}{2} v = \frac{N}{2} \log_2 N, \\ \text{complex add/subtracts} &= N v = N \log_2 N.\end{aligned}$$

In contrast, the direct DFT needs

$$\begin{aligned}\text{complex multiplications} &= N^2, \\ \text{complex additions} &= N(N-1).\end{aligned}$$

The ratio of multiplication counts (direct/FFT) is

$$\frac{N^2}{\frac{N}{2} \log_2 N} = \frac{2N}{\log_2 N},$$

which grows linearly with N .

For example, for $N = 32,768$,

$$\frac{2N}{\log_2 N} \approx 4369.$$

A machine that requires one minute for the FFT at this N would require several days for the direct DFT, using the same implementation model.

11.5 Signal-flow graphs and butterflies

The radix-2 FFT can be represented by a signal-flow graph. Nodes represent intermediate values $f_\sigma(k)$ at stage σ and index k . Directed edges connect nodes between stages.

In the radix-2 decimation-in-time structure:

- Each node in stage σ has two input edges from stage $\sigma - 1$: a *primary* input (multiplier 1) and a *secondary* input (multiplier W^p for some integer p).
- Each pair of nodes that share the same two source nodes are called a *dual-node pair* and form a butterfly.

For $N = 4$ and $N = 8$, flow graphs show:

- Stage 1: butterflies combine indices separated by $N/2$.
- Stage 2: butterflies combine indices separated by $N/4$.
- ...
- Stage ν : butterflies combine adjacent indices.

Dual-node separation

Let σ be the stage index, $1 \leq \sigma \leq \nu$, and $N = 2^\nu$.

Dual-node separation

At stage σ , the two nodes in each butterfly are separated by

$$\Delta k = \frac{N}{2^\sigma}.$$

That is, butterflies connect $f_{\sigma-1}(k)$ and $f_{\sigma-1}(k + \Delta k)$.

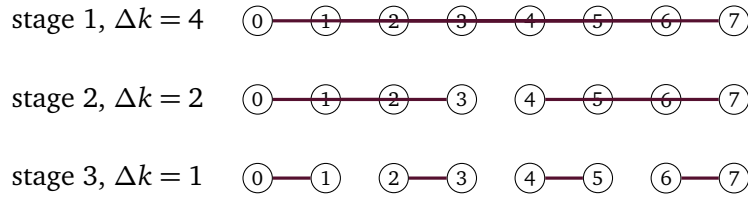


Figure 27: For $N = 8$, the radix-2 butterflies first pair samples separated by 4, then by 2, then by 1.

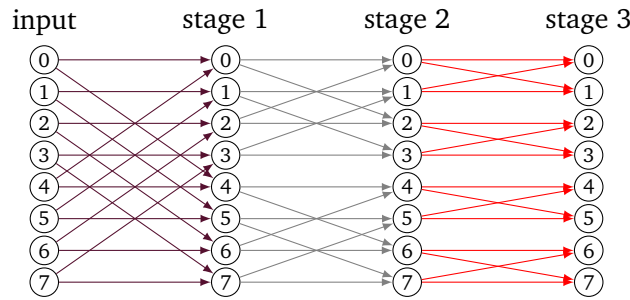


Figure 28: A compact $N = 8$ radix-2 signal-flow graph. Each pair of crossing connections is one butterfly; the separation halves from stage to stage.

Butterfly equations

Let $f_{\sigma-1}(k)$ and $f_{\sigma-1}(k + \Delta k)$ be the primary and secondary source nodes at stage $\sigma - 1$, and let p be the exponent associated with the secondary edge. The butterfly equations are

$$\begin{aligned} f_\sigma(k) &= f_{\sigma-1}(k) + W^p f_{\sigma-1}(k + \Delta k), \\ f_\sigma(k + \Delta k) &= f_{\sigma-1}(k) - W^p f_{\sigma-1}(k + \Delta k). \end{aligned} \quad (11.1)$$

Here

$$\Delta k = \frac{N}{2^\sigma}.$$

A single complex multiplication $W^p f_{\sigma-1}(k + \Delta k)$ is reused in both output expressions; the second output is a subtraction using the same product.

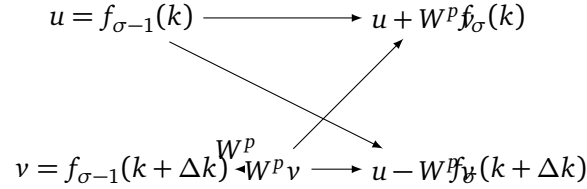


Figure 29: A radix-2 butterfly reuses one twiddle multiplication to form a sum and a difference.

For in-place computation (overlay of results on input), it is convenient to rewrite (11.1) so that the source values are not overwritten before they are needed. One practical ordering is

$$\begin{aligned} f_{\sigma}(k + \Delta k) &= f_{\sigma-1}(k) - W^p f_{\sigma-1}(k + \Delta k), \\ f_{\sigma}(k) &= f_{\sigma-1}(k) + W^p f_{\sigma-1}(k + \Delta k), \end{aligned} \quad (11.2)$$

computed in this order so that $f_{\sigma-1}(k)$ is still available when computing the second line.

Power of W in each butterfly

For stage σ , $1 \leq \sigma \leq v$, define

$$\Delta k = \frac{N}{2^\sigma}, \quad s = v - \sigma,$$

and let k be the node index (in decimal). Then

Exponent index

The exponent p associated with the secondary input of the butterfly at stage σ and index k can be obtained as follows:

1. Compute

$$q = \left\lfloor \frac{k}{2^s} \right\rfloor.$$

2. Express q in v -bit binary.
3. Reverse the order of the bits.
4. Convert back to decimal. The result is p .

This rule is equivalent to taking a truncated right-shift in binary (by s bits), then applying bit reversal on those bits.

The dual node in the same butterfly has exponent index shifted by $N/2$:

Dual-node exponent

If the upper node has multiplier W^p , the lower node has multiplier $W^{p+N/2}$.

Using $W^{p+N/2} = W^p W^{N/2}$ and $W^{N/2} = -1$ (for even N) gives

$$W^{p+N/2} = -W^p.$$

Thus one can also state:

Dual-node multiplier

If a node uses the multiplier W^P , its dual node uses $-W^P$.

Traversal and skipping inside a stage

At stage σ , butterflies are formed using blocks of length $2\Delta k$, with local indices

$$k, k + \Delta k, k + 2\Delta k, k + 3\Delta k, \dots$$

In each block, the first Δk indices serve as primary nodes for butterflies, paired with nodes Δk positions below.

Equivalently:

Stage traversal

For stage σ :

1. Set $\Delta k = N/2^\sigma$.
2. Starting from $k = 0$, apply the butterfly (11.2) for $k = 0, 1, \dots, \Delta k - 1$.
3. Skip the next Δk indices (they are dual nodes already updated).
4. Repeat the pattern “ Δk butterflies, Δk skips” until $k \geq N$.

All butterflies within a stage are independent and can, in principle, be evaluated in parallel.

11.6 Bit reversal and output permutation

The flow graph in decimation-in-time order produces outputs in an index order that is bit-reversed relative to the natural order. To obtain F_n for $n = 0, 1, \dots, N - 1$ in natural order, a final permutation is applied to the last stage vector $f_v(k)$.

Let k be the index of an element in f_v , and write its v -bit binary representation as

$$k \leftrightarrow (b_{v-1}b_{v-2} \dots b_1b_0)_2.$$

Define the bit-reversed index

$$\text{rev}(k) \leftrightarrow (b_0b_1 \dots b_{v-2}b_{v-1})_2.$$

Bit reversal

The correctly ordered DFT output is obtained by

$$F_{\text{rev}(k)} = f_v(k), \quad k = 0, 1, \dots, N - 1.$$

In an in-place implementation, this is done by swapping entries in f_v whenever necessary. To avoid undoing swaps, one uses:

Swap condition

For k from 0 to $N - 1$:

1. Compute $j = \text{rev}(k)$.
2. If $j > k$, swap $f_v(k)$ and $f_v(j)$.
3. If $j \leq k$, do nothing.

This ensures that each pair (k, j) is swapped at most once.

k	binary	reversed	output index
0	000_2	$\longrightarrow 000_2$	$\text{rev}(0) = 0$
1	001_2	$\longrightarrow 100_2$	$\text{rev}(1) = 4$
2	010_2	$\longrightarrow 010_2$	$\text{rev}(2) = 2$
3	011_2	$\longrightarrow 110_2$	$\text{rev}(3) = 6$
4	100_2	$\longrightarrow 001_2$	$\text{rev}(4) = 1$
5	101_2	$\longrightarrow 101_2$	$\text{rev}(5) = 5$
6	110_2	$\longrightarrow 011_2$	$\text{rev}(6) = 3$
7	111_2	$\longrightarrow 111_2$	$\text{rev}(7) = 7$

Figure 30: Bit reversal for $N = 8$. The FFT flow produces values in bit-reversed order, so a final permutation restores natural DFT-bin order.

11.7 Algorithm structure

A standard in-place radix-2 decimation-in-time FFT can be summarized as:

Before reading the pseudocode, match its variables to the flow graph:

- σ selects the stage.
- $\Delta k = N/2^\sigma$ is the vertical separation inside each butterfly.
- k moves from block to block.
- r walks through the butterflies inside one block.
- p selects the twiddle factor for the secondary input.

Given $N = 2^\nu$, input $f_0(k)$, $k = 0, \dots, N - 1$.

for $\sigma = 1$ to ν **do**

$\Delta k \leftarrow N/2^\sigma$.

$k \leftarrow 0$.

while $k < N$ **do**

for $r = 0$ to $\Delta k - 1$ **do**

 Determine exponent index p for node $k + r$.

$G \leftarrow W^p f_{\sigma-1}(k + r + \Delta k)$.

$u \leftarrow f_{\sigma-1}(k + r)$.

$f_\sigma(k + r + \Delta k) \leftarrow u - G$.

$f_\sigma(k + r) \leftarrow u + G$.

end for

$k \leftarrow k + 2\Delta k$.

end while

end for

Apply bit reversal on $f_\nu(k)$ to obtain F_n .

Details of computing p , the twiddle exponent, are described in the previous discussion of butterfly powers.

11.8 Cooley–Tukey derivation for $N=8$

For $N = 8$ and

$$W = e^{-i2\pi/8},$$

the DFT is

$$F_n = \sum_{k=0}^7 f_0(k) W^{nk}.$$

Split the sum into even and odd indices $k = 2r$ and $k = 2r + 1$:

$$\begin{aligned} F_n &= \sum_{r=0}^3 f_0(2r) W^{n(2r)} + \sum_{r=0}^3 f_0(2r+1) W^{n(2r+1)} \\ &= E(n) + W^n O(n), \end{aligned}$$

where

$$E(n) = \sum_{r=0}^3 f_0(2r) (W^2)^{nr}, \quad O(n) = \sum_{r=0}^3 f_0(2r+1) (W^2)^{nr}.$$

Thus E and O are 4-point DFTs (with twiddle $W^2 = e^{-i2\pi/4}$). Using their period 4:

$$F_n = E(n) + W^n O(n), \quad F_{n+4} = E(n) - W^n O(n), \quad n = 0, 1, 2, 3.$$

The second identity follows because

$$E(n+4) = E(n), \quad O(n+4) = O(n), \quad W^{n+4} = W^n W^4 = -W^n.$$

This is the top-level butterfly: one 4-point DFT of the even-indexed samples, one 4-point DFT of the odd-indexed samples, and four twiddle products $W^n O(n)$.

Each 4-point DFT is then decomposed again into even/odd parts (now 2-point DFTs). If

$$e_r = f_0(2r), \quad o_r = f_0(2r+1), \quad r = 0, 1, 2, 3,$$

then

$$W^4 = -1,$$

and splitting $r = 2q$ and $r = 2q + 1$ gives

$$\begin{aligned} E(n) &= (e_0 + (-1)^n e_2) + (W^2)^n (e_1 + (-1)^n e_3), \\ O(n) &= (o_0 + (-1)^n o_2) + (W^2)^n (o_1 + (-1)^n o_3), \end{aligned}$$

for $n = 0, 1, 2, 3$. These are exactly butterfly combinations.

So for $N = 8$, the computation factorizes into three butterfly stages:

- stage 1: stride 4 pairings,
- stage 2: stride 2 pairings,
- stage 3: stride 1 pairings,

followed by the standard bit-reversal permutation between stage ordering and natural DFT index ordering.

For $N = 8$, the stage pattern is:

stage	Δk	example pairings
1	4	(0, 4), (1, 5), (2, 6), (3, 7)
2	2	(0, 2), (1, 3), (4, 6), (5, 7)
3	1	(0, 1), (2, 3), (4, 5), (6, 7)

This table is often the easiest way to debug an implementation: each stage halves the separation between paired indices.

11.9 Using the FFT for synthesis

The DFT synthesis equation is

$$f(k) = \frac{1}{N} \sum_{n=0}^{N-1} F_n W^{-nk}, \quad k = 0, 1, \dots, N-1.$$

This can be written as

$$f(k) = \frac{1}{N} \left[\sum_{n=0}^{N-1} F_n^* W^{nk} \right]^*.$$

Thus the inverse DFT can be computed using the same FFT code used for the forward DFT, by:

1. Conjugating the input spectrum F_n .
2. Applying the forward FFT to $\{F_n^*\}$.
3. Conjugating the result and scaling by $1/N$.

11.10 Real input and composite N

When the input sequence f_k is real and N is even, one common real-FFT trick exploits conjugate symmetry to reduce computation:

- Pack the even and odd samples into one complex sequence of length $N/2$, for example $z_m = f_{2m} + if_{2m+1}$.
- Compute an FFT of length $N/2$.
- Recover the length- N spectra from the length- $N/2$ result.

This saves both time and storage in the analysis direction. For synthesis, the analogous saving applies only when the spectrum has the conjugate symmetry of a real time sequence; a general complex spectrum cannot be packed this way without extra structure.

When N is not a power of two but is composite, mixed-radix FFT algorithms can be constructed by factoring N into smaller radices and applying Cooley–Tukey style decompositions. If N is prime, no nontrivial factorization exists and one falls back to a DFT of length N (or to prime-length FFT variants such as Bluestein or Rader, which are beyond the scope of this section).

Notes on complex arithmetic and twiddle factors

Let

$$c = a + ib, \quad w = u + iv$$

be complex numbers. Then

$$\begin{aligned} cw &= (a + ib)(u + iv) = (au - bv) + i(bu + av), \\ c + w &= (a + u) + i(b + v), \\ c - w &= (a - u) + i(b - v). \end{aligned}$$

Thus a complex multiplication requires four real multiplications and two real additions if implemented naively.

For the FFT, the multiplication by a twiddle factor W^p is common. Writing

$$W^p = \cos \theta - i \sin \theta, \quad \theta = \frac{2\pi p}{N},$$

and $x(k) = a + ib$, one obtains

$$\begin{aligned} W^p x(k) &= (\cos \theta - i \sin \theta)(a + ib) \\ &= (a \cos \theta + b \sin \theta) + i(b \cos \theta - a \sin \theta), \end{aligned}$$

so real and imaginary parts can be computed from precomputed $\cos \theta$ and $\sin \theta$ using real arithmetic. Once formed, $W^p x(k)$ is reused in both outputs of the butterfly (11.2).

12 The Discrete Fourier Transform as an Estimator

In Chapter 11 the discrete Fourier transform (DFT) was evaluated by the fast Fourier transform (FFT) algorithm. In this chapter the DFT/FFT is regarded as an *estimator* for the continuous Fourier transforms (CFTs) introduced in Part I.

To keep the notation distinct, the following convention will be used throughout this chapter:

- **FFT**: the analysis/synthesis pair for the DFT.
- **CFT**: the continuous Fourier transforms:
 - complex Fourier series coefficients for periodic functions,
 - Fourier transform for finite-energy pulses.

Where needed, notation such as CFT (Fourier series) or CFT (pulse) will be used to remove ambiguity.

The point of the chapter is not that the FFT is a different Fourier transform. The point is that the FFT sum can be used as a numerical stand-in for a CFT integral or for Fourier-series coefficients, provided the scaling and grids are read correctly. There are two separate questions:

- What scale factor converts the FFT numbers into CFT estimates?
- What error is introduced because the CFT spectrum is being sampled and aliased by a finite grid?

The rectangular-rule derivation answers the first question. The aliasing derivation answers the second.

12.1 Rectangular-rule approximation and the FFT

Consider the definite integral

$$I = \int_0^T g(t) dt.$$

The rectangular rule approximates I by sampling g at N equally spaced points on $[0, T]$ and summing the areas of the corresponding rectangles.

Let

$$T_s = \frac{T}{N}, \quad t_k = kT_s, \quad g_k = g(t_k), \quad 0 \leq k \leq N-1,$$

and define

$$S_N = \sum_{k=0}^{N-1} g_k T_s.$$

Then S_N is the left-endpoint Riemann sum. Under mild regularity assumptions (e.g. g Riemann integrable on $[0, T]$),

$$\int_0^T g(t) dt = \lim_{N \rightarrow \infty} \sum_{k=0}^{N-1} g_k T_s.$$

Theorem 50 (Rectangular rule). *For each $N \in \mathbb{N}$, define $T_s = T/N$ and $t_k = kT_s$, $g_k = g(t_k)$. Then*

$$\int_0^T g(t) dt \approx \sum_{k=0}^{N-1} g_k T_s, \quad (12.1)$$

and the approximation error tends to zero as $N \rightarrow \infty$.

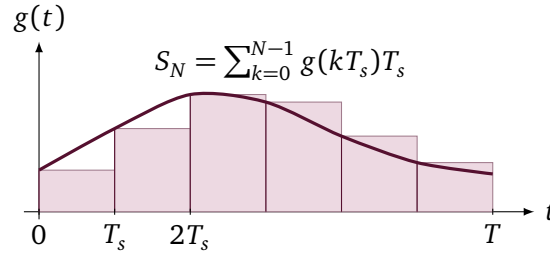


Figure 31: The rectangular rule replaces the integral of $g(t)$ by left-endpoint rectangle areas.

Proof. This is the standard convergence of left-endpoint Riemann sums to the Riemann integral. Since g is Riemann integrable on $[0, T]$, given any $\varepsilon > 0$ there exists a partition mesh size $\delta > 0$ such that for any partition with maximum subinterval length less than δ , the corresponding Riemann sum differs from the integral by less than ε . Choosing N sufficiently large ensures $T_s = T/N < \delta$, so that (12.1) differs from $\int_0^T g(t) dt$ by less than ε . $\square$

12.1.1 Application to the CFT of a pulse

Let $f(t)$ be a finite-span pulse carried by the FFT observation window $[0, T]$:

$$f(t) = 0 \quad \text{for } t < 0 \text{ and } t \geq T.$$

Its Fourier transform is

$$F(\omega) = \int_0^T f(t) e^{-i\omega t} dt. \quad (12.2)$$

Apply the rectangular rule with step $T_s = T/N$:

$$t_k = kT_s, \quad f_k = f(t_k), \quad 0 \leq k \leq N-1.$$

The right endpoint $t = T$ is not an additional FFT sample; it is the beginning of the next period in the discrete model. An isolated value assigned at $t = T$ does not change the integral, but it would create an extra impulse in the sampled comb model below, so it is excluded from the N -sample window. Then

$$F(\omega) \approx \sum_{k=0}^{N-1} f_k e^{-i\omega t_k T_s}. \quad (12.3)$$

Introduce the frequency sampling

$$\omega_n = n\omega_0, \quad \omega_0 = \frac{2\pi}{T}, \quad -\frac{N}{2} < n < \frac{N}{2}.$$

Then, for each n ,

$$\begin{aligned} F(\omega_n) &\approx \sum_{k=0}^{N-1} f_k e^{-in\omega_0 t_k T_s} = T_s \sum_{k=0}^{N-1} f_k \exp\left(-in \frac{2\pi}{T} k T_s\right) \\ &= T_s \sum_{k=0}^{N-1} f_k \exp\left(-i \frac{2\pi}{N} nk\right). \end{aligned} \quad (12.4)$$

Define the DFT (FFT analysis) of $\{f_k\}$ as

$$F_n = \sum_{k=0}^{N-1} f_k e^{-i \frac{2\pi}{N} nk}, \quad 0 \leq n \leq N-1.$$

Then (12.4) becomes

$$F(\omega_n) \approx T_s F_n = \frac{T}{N} F_n.$$

This yields:

Theorem 51 (FFT approximation of the Fourier transform). *Let $f(t)$ be a finite-span pulse carried by the sampled window $0 \leq t < T$. Sample f at*

$$t_k = kT_s, \quad T_s = \frac{T}{N}, \quad 0 \leq k \leq N-1,$$

and define $f_k = f(t_k)$. Let F_n be the DFT of $\{f_k\}$:

$$F_n = \sum_{k=0}^{N-1} f_k e^{-i \frac{2\pi}{N} nk}.$$

Let $\omega_n = n\omega_0$ with $\omega_0 = 2\pi/T$. Then

$$F(\omega_n) \approx \frac{T}{N} F_n, \quad -\frac{N}{2} < n < \frac{N}{2},$$

and, for each fixed index n , the rectangular-rule approximation error tends to zero as $N \rightarrow \infty$.

Proof. Equation (12.3) is the rectangular-rule approximation of the integral defining $F(\omega)$ in (12.2) with step $T_s = T/N$. Substituting $\omega = \omega_n = n\omega_0$ and $t_k = kT_s$ leads to (12.4). Identifying F_n as the DFT of $\{f_k\}$ yields

$$F(\omega_n) \approx T_s F_n = \frac{T}{N} F_n.$$

By Theorem 50, the rectangular rule error tends to zero as $N \rightarrow \infty$ for each fixed n , so the FFT-based approximation inherits this convergence. $\square$

In words, up to a constant scaling factor T/N , the FFT of the time samples of a pulse provides numerical estimates of the samples of its continuous Fourier transform on a uniform frequency grid.

For a pulse estimate, keep the three grids distinct:

$$\begin{aligned} t_k &= kT_s, & T_s &= \frac{T}{N}, \\ \omega_n &= n\omega_0, & \omega_0 &= \frac{2\pi}{T}, \\ \Omega_s &= N\omega_0 = \frac{2\pi}{T_s}. \end{aligned}$$

The first grid is where the pulse is sampled. The second grid is where the Fourier transform is estimated. The third number is the spacing between aliased copies of the continuous spectrum. Confusing ω_0 with Ω_s is a common source of wrong FFT scaling and wrong aliasing conclusions.

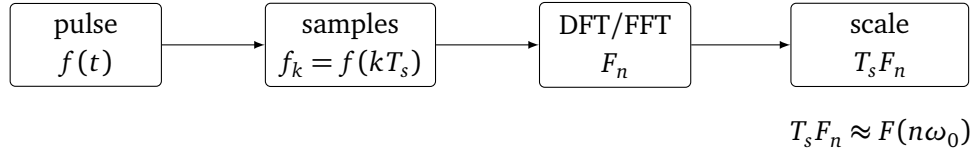


Figure 32: For a finite-span pulse, the FFT estimate is the rectangular-rule sum evaluated on the grid $\omega_n = n\omega_0$.

12.1.2 Periodic waveforms and Fourier series

Let $f_p(t)$ be a periodic waveform of period T_0 , with complex Fourier series

$$f_p(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0},$$

where

$$C_n = \frac{1}{T_0} \int_0^{T_0} f_p(t) e^{-in\omega_0 t} dt.$$

Applying the rectangular rule with $T_s = T_0/N$,

$$t_k = kT_s, \quad f_k = f_p(t_k), \quad 0 \leq k \leq N-1,$$

gives

$$\begin{aligned} C_n &\approx \frac{1}{T_0} \sum_{k=0}^{N-1} f_k e^{-in\omega_0 t_k} T_s = \frac{T_s}{T_0} \sum_{k=0}^{N-1} f_k e^{-i\frac{2\pi}{T_0} nk T_s} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-i\frac{2\pi}{N} nk}. \end{aligned}$$

Theorem 52 (FFT approximation of Fourier-series coefficients). *Let $f_p(t)$ be periodic with period T_0 . Sample f_p over one period at*

$$t_k = kT_s, \quad T_s = \frac{T_0}{N}, \quad 0 \leq k \leq N-1,$$

and define $f_k = f_p(t_k)$. Let F_n be the DFT of $\{f_k\}$:

$$F_n = \sum_{k=0}^{N-1} f_k e^{-i\frac{2\pi}{N} nk}.$$

Then

$$C_n \approx \frac{1}{N} F_n, \quad -\frac{N}{2} < n < \frac{N}{2},$$

and, for each fixed coefficient index n , the approximation error tends to zero as $N \rightarrow \infty$.

Proof. Using the rectangular rule for C_n gives

$$C_n \approx \frac{1}{T_0} \sum_{k=0}^{N-1} f_p(t_k) e^{-in\omega_0 t_k} T_s.$$

Since $T_s = T_0/N$ and $\omega_0 = 2\pi/T_0$, this becomes

$$C_n \approx \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-i\frac{2\pi}{N} nk} = \frac{1}{N} F_n.$$

For each fixed coefficient index n , the convergence follows again from Theorem 50. □

For both pulses and periodic waveforms, two implementation details are important:

- Sampling must follow left-endpoint uniform sampling, because the correspondence between the CFT and FFT is derived using that sampling scheme.
- At any sampling instant coinciding with a discontinuity of f , the half-value is used, consistent with the Fourier convergence properties.

The half-value rule is not a cosmetic convention. If a jump occurs exactly at a sample location, the Fourier series and Fourier integral recover the average of the two one-sided limits at that point. Feeding that same value to the FFT keeps the discrete estimate aligned with the CFT convention used throughout these notes.

12.2 Aliasing of Fourier spectra

For clarity, the following notation is adopted:

- $\text{per}(t)$: periodic waveform,
- $\text{PER}(n)$: its CFT line spectrum (Fourier series coefficients),
- $\text{pul}(t)$: finite-span, finite-energy pulse,
- $\text{PUL}(\omega)$: Fourier transform of $\text{pul}(t)$.

FFT line spectra obtained from samples of these functions are denoted by

- PER_n : FFT line spectrum from samples of $\text{per}(t)$,
- PUL_n : FFT line spectrum from samples of $\text{pul}(t)$.

Consider $\text{pul}(t)$ with transform $\text{PUL}(\omega)$. Define an *aliased* spectrum $\text{PUL}_a(\omega)$ by periodically replicating $\text{PUL}(\omega)$ with period Ω_a and summing:

$$\text{PUL}_a(\omega) = \sum_{m=-\infty}^{\infty} \text{PUL}(\omega - m\Omega_a).$$

Here $\Omega_a > 0$ is the *aliasing period*. The resulting aliased spectrum $\text{PUL}_a(\omega)$ is periodic with period Ω_a .

Assume $\text{pul}(t)$ is sufficiently regular (for example, absolutely integrable and piecewise smooth), so that $\text{PUL}(\omega) \rightarrow 0$ as $|\omega| \rightarrow \infty$ (typically $O(1/|\omega|)$ or faster). Therefore, for large enough Ω_a , in the central band $-\Omega_a/2 < \omega < \Omega_a/2$ the influence of remote replicas is small, and $\text{PUL}_a(\omega)$ closely follows the original $\text{PUL}(\omega)$ in that band.

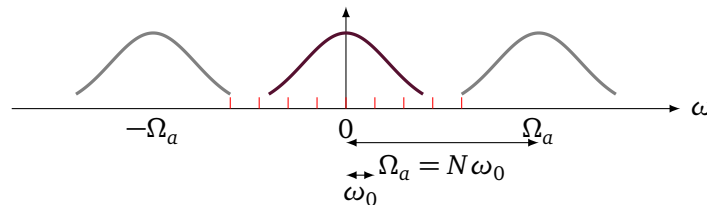


Figure 33: The estimator view first aliases the continuous spectrum with period Ω_a , then samples the aliased spectrum on the frequency grid $\omega_n = n\omega_0$.

Next, sample $\text{PUL}_a(\omega)$ at frequencies spaced by ω_0 :

$$\omega_n = n\omega_0, \quad n \in \mathbb{Z},$$

where

$$\omega_0 = \frac{\Omega_a}{N}.$$

Then

$$\text{PUL}_a(\omega_n) = \sum_{m=-\infty}^{\infty} \text{PUL}(\omega_n - m\Omega_a) = \sum_{m=-\infty}^{\infty} \text{PUL}((n - mN)\omega_0).$$

For sufficiently large Ω_a , the overlapping between neighboring replicas is small, and for each fixed index n in the central region $-N/2 < n < N/2$ one expects

$$\sum_{m=-\infty}^{\infty} \text{PUL}(\omega_n - m\Omega_a) \approx \text{PUL}(\omega_n), \quad -\frac{N}{2} < n < \frac{N}{2}, \quad (12.5)$$

with the error decreasing as Ω_a increases for that fixed index.

An analogous construction applies to the CFT line spectrum $\text{PER}(n)$ of a periodic function $\text{per}(t)$. Define

$$\text{PER}_a(n) = \sum_{m=-\infty}^{\infty} \text{PER}(n - mN),$$

which is periodic with period N in n . If $\text{per}(t)$ is piecewise smooth over one period, then $\text{PER}(n) \rightarrow 0$ as $|n| \rightarrow \infty$ (typically like $1/|n|$ for jump discontinuities, and faster for smoother signals). Taking N large then reduces overlap between replicas and yields the approximation

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) \approx \text{PER}(n), \quad -\frac{N}{2} < n < \frac{N}{2}, \quad (12.6)$$

for each fixed coefficient index n , with error decreasing as N increases.

12.3 The FFT as an estimator for CFT spectra

12.3.1 Pulses

Let $T_s > 0$ and consider the Dirac comb

$$\delta_s(t) = T_s \sum_{k=-\infty}^{\infty} \delta(t - kT_s).$$

Using the standard comb pair

$$\sum_{k=-\infty}^{\infty} \delta(t - kT_s) \longleftrightarrow \frac{2\pi}{T_s} \sum_{m=-\infty}^{\infty} \delta(\omega - m\Omega_s), \quad \Omega_s = \frac{2\pi}{T_s},$$

and multiplying by T_s , we obtain

$$\delta_s(t) \longleftrightarrow 2\pi \sum_{m=-\infty}^{\infty} \delta(\omega - m\Omega_s), \quad \Omega_s = \frac{2\pi}{T_s}.$$

Let $\text{pul}(t)$ be a finite-span pulse carried by the sampled window $0 \leq t < T$, and suppose

$$T = NT_s.$$

Define the impulse-sampled version

$$\text{pul}_s(t) = \text{pul}(t) \delta_s(t).$$

Using the sampling property of the delta distribution,

$$\text{pul}_s(t) = \text{pul}(t) T_s \sum_{k=-\infty}^{\infty} \delta(t - kT_s) = T_s \sum_{k=-\infty}^{\infty} \text{pul}(kT_s) \delta(t - kT_s).$$

Define samples

$$\text{pul}_k = \text{pul}(kT_s),$$

so that

$$\text{pul}_s(t) = T_s \sum_{k=-\infty}^{\infty} \text{pul}_k \delta(t - kT_s).$$

Method A (time-domain sampling). By shifting and scaling,

$$\text{pul}_s(t) \longleftrightarrow \sum_{k=-\infty}^{\infty} \text{pul}_k T_s e^{-i\omega kT_s}.$$

Since $\text{pul}(t)$ vanishes outside the half-open sampled window $[0, T)$, only $k = 0, \dots, N-1$ contribute, hence

$$\text{PUL}_s(\omega) = T_s \sum_{k=0}^{N-1} \text{pul}_k e^{-i\omega kT_s}. \quad (12.7)$$

Method B (multiplication $\rightarrow$ convolution). Because $\text{pul}_s(t) = \text{pul}(t) \delta_s(t)$, in the frequency domain

$$\text{PUL}_s(\omega) = \frac{1}{2\pi} (\text{PUL} * (2\pi \sum_{m=-\infty}^{\infty} \delta(\cdot - m\Omega_s))) (\omega) = \sum_{m=-\infty}^{\infty} \text{PUL}(\omega - m\Omega_s), \quad (12.8)$$

where the constants follow directly from the transform convention used throughout these notes.

Equating the two expressions for $\text{PUL}_s(\omega)$ gives

$$\sum_{m=-\infty}^{\infty} \text{PUL}(\omega - m\Omega_s) = T_s \sum_{k=0}^{N-1} \text{pul}_k e^{-i\omega kT_s}. \quad (12.9)$$

Now sample both sides at

$$\omega = \omega_n = n\omega_0, \quad \omega_0 = \frac{\Omega_s}{N},$$

so that

$$\omega_0 = \frac{2\pi}{T}, \quad T = NT_s,$$

and $T_s = 2\pi/\Omega_s$.

Substituting $\omega = \omega_n$ and $T_s = T/N$ into (12.9), the left-hand side becomes

$$\sum_{m=-\infty}^{\infty} \text{PUL}(n\omega_0 - m\Omega_s),$$

while the right-hand side becomes

$$\begin{aligned} T_s \sum_{k=0}^{N-1} \text{pul}_k e^{-i\omega_n kT_s} &= T_s \sum_{k=0}^{N-1} \text{pul}_k \exp\left(-in \frac{2\pi}{T} kT_s\right) \\ &= T_s \sum_{k=0}^{N-1} \text{pul}_k \exp\left(-i \frac{2\pi}{N} nk\right). \end{aligned}$$

Define the DFT

$$\text{PUL}_n = \sum_{k=0}^{N-1} \text{pul}_k e^{-i \frac{2\pi}{N} nk}.$$

Then the preceding right-hand side reads

$$T_s \sum_{k=0}^{N-1} \text{pul}_k e^{-i \frac{2\pi}{N} nk} = T_s \text{PUL}_n.$$

Therefore (12.9) at $\omega = \omega_n$ becomes

$$\sum_{m=-\infty}^{\infty} \text{PUL}(n\omega_0 - m\Omega_s) = T_s \text{PUL}_n. \quad (12.10)$$

Theorem 53 (Relationship between aliased CFT and FFT for pulses). *Let $\text{pul}(t)$ be carried by the sampled window $0 \leq t < T$, and let $\text{PUL}(\omega)$ denote its Fourier transform. Sample $\text{pul}(t)$ at $t_k = kT_s$ for $0 \leq k \leq N-1$, where $T = NT_s$, and form the DFT*

$$\text{PUL}_n = \sum_{k=0}^{N-1} \text{pul}_k e^{-i \frac{2\pi}{N} nk}.$$

Let $\Omega_s = 2\pi/T_s$ and $\omega_0 = \Omega_s/N = 2\pi/T$. Then, for all integers n ,

$$\sum_{m=-\infty}^{\infty} \text{PUL}(n\omega_0 - m\Omega_s) = T_s \text{PUL}_n. \quad (12.11)$$

Equivalently, the aliased, sampled CFT spectrum equals the FFT spectrum scaled by T_s .

Proof. The derivation above computes the transform of $\text{pul}(t)\delta_s(t)$ in two ways: Method A (sampling and summation), equation (12.7), and Method B (convolution in ω), equation (12.8). Equating them gives (12.9). Sampling at $\omega_n = n\omega_0$ with $\omega_0 = \Omega_s/N$ and $T_s = T/N$ yields (12.10). The identity is independent of the sign convention as long as that convention is applied consistently to all occurrences of PUL. $\square$

As observed in (12.5), for sufficiently large Ω_s one has

$$\sum_{m=-\infty}^{\infty} \text{PUL}(n\omega_0 - m\Omega_s) \approx \text{PUL}(n\omega_0), \quad -\frac{N}{2} < n < \frac{N}{2},$$

with the approximation improving for each fixed n as Ω_s increases (equivalently, T_s decreases and N increases for fixed T). Combining this with Theorem 53 gives:

Corollary 11 (Using the FFT to estimate a pulse Fourier transform). *Under the hypotheses of Theorem 53, for $-N/2 < n < N/2$,*

$$\text{PUL}(n\omega_0) \approx T_s \text{PUL}_n = \frac{T}{N} \text{PUL}_n, \quad (12.12)$$

and, for each fixed index n , the approximation error decreases as N increases (for fixed T).

Proof. The identity (12.11) holds for all n . By the aliasing approximation (12.5), in the central region

$$\sum_{m=-\infty}^{\infty} \text{PUL}(n\omega_0 - m\Omega_s) \approx \text{PUL}(n\omega_0).$$

Substituting this into (12.11) gives

$$\text{PUL}(n\omega_0) \approx T_s \text{PUL}_n.$$

Using $T_s = T/N$ yields (12.12). For each fixed n , the decay of $\text{PUL}(\omega)$ ensures that increasing Ω_s (and hence N) decreases the contribution of neighboring replicas and thus the aliasing error. $\square$

This is consistent with the result obtained from the rectangular rule in Theorem 51. The two derivations emphasize complementary aspects: one from numerical quadrature, the other from spectral aliasing.

Thus the FFT pulse estimate can be read in two equivalent ways:

- as a rectangular-rule approximation to the integral defining $F(n\omega_0)$;
- as an exact sample of an aliased spectrum, which is close to the original spectrum only when the shifted copies are small in the central band.

The second view explains why increasing N helps only when it also moves the aliases far enough away or samples a spectrum that decays fast enough.

12.3.2 Periodic functions

Now let $\text{per}(t)$ be periodic with period T_0 , and let $\text{PER}(n)$ denote its CFT (Fourier-series) coefficients. As shown in Chapter 10, if $\text{per}(t)$ is obtained by periodically repeating a pulse $\text{pul}(t)$ carried by the half-open window $[0, T_0)$, then

$$\text{PER}(n) = \frac{1}{T_0} \text{PUL}(n\omega_0), \quad \omega_0 = \frac{2\pi}{T_0}. \quad (12.13)$$

Replacing n by $n - mN$ yields

$$\text{PER}(n - mN) = \frac{1}{T_0} \text{PUL}((n - mN)\omega_0).$$

Summing over m gives

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) = \frac{1}{T_0} \sum_{m=-\infty}^{\infty} \text{PUL}((n - mN)\omega_0). \quad (12.14)$$

On the other hand, by Theorem 53 applied to $\text{pul}(t)$ and $\text{PUL}(\omega)$, one has

$$\sum_{m=-\infty}^{\infty} \text{PUL}((n - mN)\omega_0) = T_s \text{PUL}_n,$$

with $T_0 = T = NT_s$ and $\omega_0 = 2\pi/T_0$. Thus

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) = \frac{1}{T_0} T_s \text{PUL}_n = \frac{1}{N} \text{PUL}_n.$$

Now consider numerical samples of $\text{per}(t)$ over one period:

$$t_k = kT_s, \quad T_s = \frac{T_0}{N}, \quad 0 \leq k \leq N-1, \quad \text{per}_k = \text{per}(t_k).$$

Since $\text{per}(t)$ coincides with $\text{pul}(t)$ at the sampled points $t_k = kT_s$, $0 \leq k \leq N-1$, one has $\text{per}_k = \text{pul}_k$, and hence their DFTs coincide:

$$\text{PER}_n = \text{PUL}_n.$$

Therefore,

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) = \frac{1}{N} \text{PER}_n.$$

Theorem 54 (Relationship between aliased CFT and FFT for periodic functions). *Let $\text{per}(t)$ be periodic with period T_0 , and let $\text{PER}(n)$ denote its CFT coefficients. Sample $\text{per}(t)$ over one period at $t_k = kT_s$ with $T_s = T_0/N$, $0 \leq k \leq N-1$, and form the DFT*

$$\text{PER}_n = \sum_{k=0}^{N-1} \text{per}_k e^{-i \frac{2\pi}{N} nk}.$$

Then, for all integers n ,

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) = \frac{1}{N} \text{PER}_n.$$

In other words, the aliased CFT line spectrum equals the FFT spectrum scaled by $1/N$.

Proof. Since $\text{per}(t)$ is periodic with period T_0 and coincides with $\text{pul}(t)$ on the sampled window, Theorem 53 applied to $\text{pul}(t)$ yields

$$\sum_{m=-\infty}^{\infty} \text{PUL}((n - mN)\omega_0) = T_s \text{PUL}_n, \quad \omega_0 = \frac{2\pi}{T_0}.$$

Using the relation (12.13) between $\text{PER}(n)$ and $\text{PUL}(n\omega_0)$, one obtains (12.14). Combining, and using $T_0 = NT_s$, yields

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) = \frac{T_s}{T_0} \text{PUL}_n = \frac{1}{N} \text{PUL}_n.$$

Since $\text{per}_k = \text{pul}_k$ for $0 \leq k \leq N-1$, their DFTs are equal: $\text{PER}_n = \text{PUL}_n$. Substituting this gives the desired relation. $\square$

Combining Theorem 54 with the aliasing approximation (12.6),

Corollary 12 (Using the FFT to estimate Fourier-series coefficients). *Under the hypotheses of Theorem 54, for $-N/2 < n < N/2$,*

$$\text{PER}(n) \approx \frac{1}{N} \text{PER}_n, \tag{12.15}$$

and, for each fixed coefficient index n , the approximation error decreases as N increases.

Proof. By (12.6),

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) \approx \text{PER}(n), \quad -\frac{N}{2} < n < \frac{N}{2}.$$

By Theorem 54,

$$\sum_{m=-\infty}^{\infty} \text{PER}(n - mN) = \frac{1}{N} \text{PER}_n.$$

Combining the two yields (12.15). For a fixed coefficient index n , the decay of $\text{PER}(n - mN)$ as $|mN|$ grows guarantees that the error in (12.6), and hence in (12.15), decreases as N increases. $\square$

This result is consistent with Theorem 52, which was derived from the rectangular rule.

12.4 Inverting CFT spectra via the FFT

The previous sections describe how the FFT can be used to approximate CFT spectra from time-domain samples. A complementary problem is: given a CFT spectrum (either a Fourier transform $PUL(\omega)$ or a Fourier series spectrum $PER(n)$), how can one invert it numerically to the time domain using the FFT?

A direct approach would be to sample the CFT spectrum at N points, truncate, and apply the DFT synthesis formula. However, this mixes CFT-based analysis with DFT-based synthesis and introduces truncation and aliasing effects that may be difficult to control.

A more systematic approach is to construct an *approximate FFT-spectrum* by explicitly aliasing the CFT spectrum, so that the constructed discrete spectrum matches as closely as possible the FFT spectrum that would arise from sampling the corresponding time-domain signal. This is then inverted by the FFT synthesis formula in a consistent manner.

12.4.1 Fourier series spectra

Suppose $PER(n)$ is given explicitly by a CFT formula. By Theorem 54, the FFT spectrum corresponding to N time samples of $per(t)$ is

$$PER_n^{(N)} = N \sum_{m=-\infty}^{\infty} PER(n - mN), \quad n = 0, \dots, N-1. \quad (12.16)$$

In practice, the infinite sum in (12.16) is truncated:

$$PER_n^{(N)} \approx N \sum_{m=-a}^a PER(n - mN), \quad (12.17)$$

for some integer $a \geq 0$.

Definition 32 (Aliasing level). *The integer a in (12.17) is called the aliasing level. It specifies how many shifted replicas on each side of the central spectrum are included in the aliasing sum.*

Given N and a , formula (12.17) defines an approximate FFT spectrum $PER_n^{(N)}$. One then applies the FFT synthesis step to $\{PER_n^{(N)}\}_{n=0}^{N-1}$ to obtain a sampled version of $per(t)$ over one period. The accuracy is governed by:

- the decay rate of $PER(n)$ (which controls the truncation error in m),
- the choice of N (which controls the spacing of aliases).

If $PER(n)$ decays rapidly and a and N are chosen suitably, the time-domain reconstruction obtained by FFT synthesis can approximate the original periodic function closely.

12.4.2 Fourier transform spectra

A similar procedure applies to Fourier transforms. Suppose $PUL(\omega)$ is given explicitly. By Theorem 53, the FFT spectrum for a sampling frequency Ω_s and N points is

$$PUL_n^{(N)} = \frac{1}{T_s} \sum_{m=-\infty}^{\infty} PUL((n - mN)\omega_0), \quad \omega_0 = \frac{\Omega_s}{N}, \quad T_s = \frac{2\pi}{\Omega_s}.$$

Truncating the infinite sum at level a ,

$$PUL_n^{(N)} \approx \frac{1}{T_s} \sum_{m=-a}^a PUL((n - mN)\omega_0). \quad (12.18)$$

In practice one separates real and imaginary parts,

$$\text{PUL}(\omega) = A(\omega) + iB(\omega),$$

computes

$$A((n - mN)\omega_0), \quad B((n - mN)\omega_0),$$

sums over $m = -a, \dots, a$, and forms $\text{PUL}_n^{(N)}$ as in (12.18). The resulting discrete spectrum is then used as input to FFT synthesis to reconstruct samples of $\text{pul}(t)$.

Care is required in choosing T (the effective observation window) and T_s so that time-domain aliasing is controlled. In particular, the periodicity of the DFT in the time domain implies that the reconstructed time-domain signal represents a T -periodic extension of the original $\text{pul}(t)$. To avoid overlap between shifted copies in time, T must be chosen sufficiently large relative to the effective time support of $\text{pul}(t)$.

In summary, this chapter established the following points:

- The rectangular rule connects the CFT analysis integrals to the FFT analysis sums (Theorems 51 and 52).
- Aliasing of CFT spectra leads to exact identities relating aliased CFT spectra and FFT spectra (Theorems 53 and 54).
- Restricting attention to central index ranges and exploiting the decay of CFT spectra yields controlled approximations from FFT outputs to CFT spectra (Corollaries 11 and 12).
- The same aliasing mechanism can be used in the opposite direction, to construct discrete FFT spectra from closed-form CFT expressions and then invert them to the time domain by FFT synthesis.

13 The Errors in Fast Fourier Transform Estimation

13.1 Introduction

In the previous chapter, the fast Fourier transform (FFT) was related to its continuous Fourier transform (CFT) counterparts through the following aliasing identities.

Pulses. Let $f(t)$ be a finite-span pulse carried by the sampled window $[0, T)$, with Fourier transform $F(\omega)$, and let F_n be the FFT spectrum obtained from N uniform samples of $f(t)$ over $[0, T)$. Then

$$\sum_{m=-\infty}^{\infty} F[(n - mN)\omega_0] = T_s F_n, \quad \omega_0 = \frac{2\pi}{T}, \quad T_s = \frac{T}{N}, \quad (13.1)$$

where $n \in \mathbb{Z}$.

Periodic functions. Let $f_p(t)$ be periodic of period T , with complex Fourier-series coefficients $F_p(n)$, and let F_n be the FFT line spectrum obtained from N samples of $f_p(t)$ over one period. Then

$$\sum_{m=-\infty}^{\infty} F_p(n - mN) = \frac{1}{N} F_n, \quad n \in \mathbb{Z}. \quad (13.2)$$

From these alias relations, the following estimation statements were obtained in Chapter 12.

Pulse spectra. For $-N/2 < n < N/2$,

$$F(n\omega_0) \approx T_s F_n = \frac{T}{N} F_n.$$

Fourier-series coefficients. For $-N/2 < n < N/2$,

$$F_p(n) \approx \frac{1}{N} F_n.$$

Empirical rules for bounding the errors in these approximations exist in the literature. The purpose of this chapter is to derive algebraic expressions for the *relative errors* which are exact for a certain class of pulses and periodic functions, and asymptotically valid for a broader class. In addition, it is shown that the estimation errors can be reduced by choosing appropriate positions for the FFT sampling instants.

The errors in this chapter are not roundoff errors and not FFT algorithm errors. They are aliasing errors. The FFT returns the exact DFT of the samples supplied to it; the approximation enters when we interpret those DFT values as samples of a continuous Fourier transform or of an infinite Fourier-series spectrum. The remote copies of the CFT spectrum wrap back into the central range, and the relative error measures the size of those wrapped copies compared with the desired central value.

13.2 Relative error

For a generic scalar quantity, the relative error of an estimate is defined as

$$\text{relative error} = \frac{\text{estimate} - \text{exact}}{\text{exact}}.$$

Pulse spectra. For the pulse case, with exact spectrum $F(\omega)$ and FFT-based estimate $T_s F_n$ at $\omega = n\omega_0$, define

$$E_n^{(\text{pulse})} = \frac{T_s F_n - F(n\omega_0)}{F(n\omega_0)}.$$

Using (13.1), this can be written as

$$E_n^{(\text{pulse})} = \frac{\sum_{m=-\infty}^{\infty} F[(n-mN)\omega_0] - F(n\omega_0)}{F(n\omega_0)}.$$

The $m = 0$ term cancels, leaving

$$E_n^{(\text{pulse})} = \frac{\sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} F[(n-mN)\omega_0]}{F(n\omega_0)}. \quad (13.3)$$

Periodic spectra. Similarly, for the periodic case, with exact coefficient $F_p(n)$ and FFT-based estimate F_n/N , define

$$E_n^{(\text{per})} = \frac{F_n/N - F_p(n)}{F_p(n)}.$$

Using (13.2),

$$E_n^{(\text{per})} = \frac{\sum_{m=-\infty}^{\infty} F_p(n-mN) - F_p(n)}{F_p(n)} = \frac{\sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} F_p(n-mN)}{F_p(n)}.$$

In both cases it is understood that the denominators $F(n\omega_0)$ and $F_p(n)$ are nonzero in the range of interest.

13.3 Canonical pulses and order of continuity

Definition 33 (Canonical- k pulse). Let $f(t)$ be a finite-span pulse sampled with left-endpoint uniform sampling at step T_s , and suppose:

- all sampling instants are $t_j = jT_s$, $0 \leq j \leq N-1$;
- at any sampling instant coinciding with a discontinuity of f , the sample value supplied to the FFT is the average of the one-sided limits (half-value);
- the sample at $t = 0$ is the average of $f(0)$ and $f(T)$.

If the $(k+1)$ st derivative $f^{(k+1)}(t)$ consists only of Dirac deltas, each located at a sampling instant $t = g_j T_s$ with integer g_j , then f is called canonical- k .

This definition packages two requirements. First, the waveform is simple enough that enough differentiations reduce it to impulses. Second, the impulse locations land exactly on FFT sample instants. That alignment is what makes the alias terms share the same exponential factors and eventually collapse to a simple function of $Z = n/N$.

Example 12. Any rectangular pulse $\text{Rect}((t - t_0)/T)$ whose endpoints lie exactly at sampling instants is canonical-0, since its first derivative consists only of jumps at those instants and hence is a finite sum of Dirac deltas located at sampling instants.

Example 13. A piecewise-linear pulse made of ramps and constant segments such that all break points (changes of slope) lie at sampling instants, and whose second derivative consists of Dirac deltas at those break points, is canonical-1.

Any periodic function $f_p(t)$ of period T can be viewed as the periodic repetition of a single-period defining pulse $f(t)$ on the half-open interval $[0, T)$. Accordingly, f_p is called canonical- k if its defining pulse f is canonical- k in the sense of Definition 33.

For intuition, canonical-0 means the signal itself jumps; canonical-1 means the signal is continuous but its slope jumps; canonical-2 means the slope is continuous but the curvature jumps, and so on. Each higher order usually pushes more spectral energy toward low frequencies.

Definition 34 (Order of continuity). A pulse $f(t)$ is said to be continuous of order k (or continuous- k) if f and its derivatives up to order $k - 1$ are continuous everywhere, while the k th derivative has discontinuities.

Example 14.

- Rectangular pulses are continuous-0.
- Canonical- k pulses are continuous- k .
- A triangular pulse is continuous-1: f is continuous, while $f'(t)$ has jump discontinuities.

For continuous-0 pulses, one typically has $|F(\omega)| = O(1/|\omega|)$ as $|\omega| \rightarrow \infty$; for continuous-1 pulses, typically $|F(\omega)| = O(1/\omega^2)$.

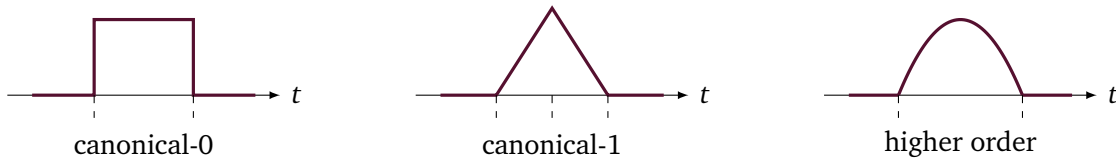


Figure 34: Canonical functions have break points aligned with the sampling grid. Higher continuity usually gives faster spectral decay and smaller aliasing error.

13.4 Error expressions for canonical pulses and periodic functions

13.4.1 Canonical- k pulses

Let $f(t)$ be a canonical- k pulse with $(k + 1)$ st derivative

$$f^{(k+1)}(t) = \sum_{j=1}^r p_j \delta(t - g_j T_s),$$

where $p_j \in \mathbb{C}$ are the weights, $g_j \in \{0, 1, \dots, N - 1\}$ are the sample indices at which the Dirac deltas occur, and r is finite. Applying the Fourier transform yields

$$(i\omega)^{k+1} F(\omega) = \sum_{j=1}^r p_j e^{-i\omega g_j T_s},$$

so that

$$F(\omega) = \frac{1}{(i\omega)^{k+1}} \sum_{j=1}^r p_j e^{-i\omega g_j T_s}. \quad (13.4)$$

Substitute (13.4) into the relative-error expression (13.3) for pulses:

$$\begin{aligned} E_n^{(\text{pulse})} &= \frac{\sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} F((n-mN)\omega_0)}{F(n\omega_0)} \\ &= \frac{\sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{1}{(i(n-mN)\omega_0)^{k+1}} \sum_{j=1}^r p_j e^{-i(n-mN)\omega_0 g_j T_s}}{\frac{1}{(in\omega_0)^{k+1}} \sum_{j=1}^r p_j e^{-in\omega_0 g_j T_s}}. \end{aligned} \quad (13.5)$$

Note that

$$(n-mN)\omega_0 g_j T_s = (n-mN) \frac{2\pi}{T} g_j T_s = (n-mN) \frac{2\pi}{N} g_j = n \frac{2\pi}{N} g_j - mN \frac{2\pi}{N} g_j = n \frac{2\pi}{N} g_j - 2\pi m g_j.$$

Hence

$$e^{-i(n-mN)\omega_0 g_j T_s} = e^{-in\omega_0 g_j T_s} e^{+i2\pi m g_j} = e^{-in\omega_0 g_j T_s},$$

since $e^{i2\pi m g_j} = 1$ for integers m, g_j . Therefore, the exponential factors in numerator and denominator of (13.5) cancel, giving

$$E_n^{(\text{pulse})} = \frac{\sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{1}{(i(n-mN)\omega_0)^{k+1}}}{\frac{1}{(in\omega_0)^{k+1}}} = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{n}{n-mN} \right)^{k+1}. \quad (13.6)$$

This cancellation is the reason canonical errors are universal. Once the break points sit on the sampling grid, the detailed pulse heights and break locations drop out of the relative error. What remains is only the smoothness order k and the normalized bin location $Z = n/N$.

13.4.2 Canonical- k periodic functions

Let $f_p(t)$ be periodic with period T , and define its single-period restriction as $f(t)$ on $[0, T)$. Assume $f(t)$ is canonical- k and denote by $F(\omega)$ its Fourier transform. The CFT Fourier-series coefficients of $f_p(t)$ are

$$F_p(n) = \frac{1}{T} F(n\omega_0), \quad \omega_0 = \frac{2\pi}{T}.$$

Repeating the same steps for $F_p(n)$ as for $F(\omega)$, one obtains the same error expression (13.6) for $E_n^{(\text{per})}$.

We summarize these observations as follows.

Theorem 55 (Error expression for canonical- k functions). *Let either*

- $f(t)$ be a canonical- k pulse carried by the sampled window $[0, T)$, or
- $f_p(t)$ be a periodic function of period T whose defining pulse on $[0, T)$ is canonical- k .

Let $F(\omega)$ be the CFT of $f(t)$ and $F_p(n)$ the CFT coefficients of $f_p(t)$. Let F_n denote the FFT spectrum obtained from N left-endpoint uniform samples, with half-values supplied at discontinuities. Assume $F(n\omega_0) \neq 0$ (pulse) or $F_p(n) \neq 0$ (periodic) in the range of interest. Then, for both pulses and periodic functions, the relative error in the n th spectral estimate is

$$E_n = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{n}{n-mN} \right)^{k+1}, \quad -\frac{N}{2} < n < \frac{N}{2}. \quad (13.7)$$

At $n = 0$, this formula is interpreted by its limiting value, which is zero when the exact DC denominator is nonzero.

13.5 Properties of the error for canonical- k functions

Introduce the normalized index

$$Z = \frac{n}{N}.$$

The central index range $-N/2 < n < N/2$ corresponds to $|Z| < \frac{1}{2}$.

From (13.7),

$$E_n = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{n}{n-mN} \right)^{k+1} = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{Z}{Z-m} \right)^{k+1} \equiv E(Z, k). \quad (13.8)$$

We now record some basic properties, focusing first on small values of k .

Canonical-0 case. For $k = 0$,

$$E(Z, 0) = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{Z}{Z-m}, \quad |Z| < \frac{1}{2}. \quad (13.9)$$

This series behaves like $\sum_m Z/m$ for large $|m|$, so the individual terms decay as $1/m$ and the symmetric summation must be interpreted in the principal-value sense. Rewriting symmetrically,

$$E(Z, 0) = 2Z^2 \sum_{m=1}^{\infty} \frac{1}{Z^2 - m^2}, \quad |Z| < \frac{1}{2},$$

and the terms then decay as $1/m^2$.

Canonical-1 case. For $k = 1$,

$$E(Z, 1) = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{Z}{Z-m} \right)^2.$$

Canonical-2 and canonical-3. Similarly, for $k = 2$ and $k = 3$ one has

$$E(Z, 2) = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{Z}{Z-m} \right)^3,$$

$$E(Z, 3) = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{Z}{Z-m} \right)^4.$$

In all cases, the series converge absolutely for $|Z| < \frac{1}{2}$ when written in symmetric form (pairing $\pm m$).

From these series representations one can infer:

- For $k = 0, 1$, the series converge essentially like $\sum 1/m^2$ once written in symmetric form, and for $k \geq 2$ the decay is at least $\sum 1/m^3$.
- The errors $E(Z, k)$ are real for canonical- k pulses and periodic functions.
- $E(0, k) = 0$ for all k by the limiting form of the alias sum, so the error in the DC spectral element is zero when the exact DC denominator is nonzero.
- For even k , $E(Z, k)$ is negative; for odd k , $E(Z, k)$ is positive (for $0 < Z < \frac{1}{2}$).
- $E(Z, k)$ is an even function of Z , hence $E(-Z, k) = E(Z, k)$.

13.6 Closed-form Z-curves via trigonometric identities

The preceding series representations can be expressed in closed form using classical identities from complex analysis. A standard result (see, e.g., Ahlfors or MacRobert) is

$$\sum_{m=1}^{\infty} \frac{2Z^2}{Z^2 - m^2} = \pi Z \cot(\pi Z) - 1, \quad Z \notin \mathbb{Z}.$$

This series is precisely the symmetric version of (13.9), so

$$E(Z, 0) = \pi Z \cot(\pi Z) - 1, \quad |Z| < \frac{1}{2}. \quad (13.10)$$

Using the standard partial-fraction identity

$$\sum_{m=-\infty}^{\infty} \frac{1}{(Z - m)^2} = \frac{\pi^2}{\sin^2(\pi Z)},$$

and removing the $m = 0$ term gives

$$\sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{1}{(Z - m)^2} = \frac{\pi^2}{\sin^2(\pi Z)} - \frac{1}{Z^2}.$$

In terms of $E(Z, 1)$,

$$E(Z, 1) = Z^2 \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{1}{(Z - m)^2} = \frac{\pi^2 Z^2}{\sin^2(\pi Z)} - 1, \quad |Z| < \frac{1}{2}.$$

Further differentiation of the full sums, followed by removal of the $m = 0$ term, yields the higher-order sums:

$$\begin{aligned} E(Z, 2) &= \pi^3 Z^3 \csc^2(\pi Z) \cot(\pi Z) - 1, \quad |Z| < \frac{1}{2}, \\ E(Z, 3) &= \pi^4 Z^4 \csc^2(\pi Z) \left[\cot^2(\pi Z) + \frac{1}{3} \right] - 1, \quad |Z| < \frac{1}{2}. \end{aligned}$$

Theorem 56 (Closed-form error expressions for canonical- k functions). *For canonical- k pulses and periodic functions, the relative error $E(Z, k)$, defined in (13.8), admits the following closed forms for $|Z| < \frac{1}{2}$:*

$$E(Z, 0) = \pi Z \cot(\pi Z) - 1,$$

$$E(Z, 1) = \frac{\pi^2 Z^2}{\sin^2(\pi Z)} - 1,$$

$$E(Z, 2) = \pi^3 Z^3 \csc^2(\pi Z) \cot(\pi Z) - 1,$$

$$E(Z, 3) = \pi^4 Z^4 \csc^2(\pi Z) \left[\cot^2(\pi Z) + \frac{1}{3} \right] - 1.$$

In practice, these formulas are useful for plotting *log-linear* Z -curves of $|E(Z, k)|$ versus Z for $k = 0, 1, 2, 3$. Such curves can be used to read off the relative error corresponding to a given ratio $Z = n/N$.

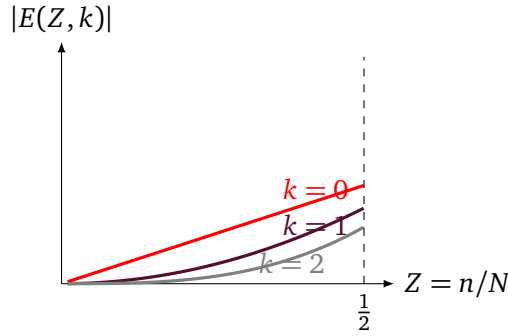


Figure 35: Qualitative Z -curves: for fixed $Z = n/N$, smoother canonical functions have smaller relative aliasing error.

Some qualitative features observed from these curves include:

- $|E(Z, 1)|$ is slightly larger than $|E(Z, 0)|$ over much of the interval $0 < Z < \frac{1}{2}$, despite faster decay of the corresponding CFT spectrum for canonical-1 functions.
- If n is fixed and N increases, then $Z = n/N \rightarrow 0$ and $E(Z, k) \rightarrow 0$. Hence the error in the n th spectral element can be made arbitrarily small by increasing N for fixed n .
- If $Z = n/N$ is kept fixed while N increases, the error $E(Z, k)$ remains fixed; enlarging N does not reduce the relative error for that fixed ratio Z . For example, as $Z \rightarrow \frac{1}{2}^-$:

$$\begin{aligned} E\left(\frac{1}{2}, 0\right) &\rightarrow -100.00\%, & E\left(\frac{1}{2}, 1\right) &\rightarrow 146.74\%, \\ E\left(\frac{1}{2}, 2\right) &\rightarrow -100.00\%, & E\left(\frac{1}{2}, 3\right) &\rightarrow 102.94\%. \end{aligned}$$

13.7 Asymptotic behavior for noncanonical functions

The previous results were derived for canonical functions. Many practically relevant signals are not canonical but satisfy weaker smoothness conditions.

Theorem 57 (Asymptotic error for continuous-0 pulses). *Let $f(t)$ be finite-span and piecewise continuous of order 0 on the sampled window $[0, T)$, with the following properties:*

- $f(t) = 0$ outside the sampled window $[0, T)$.

- $f(t)$ is twice continuously differentiable on each open interval of continuity.
- All discontinuities (break points) lie at sampling instants $t = kT_s$.

Let $F(\omega)$ be the Fourier transform of $f(t)$, and suppose $F(n\omega_0) \neq 0$ in the range of interest. Let $E_n(Z)$ be the relative error in the FFT estimate of $F(n\omega_0)$ when N samples are used, with $Z = n/N$ fixed. Then, for each fixed Z ,

$$\lim_{N \rightarrow \infty} E_n(Z) = E(Z, 0),$$

where $E(Z, 0)$ is the canonical-0 error function from Theorem 56.

Thus, for noncanonical but continuous-0 pulses (or periodic functions with continuous-0 defining pulses), the canonical-0 error curve still describes the asymptotic behavior of the relative error as $N \rightarrow \infty$, provided $Z = n/N$ is held fixed.

The same phenomenon occurs for higher orders of continuity.

Theorem 58 (Asymptotic error for continuous- k pulses). *Let $k \geq 1$. Let $f(t)$ be finite-span and continuous of order k on the sampled window $[0, T)$, with the following properties:*

- $f(t) = 0$ outside the sampled window $[0, T)$.
- The $(k-1)$ st derivative $f^{(k-1)}(t)$ is $(k+2)$ times continuously differentiable on each open interval of continuity.
- All break points lie at sampling instants $t = \ell T_s$ for integers ℓ .

Let $F(\omega)$ be the Fourier transform of $f(t)$, and suppose $F(n\omega_0) \neq 0$ in the range of interest. Let $E_n(Z)$ be the relative error in the FFT estimate at $\omega = n\omega_0$ with $Z = n/N$ fixed. Then

$$\lim_{N \rightarrow \infty} E_n(Z) = E(Z, k),$$

where $E(Z, k)$ is the canonical- k error function defined in (13.8) and given in closed form by Theorem 56 for $k = 0, 1, 2, 3$.

In both theorems, the statements extend to periodic functions by interpreting $f(t)$ as the single-period defining pulse and $F(\omega)$ as its finite-interval Fourier transform.

13.8 Sampling when break points are not at sampling instants

The previous asymptotic results assumed that all break points lie at sampling instants. In many practical scenarios, break points fall in the interior of sampling intervals. This modifies the error expressions.

A representative effect can be illustrated as follows. For a canonical- k function, the error series has the form

$$E(Z, k) = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \left(\frac{Z}{Z-m} \right)^{k+1}.$$

If the sampling grid is shifted such that break points lie at the *centers* of sampling intervals rather than at sampling instants, the contribution of each alias term acquires an additional factor $(-1)^m$. The error then becomes

$$\tilde{E}(Z, k) = \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} (-1)^m \left(\frac{Z}{Z-m} \right)^{k+1}.$$

In symmetric form,

$$\tilde{E}(Z, k) = Z^{k+1} \sum_{m=1}^{\infty} (-1)^m \left[\frac{1}{(Z-m)^{k+1}} + \frac{1}{(Z+m)^{k+1}} \right],$$

and the summand now alternates in sign. This reduces the absolute magnitude of the sum compared to the nonalternating case. Consequently, placing all break points at the centers of sampling intervals systematically reduces the relative error.

The same mechanism appears in concrete examples such as a gated cosine or a rectangular pulse, where numerical evaluations show that shifting the sampling grid so that discontinuities fall at midpoints can reduce error magnitudes by roughly a factor of two over a broad range of Z .

13.9 Error correction for canonical functions

For canonical functions the error expressions are exact, so one can invert them to correct FFT estimates and recover the exact CFT values.

13.9.1 Rectangular pulse

Consider a rectangular pulse of width τ inside an FFT observation window of length T :

$$f(t) = \text{Rect}\left(\frac{t-t_c}{\tau}\right), \quad 0 < \tau < T,$$

with its endpoints $t_c - \tau/2$ and $t_c + \tau/2$ lying on sampling instants. Its Fourier transform is

$$F(\omega) = \tau e^{-i\omega t_c} \text{sinc}\left(\frac{\omega\tau}{2}\right).$$

Here T determines the FFT grid $\omega_0 = 2\pi/T$, while τ is the pulse width. Keeping these two lengths separate avoids confusing the observation window with the physical width of the pulse. For a canonical-0 sampling (endpoints at sampling instants, half-values used), the alias relation (13.1) and the canonical-0 error expression (13.10) imply

$$T_s F_n = F(n\omega_0) [1 + E(Z, 0)] = F(n\omega_0) [\pi Z \cot(\pi Z)],$$

so that

$$F(n\omega_0) = \frac{T_s F_n}{\pi Z \cot(\pi Z)}, \quad Z = \frac{n}{N}. \quad (13.11)$$

At $Z = 0$, the denominator is interpreted by its limiting value 1. In words, dividing the FFT estimate $T_s F_n$ by $\pi Z \cot(\pi Z)$ yields the exact value of $F(n\omega_0)$ at each n for which $F(n\omega_0) \neq 0$.

In numerical experiments, even for relatively small N (e.g. $N = 64$), applying (13.11) recovers the exact transform samples of such a grid-aligned rectangular pulse to machine precision, apart from bins where the exact transform sample is zero and relative error is undefined.

13.10 Effect of zero padding on estimation errors

In many applications, the number of available samples is not a power of two. A common practice is to append zeros to the data in order to reach the next power of two for efficient FFT computation. This operation is often referred to as *zero padding* or *adding white space*.

Zero padding also refines the frequency sampling grid and can provide a smoother visual representation of the spectrum. However, when interpreting FFT outputs as *estimates* of continuous Fourier transforms, the following fact must be noted:

Zero padding does not add spectral information; it only changes the frequency grid on which the same underlying estimator is sampled.

In the error formulas of this chapter, the relative error is governed by the normalized index $Z = n/N$. Therefore, at fixed Z the canonical error curves $E(Z, k)$ are unchanged by zero padding. What changes is the set of available frequency bins (finer spacing in ω), not the intrinsic estimation law.

Thus, while zero padding is useful for display and interpolation of discrete spectra, it should be used with care if the goal is to approximate continuous-time Fourier transforms with controlled relative error.

The practical rule is:

Use zero padding to see more points between existing spectral samples, not to claim that the original CFT estimate has become more accurate at a fixed normalized location Z .

To reduce the actual CFT estimation error at a particular physical frequency, one must improve the sampling model itself: increase the observation window or sampling rate in a way that reduces the relevant aliasing terms, or use an appropriate correction formula when the signal is canonical.

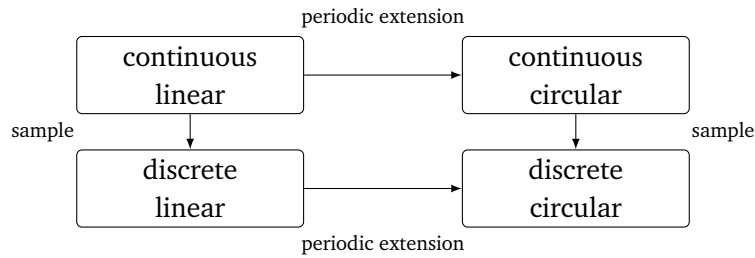
14 The Four Kinds of Convolution

14.1 Introduction

The convolution integral from Chapter 7 is only one member of a larger family. When FFTs enter the calculation, it is important to know which kind of convolution is being performed, because the FFT is inherently periodic in its discrete index.

There are four useful cases:

1. *continuous linear convolution*: finite-span continuous pulses,
2. *continuous circular convolution*: periodic continuous waveforms,
3. *discrete linear convolution*: finite-length discrete pulses,
4. *discrete circular convolution*: periodic discrete sequences.



The FFT computes the lower-right case directly.

Figure 36: The four convolution settings. FFT convolution is discrete circular convolution; the other cases are obtained from it only after the correct sampling, padding, and scaling choices.

14.2 The Four Definitions

The four cases differ in two independent ways: whether the variable is continuous or discrete, and whether the data live on a line or on a circle. The formulas are:

$$\text{continuous linear:} \quad f(t) = \int_{-\infty}^{\infty} g(\tau)h(t-\tau) d\tau,$$

$$\text{continuous circular:} \quad f_p(t) = \int_0^{T_0} g_p(\tau)h_p(t-\tau) d\tau, \quad g_p, h_p \text{ } T_0\text{-periodic},$$

$$\text{discrete linear:} \quad f_k = \sum_{m=-\infty}^{\infty} g_m h_{k-m},$$

$$\text{discrete circular:} \quad f_k = \sum_{m=0}^{N-1} g_m h_{k-m}, \quad \text{subscripts read modulo } N.$$

The word *linear* means that shifts happen along an ordinary line. The word *circular* means that the index wraps around a period. This distinction is not cosmetic: the FFT always works with periodic index sets, so the raw convolution computed by an FFT is circular even when the desired mathematical operation is linear.

An easy way to choose the correct case is to ask two questions:

1. Are the original signals continuous functions or sampled sequences?
2. Are they finite-span objects on a line, or one period of periodic objects?

The answers select one of the four formulas above. If the desired answer is linear but the computation is performed by FFT, then padding and the span restriction are the bridge from the lower-right case back to the linear case.

14.3 Span and FFT Window Length

For a finite-span pulse, the *span time* is the elapsed time from the beginning of the pulse to the end of the pulse. For a finite-length sequence, the *span count* is the number of samples from the first sample in the pulse to the last, including any internal zeros. We use T_σ for span time and C_σ for span count.

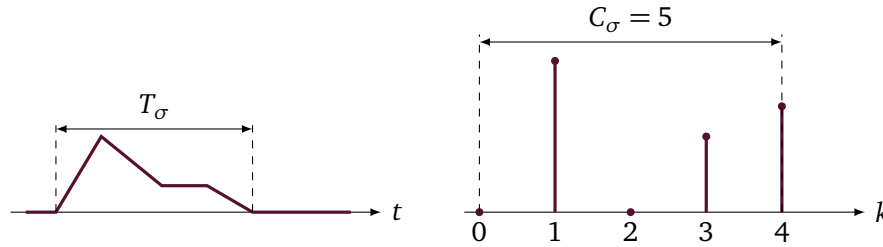


Figure 37: Span time for a continuous pulse and span count for a discrete pulse. Internal zeros belong to the span count.

For a periodic discrete sequence, the span count is measured around the circle. If a period has N slots and its longest uninterrupted run of zeros has length Z , then the circular span is

$$C_\sigma = N - Z.$$

Zeros at the end and beginning of the written period count as one uninterrupted run, because those ends touch on the circle. This is why the same displayed list of samples can have a different span depending on whether it is being treated as linear or periodic.

If a finite-span pulse is sampled at spacing T_s and placed in an FFT data window of length N , the associated time window is

$$T = NT_s.$$

Samples beyond the pulse support are zero padding, or white space. The padding is not extra information; it gives circular convolution enough room to avoid wrapping the output back onto itself.

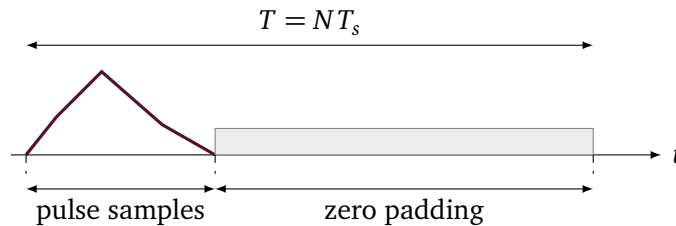


Figure 38: An FFT window for a finite-span pulse contains the pulse samples plus enough zero padding to make the circular computation behave like a linear one.

14.4 Linear and Circular Discrete Convolution

For finite or absolutely summable sequences, the discrete linear convolution is

$$f_k = \sum_{m=-\infty}^{\infty} g_m h_{k-m}.$$

The sequence h_{k-m} is a reversed and shifted copy of h_m , so the calculation is the discrete version of sliding one pulse past another and adding the pointwise products.

For N -periodic sequences, the discrete circular convolution is

$$f_k = \sum_{m=0}^{N-1} g_m h_{k-m}, \quad k = 0, 1, \dots, N-1, \quad (14.1)$$

where every subscript is interpreted modulo N . Graphically, the samples lie on a circle rather than on an infinite line.

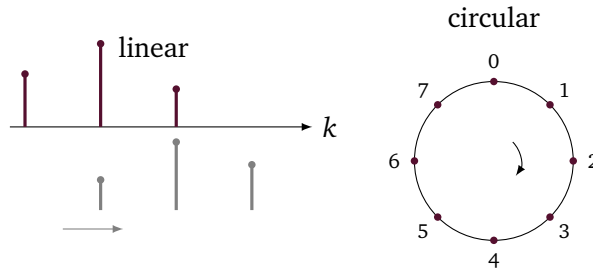


Figure 39: Linear convolution slides along a line; circular convolution rotates around a period.

Example 15 (Linear convolution versus circular wraparound). *Let*

$$g = (1, 2, 1), \quad h = (1, 1),$$

with all omitted samples equal to zero. The ordinary discrete linear convolution is

$$f = (1, 3, 3, 1),$$

since

$$f_0 = 1, \quad f_1 = 1 \cdot 1 + 2 \cdot 1 = 3, \quad f_2 = 2 \cdot 1 + 1 \cdot 1 = 3, \quad f_3 = 1.$$

The span count is $3 + 2 - 1 = 4$. If these same samples are placed in a period of length $N = 3$, the last value wraps into the first slot, giving

$$f^{(3)} = (2, 3, 3).$$

This is aliasing: the circular result is not the desired linear result. If instead $N = 4$, the circular period has enough room and the FFT circular convolution gives

$$f^{(4)} = (1, 3, 3, 1),$$

which agrees with the linear convolution over one period.

k	0	1	2	3
linear f_k	1	3	3	1
$N = 3$ circular	$1 + 1 = 2$	3	3	
$N = 4$ circular	1	3	3	1

Theorem 59 (DFT product rule for circular convolution). *Let f_k be the discrete circular convolution of g_k and h_k as in (14.1). If F_n , G_n , and H_n are their DFTs using the convention of Chapter 10, then*

$$F_n = G_n H_n, \quad n = 0, 1, \dots, N-1.$$

Proof. Starting from the DFT of f_k ,

$$F_n = \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} g_m h_{k-m} \right) W^{nk}.$$

Interchange the sums and set $r = k - m$, with indices understood modulo N :

$$F_n = \sum_{m=0}^{N-1} g_m W^{nm} \sum_{r=0}^{N-1} h_r W^{nr} = G_n H_n.$$

The modulo- N interpretation is what allows the shifted index $r = k - m$ to run over exactly one full period. $\square$

14.5 The Span Restriction

Circular convolution wraps around after N samples. If the desired output is a linear convolution, this wraparound must not overlap the main output.

If g_k has span count C_g and h_k has span count C_h , then their linear convolution has span count

$$C_f = C_g + C_h - 1.$$

Therefore an N -point circular convolution can reproduce the linear convolution without aliasing only if

$$C_g + C_h - 1 \leq N. \quad (14.2)$$

Equivalently,

$$C_g + C_h \leq N + 1.$$

This is the book's span restriction. It is not a numerical accuracy condition; it is a geometry condition. If it fails, the circular answer is still the correct circular convolution, but it is the wrong answer for linear convolution because a tail has wrapped into the front of the period.

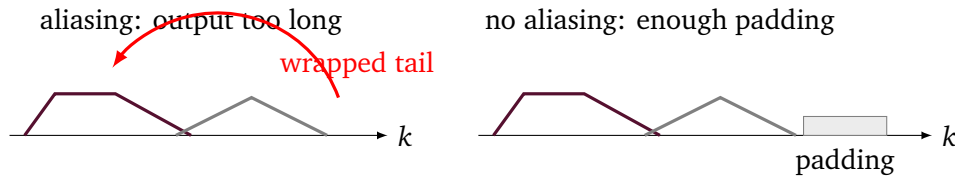


Figure 40: The span restriction prevents the circularly wrapped tail from overlapping the desired linear convolution output.

For continuous pulses supported in $[0, T_g]$ and $[0, T_h]$, the continuous linear convolution is supported in $[0, T_g + T_h]$. Thus the sampled FFT window must satisfy

$$T_g + T_h \leq T = NT_s$$

if the FFT computation is to emulate continuous linear convolution without time-domain wraparound.

14.6 Using the FFT for Convolution

The practical FFT recipe follows directly from Theorem 59. For discrete circular convolution:

1. compute $G_n = \text{DFT}\{g_k\}$ and $H_n = \text{DFT}\{h_k\}$;
2. form $F_n = G_n H_n$;
3. compute $f_k = \text{IDFT}\{F_n\}$.

For finite discrete pulses, add a preliminary step: choose N large enough so that $C_g + C_h - 1 \leq N$. Without that step, the FFT recipe is still correct, but it computes the circular convolution rather than the desired linear convolution.

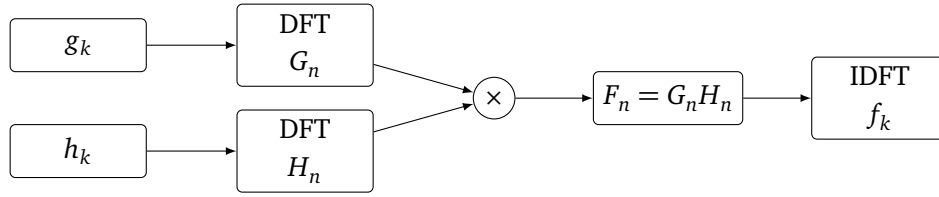


Figure 41: FFT convolution: transform both inputs, multiply the spectra term by term, then invert. The raw result is a discrete circular convolution.

If the inputs are finite discrete pulses and the span restriction (14.2) is satisfied, one period of the circular result equals the ordinary discrete linear convolution.

14.7 Continuous Convolution from FFT Samples

For continuous linear convolution of finite-span pulses,

$$f(t) = \int_0^T g(\tau)h(t-\tau)d\tau,$$

where T is the FFT observation window. With $t_k = kT_s$ and $\tau_m = mT_s$, the rectangular rule gives

$$f(t_k) \approx T_s \sum_{m=0}^{N-1} g_m h_{k-m}.$$

Thus the IDFT output from the FFT convolution must be multiplied by

$$T_s = \frac{T}{N}$$

to estimate the continuous convolution values. The factor T_s is the rectangle width from the Riemann sum. Forgetting it gives the right shape but the wrong physical scale.

For continuous circular convolution of T_0 -periodic functions,

$$f_p(t) = \int_0^{T_0} g_p(\tau)h_p(t-\tau)d\tau.$$

If

$$g_p(t) = \sum_{n=-\infty}^{\infty} G_p(n)e^{in\omega_0 t}, \quad h_p(t) = \sum_{n=-\infty}^{\infty} H_p(n)e^{in\omega_0 t},$$

then

$$F_p(n) = T_0 G_p(n) H_p(n). \quad (14.3)$$

The same time-domain scale $T_s = T_0/N$ converts the raw IDFT convolution result into samples of $f_p(t)$. For displaying a spectrum as Fourier-series coefficients, the FFT product $G_n H_n$ must be scaled by T_0/N^2 , because each raw FFT spectrum contributes a factor $1/N$ and the continuous circular convolution contributes the factor T_0 in (14.3).

14.8 Operation Count

Suppose two finite sequences each have span count L . Direct linear convolution requires L^2 complex multiplications and $(L-1)^2$ additions in the worst case. Using the FFT requires two DFTs, one inverse DFT, and N pointwise spectral multiplications. With an N -point radix-2 FFT this is approximately

$$\frac{3}{2}N \log_2 N + N$$

complex multiplications. Taking $N \approx 2L$ to satisfy the span restriction gives

$$C_{\text{FFT}} \approx L(3 \log_2 L + 5).$$

L	direct L^2	FFT estimate $L(3 \log_2 L + 5)$	direct/FFT
16	256	272	0.94
64	4096	1472	2.78
256	65536	7424	8.83
1024	1048576	35840	29.26

For short sequences, the overhead of several FFTs may not be worth it. For long sequences, the L^2 direct count grows much faster than the FFT-based count, which is why FFT convolution is so important in numerical signal processing.

15 Emulating Dirac Deltas and Differentiation on the FFT

15.1 Time-Domain Dirac Deltas on the FFT

A true Dirac delta has no finite sample value at its location, but an FFT data vector can emulate its two useful actions:

- it is zero away from its location;
- under a sampled integral, it contributes a prescribed weight at its location.

Let $T_s = T/N$. To emulate the weighted impulse

$$\mu \delta(t - k_0 T_s),$$

load the time vector with

$$x_k = \begin{cases} \mu/T_s = \mu N/T, & k = k_0, \\ 0, & k \neq k_0. \end{cases}$$

The scale $1/T_s$ is forced by the rectangular-rule interpretation:

$$T_s \sum_{k=0}^{N-1} x_k \phi(k T_s) = T_s \frac{\mu}{T_s} \phi(k_0 T_s) = \mu \phi(k_0 T_s).$$

Thus the sampled object has the same area action as the continuous impulse.

This is the most important point: the height of the loaded sample is not the impulse weight. The sample height is the impulse weight divided by the grid spacing. The product “height times sample width” is the weight.

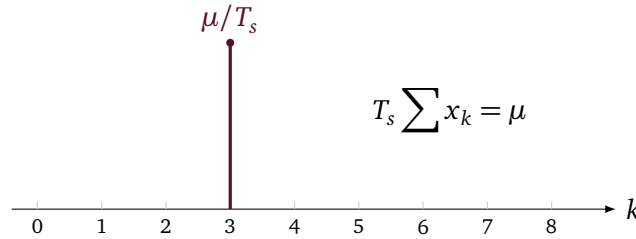


Figure 42: A time-domain Dirac delta is represented on the FFT grid by one nonzero sample of height μ/T_s , so that its sample area is μ .

The DFT of this vector is

$$X_n = \frac{\mu}{T_s} W^{nk_0}.$$

Multiplying by T_s to interpret the DFT as a continuous Fourier-transform estimate gives

$$T_s X_n = \mu W^{nk_0} = \mu e^{-i(2\pi n/T)k_0 T_s},$$

which is exactly the transform sample of $\mu \delta(t - k_0 T_s)$.

15.2 Frequency-Domain Dirac Deltas on the FFT

The dual construction represents an impulse in the frequency domain. Let the frequency grid spacing be

$$\Delta\omega = \omega_0 = \frac{2\pi}{T}.$$

To emulate

$$\mu \delta(\omega - n_0\omega_0),$$

it is useful to distinguish two related numbers. The continuous-spectrum bin height used in a Riemann-sum plot is

$$\hat{F}_n = \begin{cases} \mu/\Delta\omega = \mu T/(2\pi), & n = n_0, \\ 0, & n \neq n_0. \end{cases}$$

This is the same area logic as before, now on the frequency grid:

$$\Delta\omega \sum_n \hat{F}_n \Psi(n\omega_0) = \mu \Psi(n_0\omega_0).$$

The raw DFT/IDFT bin has a different scale, because the IDFT includes the factor $1/N$ rather than the continuous inverse-transform factor $1/(2\pi)$. Since $T = NT_s$, the raw bin to load is

$$F_{n_0}^{\text{FFT}} = \frac{1}{T_s} \hat{F}_{n_0} = \frac{\mu}{T_s \Delta\omega} = \frac{N\mu}{2\pi}.$$

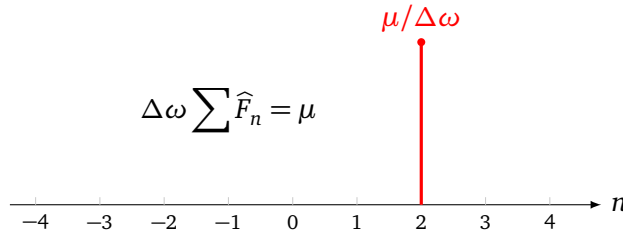


Figure 43: A frequency-domain Dirac delta has plotted continuous-spectrum bin height equal to impulse weight divided by bin spacing. The raw FFT bin has the additional IDFT scale shown in the text.

For example, the spectrum $2\pi A \delta(\omega - \omega_0 n_0)$ synthesizes to

$$Ae^{in_0\omega_0 t}.$$

The plotted continuous-spectrum height is $2\pi A/\Delta\omega = AT$. The raw IDFT bin must be NA , because

$$\frac{1}{N} NAe^{i2\pi n_0 k/N} = Ae^{i2\pi n_0 k/N}.$$

The same warning applies here: the plotted bin height is not the spectral impulse weight. The product “plotted bin height times bin spacing” is the impulse weight, and the raw FFT bin includes the extra $1/T_s$ scale.

15.3 Differentiation on the FFT

Continuous differentiation has the exact transform rule

$$\frac{df}{dt} \longleftrightarrow i\omega F(\omega).$$

On a finite FFT grid, the data are periodic and discrete, so ordinary derivatives are replaced by finite differences. Because shifts have simple DFT multipliers, these finite differences can be applied directly to the FFT spectrum.

With $W = e^{-i2\pi/N}$, the shift rule is

$$f_{k-m} \longleftrightarrow W^{nm} F_n.$$

Therefore the main difference formulas have the following spectral multipliers:

operation	time-domain formula	multiply F_n by
backward first difference	$\frac{f_k - f_{k-1}}{T_s}$	$\frac{1 - W^n}{T_s}$
forward first difference	$\frac{f_{k+1} - f_k}{T_s}$	$\frac{W^{-n} - 1}{T_s}$
central first difference	$\frac{f_{k+1} - f_{k-1}}{2T_s}$	$\frac{W^{-n} - W^n}{2T_s} = \frac{i \sin(2\pi n/N)}{T_s}$
central second difference	$\frac{f_{k+1} - 2f_k + f_{k-1}}{T_s^2}$	$\frac{W^{-n} - 2 + W^n}{T_s^2} = -\frac{4 \sin^2(\pi n/N)}{T_s^2}$

For small $|n|$, the central first-difference multiplier is close to

$$\frac{i \sin(2\pi n/N)}{T_s} \approx i \frac{2\pi n}{NT_s} = i \frac{2\pi n}{T},$$

which is the continuous multiplier $i\omega_n$.

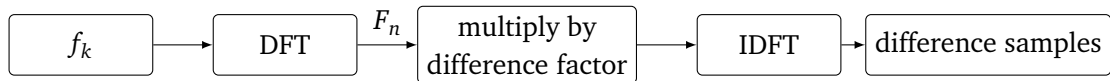


Figure 44: FFT differentiation applies a finite-difference multiplier in the frequency domain, then transforms back.

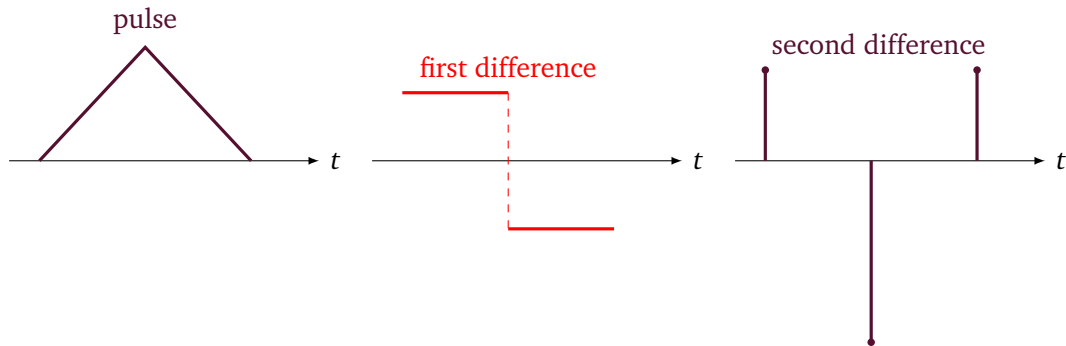


Figure 45: A triangular pulse differentiates to a two-level pulse; differentiating again produces a set of impulse-like samples at the break points.

Because the FFT is periodic, every finite-difference rule above is also periodic. A jump at the end of the data window is treated as adjacent to the beginning of the window. When the data represent a nonperiodic pulse, enough zero padding is needed to keep this artificial endpoint connection from contaminating the desired region.

For implementation, the chapter can be reduced to the following checks:

- To emulate a time impulse of weight μ , load height μ/T_s .
- To emulate a frequency impulse of weight μ , use plotted height $\mu/\Delta\omega$, but load raw IDFT bin height $N\mu/(2\pi)$.
- To differentiate FFT data, decide which finite difference is intended and multiply by its exact discrete multiplier, not by $i\omega$ unless a continuous spectral derivative is deliberately being approximated.
- If the data are not truly periodic, pad or window them so the endpoint connection does not create a false jump.